

**An open source software package to compare and extend effective mass models
of the phosphorous donor in silicon.**

by

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January 23, 2024

A dissertation submitted to the faculty of the Graduate School of the University at
Buffalo, The State University of New York in partial fulfillment of the requirements
for the degree of

Doctor of Philosophy
Department of Physics

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I dedicate this work to those who have been with me every step of the way in this process, holding me up and pushing me forward. I have benefited greatly from the support of my father and mother, as well as the support of my sister, Katie and my brother-in-law, Jonathan. There is of course, a special dedication to be made. Without the endless support and love of my wife, Aniqua Hasan, I don't think this dissertation would ever have been finished and defended.

Acknowledgements

First, I need to acknowledge the steady support of my PhD advising committee. In my principal advisor, Dr. Xuedong Hu, I found a strong voice to guide me through the substantial challenges I've had as I've worked to complete this PhD. Dr. Hu's substantial patience and insight regarding my personal and work habits is a large part of what brought this dissertation work to completion. I want to thank Dr. Doreen Wackeroth for her patient enthusiasm and her physics perspective from outside solid state. Finally, Dr. Jong Han was invaluable for his hard questions and challenging insights.

The other physics students I've met at UB have been an important source of perspective and wisdom. Science is inherently a communal and human pursuit. I especially want to call out the advice from and conversations with Omar Elsherif, Courtney Fitzgerald, John Truong, Bilal Tariq, Lauren Hay, Xinyu Zhao, Jiawei Wang, and Zongye Wang.

Abstract

We have put together *Cantankerous Impurity*, an open source software package to explore the limits of the effective mass model. Evaluating effective mass models of an isolated phosphorus donor in silicon requires disentangling various sources of error. In an effort to suppress the error resulting from state approximation, we construct states using envelope functions expanded in freely extensible basis sets equipped with tunable parameters. Additionally, a full consideration of the effect of symmetry is applied to this basis set. This robust symmetrized basis set allows us, in principle, to compute arbitrarily precise approximate eigenstates of the effective mass Hamiltonian. We can, in principle, closely approximate the exact donor electron wavefunction for a broad class of model potentials, which in turn allows us to evaluate and compare extant and new effective potential models.

In this dissertation, we'll describe the history of the effective mass model used to describe the motion of an excess electron around an isolated substitutional impurity within silicon. We'll use that history to construct a suite of software that will allow us to explore a very broad class of model potentials. We reveal and explain some of the features of the software, and present a few benchmarking results.

All materials related to *Cantankerous Impurity* can be found [online](#).

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Chapter 1

Context

1.1 Overview of Quantum Computing

1.1.1 Why quantum computing?

Interest in quantum computing is high, reaching well beyond academic interest, with many extant public and private sector projects seeking to develop, display, and eventually exploit quantum computation. This interest motivates a great deal of intellectual and economic investment in physics, computation, and information science. On the commercial side IBM, NTT, Intel, and Toshiba lead the way in patent applications as of 2020[13]. Additionally, governments around the world have invested the equivalent of billions of USD in initiatives on quantum technologies, such as the European Union, France, Germany, Japan, Canada, and the US among others[3].

Attempting to prophesy on the future feasibility of useful quantum computation is well beyond the scope of this document and the author's knowledge and abilities, of course. John Preskill, a leading researcher in this field, labels the current era of quantum computing to be the NISQ era, the era of *noisy intermediate-scale quantum* computing[14]. As such we are not yet able to leverage quantum computing to accomplish many of the most referenced applications of quantum computing.

However, the author feels that such global interest in the fundamental ideas of quantum mechanics and in understanding of computation and information beyond the classical represents an exciting chance to learn ever more about the deeper nature of our universe and reality.

Our understanding of the world is not well posed to process quantum information. To some degree, this can be reflected in our idea of "measurement" within quantum mechanics. Consider the Stern-Gerlach experiment[6]. After a stream of silver atoms are passed through an appropriately directed inhomogenous magnetic field, the path of the silver atoms will be deflected according to relative orientation between the magnetic field and the atom's magnetic moment. The path of the atom may be recorded as a spot on the receiving plate. When the experiment is performed many times two clear clusters of spots emerge, corresponding respectively to the orientation of the atom's magnetic moment completely along or against the external magnetic field. This was an early success of quantum mechanics, as the theory predicted such an outcome would be seen. Both quantum mechanics and classical mechanics predict that, prior to the interaction of the silver atom with the external magnetic field, the atom exists in a state of arbitrary orientation. Classically, an atom in such a state would have a component of its magnetic moment lying along or against the external magnetic field. The magnitude of this component is arbitrary, and so is the degree of deflection. Quantum mechanics, on the other hand, predicts the arbitrary initial state of the magnetic moment to take the form of a superposed wavefunction with components lying entirely along or against the external magnetic field. Such a two-level system is important to quantum computing, as we will describe later, and is conventionally represented as

$$|\Psi\rangle = \alpha |0\rangle + \beta |1\rangle \tag{1.1}$$

As such, the effect of the external magnetic field upon the moment of the atom is

discrete, as should be the position of the atom upon the detector plate. But when does the superposition of states become one or the other? We can easily formulate the two recorded positions of the atom on the detector plate as being in a superposition. Does the resolution of the superposition happen when the atom interacts with the magnetic field. Perhaps it occurs in the human eye recording the spot on the plate, or perhaps even further as result of visual information processing in the human brain, or perhaps we have to resort to a semi-mystical idea of a "consciousness" that measures the state of a human's physical brain?

Such debates are well beyond the scope of this document, the author, and unless clear experiments can be devised to test the implications of such interpretations, they are firmly beyond the scope of physics. For our purposes, we informally term "measurement" as the process by which classical information is extracted from a quantum system. Perhaps anticipating the adage "shut up and calculate!" variously attributed to Dirac or Feynman, Von Neumann discussed this process and concluded that the exact place where the "measurement" takes place has no impact upon the fundamental mathematics of the theory[21].

Much of quantum computing's promise comes from the idea that quantum information is a more robust concept than classical information, where an exchange of information between particles at a point in time allows for more correlation between those particles' states when later examined than is possible in a classical setting, as discussed by Bell in his famous paper on Bell's inequalities[1]. This property, known as *entanglement*, can be interpreted as an exchange of classical information in a non-local fashion, or an example of reality's failure to be governed purely by classical information. This indescribability of the quantum universe in classical terms may be asserted in the form of a principle that might be called "counterfactual definiteness" or in the idea associated with the so-called Everett Many-world hypothesis that measurement fundamentally doesn't physically exist. A result to a classical entity that has

a single classical outcome has, in fact, many classical outcomes that exist as part of the superposition of a larger quantum system including the system at hand, its environment, and the observer. While the present author subscribes somewhat to the "shut up and calculate!" adage, in the interests of full disclosure, the present author has found the most conceptually useful philosophical framework for anchoring his mathematical and physical work to be Everett's many worlds with decoherence models for wave functions collapse. We disclose this not to firmly assert a "correct belief" or factional affiliation but because we want to aid communication with the reader. We anticipate that while such philosophical sympathies might color our mode of description we believe the results stand on their own physically and mathematically.

One of the ways in which quantum information might be said to be more robust than classical information lies in the possibilities of encoding classical information. Firstly, quantum particles exist in a superposition of classically ponderable states. We saw this above when we considered the state of the Stern-Gerlach particle prior to measurement. To that end, we will formally define the *qubit*. Quantum error correction as a field recognizes a difference between the simple physical qubit and the logical qubit. The logical qubit is an assemblage of associated physical qubits working together to produce a qubit of information that may be robustly involved in quantum computation without unacceptable error. We focus in this document on the physical qubit. At its core the physical qubit is a two-level system that may be described by a two-term wavefunction of the form introduced above

$$|\Psi\rangle = \cos(\theta/2)|\mathbf{0}\rangle + \sin(\theta/2)e^{i\phi}|\mathbf{1}\rangle. \quad (1.2)$$

With $\theta \in [0, \pi)$ and $\phi \in (-\pi, +\pi]$ we can compare this to classical physical bits, which are likewise two-level physical systems, but in the bit the system is either in the state $|\mathbf{0}\rangle$ or $|\mathbf{1}\rangle$. We can already see that storing the precise state of a two-level quantum system in a classical computer will require two complex numbers, or about 256 bits

of storage. For the qubit and other two-level systems, asserting a normalization condition and dropping an overall phase factor allows us to represent a qubit using two real numbers θ and ϕ as above. Two real numbers are about 128 bits in most computing systems. Already we can see a clear difference between qubit and bit, but for one qubit it is not so difficult to find 128 spare bits to work with. We might be tempted to imagine that a system of bits could be represented with two complex numbers for each qubit. This is not the case. A linear increase in the number of qubits requires an exponential increase in the number of bits due to the aforementioned entanglement. The arbitrary states of a two-level quantum system are described by a wavefunction of the form

$$|\Psi\rangle = a_{00} |00\rangle + a_{01} |01\rangle + a_{10} |10\rangle + a_{11} |11\rangle \quad (1.3)$$

which requires four complex numbers. Asserting normalization and arbitrary overall phase only reduces the representation budget by 128 bits, and becomes increasingly irrelevant as system size increases. As we examine the wavefunctions of larger and larger systems of qubits, we can see that the number of complex numbers needed to represent the system goes as $\approx 2^N$. This makes many-body systems with many quantum degrees of freedom very difficult to simulate classically in general, though perhaps for individual problems simplifications and analytic shortcuts can be found to reduce the computational overhead. Still, bespoke solutions to computational expense just swaps computer workhours for human ones, and generally human work hours are more expensive.

This then, is the most exciting prospect of quantum computing for the present author. A truly digital, general quantum computer will make entire domains of problems much more amenable to simulation and free some of those expensive human workhours to survey much broader swathes of physics and physical understanding.

Beyond the promise of quantum simulation, the quantum computing project is fundamentally a practical one, with much of its motivation focused on the applica-

tions of a quantum computing device to specific problems. Some of the well-known algorithms that are used to demonstrate a quantum advantage include Shor's algorithm[18] and Grover's search algorithm[7]. Shor's algorithm is a way to factor numbers, such as 15 into 3 and 5, in polynomial time. This is far from a trivial application – easy factoring of numbers has dramatic practical implications, particularly in RSA cryptography[15]. For RSA cryptography, a key component involves the idea that the complexity for factoring a number scales exponentially with the size of the number. To date, no polynomial time algorithm to factor numbers for classical computers has not been found. While it has been proven that classical polynomial factoring is impossible, it is not expected to be found anytime in the near future. This is not the case for Shor's algorithm, which can factor in polynomial time. As easy factoring would break RSA encryption, and as RSA encryption underlies much of the secure communication over the internet, a practical implementation of Shor's algorithm would have devastating security consequences as a zero day exploit.

1.1.2 Implementing quantum computing

In 2000, DiVincenzo presented a set of five criteria a physical system would need to be able to demonstrate to be considered a workable implementation of quantum computing.

1. A scalable physical system with well characterized qubits.
2. The ability to initialize the state of the qubits.
3. Long relevant decoherence times, much longer than the gate operation time.
4. A "universal" set of quantum gates.
5. A qubit-specific measurement capability.

The proposal for quantum computing currently seeing the most interest in the field is superconducting qubits. These qubits use superconducting Josephson junctions to create and address two-level systems. These two-level systems are generated, in essence, through anharmonic charge/current quantum oscillations produced as a result of the nonlinear inductance of the Josephson junction. The anharmonicity is necessary to lift the equal level spacing between states of the linear harmonic oscillator. Thus, such anharmonicity allows the levels acting as $|0\rangle$ and $|1\rangle$ to separately address from the other oscillator levels[19, 9]. As these qubits are produced using extant solid state techniques, they can take advantage of the preexisting techniques and community that has developed around micro- and nanomanufacturing of silicon for the classical computing industry. However, the superconducting nature of these qubits require substantial cooling apparatus, which increases the physical footprint of each qubit substantially. This large physical footprint makes scalability and practical commercial production difficult. Additionally, since the qubits are manufactured, their energy levels and other characteristics are not consistent from qubit to qubit. To ensure desired and predictable operation of the quantum computer, many control parameters have calibrated, and the number of control parameters increases superlinearly as the number of qubits increases[10, 22].

Another popular proposal involves using ionized atoms laser-cooled and suspended by electric fields. These ions are held in linear chains. The qubits are encoded on two atomic levels, with either hyperfine separation in the ground state or optical energy separation between the atomic ground and first excited state. Unlike with superconducting qubits, characterization and calibration are not a potential problem; The properties of each ion are essentially identical. However the trapped ion qubits display problems with scalability – it is difficult to create longer chains of trapped ions. The problem is not with trapping the ions – many numbers of ions can be trapped at a time, but such large numbers of ions do not permit individual addressability and

readout. The bottleneck lies in developing the optical and electronic control of the trapped ions as the chains increase in size.

In addition to the superconducting and trapped ion qubits, there are more technologies being explored. Of some interest in the field are qubits hosted in semiconductor systems without superconductivity, and so without the cooling requirements of the superconducting qubit. Our interest in this field are specifically in the proposals that use systems embedded in silicon crystals. Silicon is an economically significant material, with it being used in the construction of integrated circuits. The importance of the integrated circuit as well as other semiconductor components to modern life cannot easily be overstated, with electronic devices made up of semiconductor components present in some fashion in every sector of the world economy. Proposals for quantum computing that focus on silicon recognize that humanity, as a whole, has developed substantial skills in understanding and manipulating silicon. The previous quantum computing proposals come with not-insubstantial requirements for accessory technologies, with much development time likely required to reduce the size and both initial and ongoing cost of these secondary components.

Due to the importance and maturity of the extant semiconductor electronics industry, the requirements for the secondary accessory technologies to create silicon quantum computers are expected to be much less strenuous as compared to those of the superconducting and trapped ion qubits. However, silicon quantum computing still needs to show the ability to satisfy the DiVincenzo criteria. In particular, groups struggle to make high-fidelity two-qubit gates, which has implications for scalability. However quantum algorithms have been implemented on a programming two-qubit processor.

1.2 Historical Narrative of EMA

1.2.1 Kohn-Luttinger and donor states in silicon

In a 1955 paper[12] W. Kohn and J.M Luttinger derived an expression for how electrons and holes would move in the presence of a localized perturbation to the periodic field, in particular the periodic fields of the Si and Ge semiconductors. These semiconductors do not have a simple conduction band with a single energy minima – instead they have 6(Si) or 8(Ge) equivalent minima lying along different directions in $\mathbf{k}$ -space. In a later paper that year[11], Kohn and Luttinger (hereinafter referred in this section as KL) presented an application of the effective mass theory to donor states in silicon. An excess electron, bound by an impurity in silicon with an excess positive charge, becomes trapped in the crystal. Such a system can be referred to as an electron donor, conventionally just called a donor. In silicon, the focus is on the Group V substitutional impurities, such as phosphorous, which feature an outer electron shell that matches the outer shell of the silicon atom. This allows the phosphorous to participate in the crystal structure as though it were silicon. The net effect to the system through the introduction of this impurity is some minor deformation of the lattice at or near the donor site, and most importantly that such an atom participating in silicon bonds will have an excess positive charge that isn't satisfied by the electrons from its crystalline neighbors. This creates the situation above, where an excess electron is localized in the region of the donor. While deep donors exist, we limit our focus here to "shallow" donors. We will save further discussion of "shallowness" and its impact on the definition of the effective mass approximation in Section 2.1.1.

KL examined a single-valley model of the donor electron wavefunction that went as so

$$\Psi_{\mathbf{r}} = F_{\mathbf{r}} e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{0\mathbf{k}_q; \mathbf{r}} \quad (1.4)$$

Above $\mathbf{k}_q$ is the point in k -space where the energy minima of the lowest conduction

band are located. F represents an envelope function that introduces spatial localization of the wavefunction. We can see that the wavefunction takes the form of a conduction band wavefunction attenuated and made a bound state by a leading envelope function factor.

KL did not extend their model to consider intervalley coupling, and so this model's ground state manifold is 6-fold degenerate. In actuality, intervalley couplings lift this 6-fold degeneracy to create a 1-fold A_1 state, 2-fold E state, and a 3-fold T_2 state, with A_1 , E and T_2 referring to the particular symmetries of the overall state. For now, we will table the discussion of multi-valley states. We will return to this topic in the next section. KL asserted that in the "shallow" limit when the effective mass approximation can be applied, F will obey the following equation.

$$\left[-\frac{\hbar^2}{2m_{\parallel}} \frac{\partial^2}{\partial x^2} - \frac{\hbar^2}{2m_{\parallel}} \frac{\partial^2}{\partial y^2} - \frac{\hbar^2}{2m_{\perp}} \frac{\partial^2}{\partial z^2} - \frac{e^2}{4\pi\epsilon\sqrt{x^2+y^2+z^2}} \right] F_{\mathbf{r}} = E F_{\mathbf{r}} \quad (1.5)$$

This is a nonseparable partial differential equation, and no closed form exact solution is known. However, we can make an educated guess as to the form of the envelope, and KL predicted that the envelope could be approximated as

$$F_{\mathbf{r}} \approx F'_{ab;\mathbf{r}} \equiv \frac{1}{\sqrt{\pi a^2 b}} e^{-\sqrt{\frac{x^2+y^2}{a^2} + \frac{z^2}{b^2}}}. \quad (1.6)$$

It is convenient to assert the above function is normalized in the sense that

$$\int d\mathbf{r} |F'_{ab;\mathbf{r}}|^2 = 1. \quad (1.7)$$

With a trial function such as this, one which includes arbitrary parameters, one can compute the expectation energy manifold as a function of the arbitrary parameters.

KL formally combined (1.5) and (1.6) to create an effective energy function.

$$E_{ab} = \frac{\hbar^2}{2m_{\parallel}} \int d\mathbf{r} \tilde{F}'_{ab;\mathbf{r}} \left[-\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \gamma \frac{\partial^2}{\partial z^2} - \frac{e^2}{4\pi\epsilon r} \right] F'_{ab;\mathbf{r}}, \quad (1.8)$$

$$= \frac{\mathfrak{E}\mathfrak{r}^2}{3a^2} \left[2 + \gamma \left(\frac{a}{b} \right)^2 - 6 \left(\frac{a}{b} \right) \left(\frac{a}{\mathfrak{r}} \right) \frac{\text{ArcTan} \left(\sqrt{\left(\frac{a}{b} \right)^2 - 1} \right)}{\sqrt{\left(\frac{a}{b} \right)^2 - 1}} \right]. \quad (1.9)$$

The quantities $\mathfrak{E}$ and $\mathfrak{r}$ are system units. Their definition and values may be found in Appendix A. $\gamma = m_{\parallel}/m_{\perp}$ is the effective mass anisotropy. We may implement a and b as unitful or dimensionless variational parameters. If a dimensionless representation is desired, we may take $a, b \rightarrow \mathfrak{r}a, \mathfrak{r}b$. When that is done, the $\mathfrak{r}$ drops out of the equation above and the minimization problem becomes purely numerical.

It can be shown by applying the calculus of variations to quantum mechanics that an energy eigenstate is an energetic stationary point in state space, and in particular is a minimum of energy. If selecting from a parameterized subspace, the problem goes from the calculus of variations to normal real number calculus. In the *variational method*, we can extend this principle and suggest that the best approximation to the energy eigenstate consists of selecting a trial wavefunction from a parameterized set by optimizing the parameters to minimize the energy.

Using a value of $\gamma = 0.207901$, we can minimize (1.9) by varying a and b . If we do so, we find

$$E_{\min} = -1.5653\mathfrak{E}, \quad (1.10)$$

$$a_{\min} = 0.74832\mathfrak{r}, \quad (1.11)$$

$$b_{\min} = 0.43001\mathfrak{r}. \quad (1.12)$$

This is a ground state energy of $E_{\min} = -31.1$ meV. Since the KL model does not include intervalley coupling, the model predicts a six-fold degenerate ground state. In Chapter 4 we will explore in greater detail how our software package can directly repli-

cate and freely extend these results. In particular, we wish to examine the implications of introducing multi-valley interactions.

In their paper, KL suggest a few more trial wavefunctions to approximate various excited states of the silicon donor. KL found reasonable, but not exact, agreement with the excited states. We believe that it is worthwhile to consider the sources of error within the KL model and procedure, especially as one considers an approach to improve or replace parts of the model.

As an initial examination, we might consider, compare, and contrast the following two sources of error. To begin with, Let's consider the effective mass approximation itself. The singular nature of the positive charge center of the donor is likely to cause some degree of interband coupling, leading to a degree of divergence between the predictions of the model and the values obtained via experiment. However, it seems uncertain to this author what degree of violation of the EMA is induced by the positive charge center. Additionally, the KL model neglects intervalley contributions to the expectation energy.

Consider the process proceeding from choosing initial assumptions and model constructions, all the way to error analysis of the final results. We can imagine that introducing corrections to the EMA model or using a different model will introduce a change to that process very early on, likely introducing many downstream effects. If we were wanting to improve our predictions in a timely manner, we might instead explore a refinement to the prediction process farther down stream.

As an example of such a downstream error that might be examined is the error introduced by the use of a variational wavefunction. A variational wavefunction, in general, is an approximation. The quality of that approximation depends on many factors, and so cannot be easily refined to produce high-quality improvements to the wavefunction.

An alternative we can imagine is to express the wavefunction using a functional

basis spanning the Hilbert space of the system in question. Additionally, we can imagine this basis itself being equipped with variational parameters allowing for refinement and optimization. Such an approach, combining a basis representation of the wavefunction with the variational method, is known as the *Ritz method*.

We can imagine that resolving these two sources of error will each introduce very different costs. The error that lies farther upstream in the prediction process will cost more analytical and software development time. In a broader collaboration, we can also imagine that such an error will require response from and delegation of work to more parts of the collaboration, introducing additional time cost in communication and management. An error farther downstream in the prediction process, on the other hand, will incur far less time and work costs. We can also imagine that development of such an improvement could be isolated to smaller working groups of the collaboration, possibly even a single individual.

Here the authors assert that here we are beginning to see a distinction in sources of error between what we are calling *technical* and *model* error. We will explore in more detail what is meant by our usage of these terms, but as a starting point we can loosely identify that errors lying farther up the prediction process are *model* errors while those lying farther down the prediction process are *technical* errors.

The exact line between these two categories is contextual and somewhat arbitrary, and indeed it might be more philosophically accurate to identify the two as poles in an interpretational spectrum. Regardless of philosophical or formal ambiguity, we still find the above a useful distinction. As we identified earlier, this usefulness is particularly pointed when we consider how error management will be allocated amongst researchers and technicians in a larger academic or commercial research collaboration. Even the individual researcher might find themselves in a collaboration of sorts, even if the only members of such a collaboration are future and past versions of themselves. For now, we will table the discussion of error analysis or improvement of the KL

model and revisit it as part of the historical narrative of this document.

While we can continue examining the other variational wavefunctions employed by KL in their exploration of the excited donor states of silicon, we find it convenient at this juncture to move forward. In the next section, we will examine a significant improvement upon the bespoke single-state variational method and discuss how R.A. Faulker used the Ritz method.

1.2.2 Faulkner and the Ritz method

In a 1969 paper[2], R.A. Faulkner took KL's idea and applied the Ritz method. The Ritz method consists of the observation that a trial function of the form

$$|\Psi\rangle = \sum_i^N c_i |\Psi_i\rangle, \quad (1.13)$$

where Ψ_i comes from a finite set of known functions. They are not required to be orthogonal. The expectation energy of this state has the form

$$\varepsilon = \frac{\sum_{ij} c_i^* c_j \langle \Psi_i | \hat{H} | \Psi_j \rangle}{\sum_{ij} c_i^* c_j \langle \Psi_i | \Psi_j \rangle}. \quad (1.14)$$

It can be shown that applying the variational method to the c coefficients generates the following result

$$|\mathcal{H} \mathbf{c} - \varepsilon \mathcal{S} \mathbf{c}| = 0. \quad (1.15)$$

where

$$\mathcal{H}_{ij} = \langle \Psi_i | \hat{H} | \Psi_j \rangle, \quad (1.16)$$

$$\mathcal{S}_{ij} = \langle \Psi_i | \Psi_j \rangle. \quad (1.17)$$

We can state this conclusion another way; namely, the variational method applied to a state constructed from a finite non-orthogonal basis is equivalent to the *generalized eigenvalue problem* of the form

$$\mathcal{H} \mathbf{c} = \varepsilon \mathcal{S} \mathbf{c}. \quad (1.18)$$

Faulkner added an additional feature to this model. He added additional variational parameters to the trial wavefunctions themselves, which allows the Ritz eigenvalues to be optimized. Working from the same model of the single-valley equation for the envelope function that KL used, namely (??), Faulkner postulated a trial wavefunction of the form

$$F_{\mathbf{r}} = \sum_{nlm} c_{nlm} \left(\frac{\beta}{\gamma} \right)^{\frac{1}{4}} \Psi_{nlm}(\alpha_{lm} x, \alpha_{lm} y, \alpha_{lm} (\beta/\gamma)^{1/2} z). \quad (1.19)$$

with $\gamma = m_{\perp}/m_{\parallel}$. Faulkner chose the bound states of the hydrogen atom as his basis, which we denote as Ψ_{nlm} . Faulkner introduced variational parameters, an overall anisotropy parameter β as well as a radial parameter of the form $r \rightarrow r \alpha_{lm}$.

This method proved quite effective at matching the energies of the excited states of donors in both silicon and germanium, but we see three primary problems with Faulkner's methodology.

1. As before, by neglecting the multi-valley effects, the predicted ground state is 6-fold degenerate rather than being separated into the A_1 , E and T_2 states.
2. The bound states of the hydrogen donor do not constitute a complete set of functions spanning the separable Hilbert space. This is somewhat relieved by allowing the α radial parameter to rescale the hydrogen wavefunctions, meaning it can capture smaller features. However, to maintain orthonormality, all α_{ml} parameters of functions with the same m and l are set equal. This limits the degree of angular structure that can be present within the function.

3. We have found it practically difficult to find a single set of orthonormal basis functions that converge well to all desired states. In fact, we might suggest that the only other such orthonormal complete set that might suffice with great efficiency to cover all desired eigenstates is the actual energy eigenstates themselves, though we cannot explicitly prove this point. For our purposes, we consider joined sets of multiple orthonormal bases with independent variational parameters. We term the orthonormal subsets the *subbases*. Each subbasis has its own set of anisotropic and radial variational parameters. The collected joined non-orthonormal set of all basis functions is termed the *superbasis*.

1.2.3 Shindo-Nara and the multi-valley EMA equation

The first known attempt to extend the single-valley Kohn-Luttinger theory of silicon donors can be found in an appendix to a 1962 paper by H. Fritzsche[4]. The appendix was authored by W. D. Twose[20]. This multi-valley EMA equation was used by some authors as they explored the best form of the impurity potential to get good quantitative results that match experiment. In 1976, Shindo and Nara[17] presented a paper that noted the Twose EMA equation made two assumptions in its derivation, namely

$$\frac{1}{(2\pi)^3} \int d\mathbf{k} e^{i\mathbf{k}(\mathbf{r}-\mathbf{r}')} u_{\mathbf{k}_q\mathbf{r}} \rightarrow u_{\mathbf{k}_q\mathbf{r}}. \quad (1.20)$$

and that

$$u_{\mathbf{k}_p\mathbf{r}}^* u_{\mathbf{k}_q\mathbf{r}} = 1. \quad (1.21)$$

Even at the time, Twose noted that these assumptions would likely need to be abandoned as central cell corrections to the impurity potential were introduced, especially since the strongly real-space localized central corrections would be quite broad in $\mathbf{k}$ -space.

We shall henceforth refer to Shindo and Nara collectively as SN in this section. To

move beyond the assumptions above, SN introduced and derived a new multi-valley expression that included the Bloch functions, including their complicated periodic factors, explicitly. For our own work, we find including as much detail about the Bloch functions as possible within the initial derivation of our model to be the most useful way to construct the model. If computational resources are limited, the Bloch functions can always be approximated by simplified or truncated forms at a later point in the model derivation or in the explicit construction of the overlap, Hamiltonian, and Hamiltonian-squared matrix elements. We replicate this feature within our software, with the ability to turn on or off the Bloch function contributions selectively.

The equation we derive in Chapter 2 owes heavily to SN’s derivation, though we expand the derivation by computing an EMA expression for the Hamiltonian-squared matrix, and by extension, an EMA estimate of the variance of our approximate energy eigenstates. We hope to use the variance in the future to provide an additional metric on the quality of our approximate eigenstates, in addition to spectrum replication.

Another way in which our derivation differs from SN lies in that SN remove intervalley terms involving the effective kinetic operator. Our own explorations with the variance of our states suggested dropping these terms introduces negative variances. While we agree that in computations, we should expect that these terms will be very small compared to the other contributions, we believe that these terms should be minimized by an appropriate choice of impurity potential and center cell correction, rather than being removed entirely. As it is conventional to drop these terms in most explorations of the EMA, we implemented a toggle within the code to allow the user to include the intervalley kinetic terms if desired.

1.2.4 Gamble and a tetrahedral potential

In a 2015 paper[5], J.K. Gamble et al. employed the Shindo-Nara EMA equations for the silicon donor and used Fourier expansions of the Bloch function. Gamble found

that including these contributions from the Bloch functions necessitates including a tetrahedral model potential as a central cell correction to the screened Coulomb potential of the excess positive charge of the substitutional donor atom. This central cell correction, by employing potential "wings" lying along the bond directions as well as a central potential, is seen to have the same T_d point symmetry as the donor point-defect system. Gamble applied this tetrahedral model potential and employed a Ritz basis of Gaussian trial wavefunctions to examine the spatial energy wavefunctions of the donor electron. A recent direct measurement of the donor [16] wavefunctions found agreement with the predictions of another method to model the donor system, that of atomistic tight-binding simulations.

The particular interest here of Gamble et al. was examining the factors under which a particular EMA model of the donor wavefunction could be made to agree with the tight-binding results and the measured reality. Of particular note, the Gamble method involved optimization of the parameters of the central cell correction to fix the resultant energies of the donor model to the experimentally measured energies. In a 2015 paper [5], J.K. Gamble et al. employed the Shindo-Nara EMA equations for the silicon donor and used Fourier expansions of the Bloch function. Gamble found that including these contributions from the Bloch functions necessitates including a tetrahedral model potential of the same symmetry as the donor point-defect system.

To the present author, Gamble et al.'s largest contribution to this work consisted in recognizing the importance of inclusion of the full Bloch functions in computations involving the EMA and the importance of symmetry upon the central cell corrections. As well, the Gamble paper's commentary on the ways in which small differences or different approximations employed in the EMA model can lead to very different quantitative predictions inspired the present author's own research along these lines. It led us to focus instead on capturing the broader landscape of theories that may exist in a single computational tool. As well, it inspired us to consider the distinction between

technical error, and *model* error. We will define what we mean by these terms below. However, the distinction is fundamentally prompted by these questions; "Can EMA models be brought in line with experimental data? If they cannot, what does that tell us about what models can be used to analyze a shallow donor system?"

1.2.5 Modelling and sources of error

In the previous section, we discussed a distinction between *technical* and *model* error. We consider a *technical* error to result from computation, mathematical, or technical concerns. For example, the error that may be incurred as a result of employing numerical integration, or the error that results from using a variational wavefunction, or from truncating an infinite basis to allow conventional numerical eigensolver methods to be employed. We expect this error to be, by definition, able to be reduced by the allocation of more resources. The numerical integrator's error tolerance can be lowered, or the size of the finite basis can be expanded. These kinds of errors may not be practical to reduce, but we have designed our software to be scalable in principle so more resources can be delegated to reduce these technical errors.

This is to be distinguished from *model* error, which we consider to be those errors inherent to the set of approximations made in deriving the Shindo-Nara equations. If the reality of the donor wavefunction is that its $\mathbf{k}$ -space wavefunction is not confined, in some sense, to the conduction band minima, we will see significant deviations between the results of the model and actual experiment as conducted on the donor.

Of course, in some cases, we expect the division between these errors to be unclear, or possibly subjective. For example, the present author considers the best effective potential to use within the model to be an open question. It has been well-known that the screened Coulomb effective potential becomes increasingly inaccurate in the local environment of the donor, where the assumptions of the screening from the silicon crystal breaks down. It is thought by many that this potential model

can be corrected by adding a central cell correction. Finding the best possible effective potential is an example of reducing technical error. Any errors that remain even when employing this hypothetical best potential, perhaps due to many-body effects that can't be represented with a mean field approximation, should be considered an example of model error.

Additionally, by focusing on comparing the EMA wave functions with other methods and with experiment, Gamble et al. implicitly suggested that additional metrics are necessary to evaluate a particular model. This inspired us to compute a Shindo-Nara effective mass equation for the Hamiltonian-squared matrix. This allows us to estimate the variance of our approximate eigenfunctions, which will tend towards zero as our states becomes a better approximation for a particular eigenstate. We can compare and contrast this with the results obtained for the expectation energy of the same to get better insight into how these states converge to the exact wavefunctions predicted within the EMA.

Chapter 2

Analytical Model

2.1 Derivation of EMA Energy Functional

2.1.1 Localized shallow semiconductor heterostructures in the bulk

The interruption of the crystalline structure of the semiconductor, as with impurities or heterostructures, can serve to capture excess electrons within the semiconductor. We wish to examine the physics of these captured electrons. The Hamiltonian of such a system may be written as

$$\left[\frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 + \hat{U} - E \right] |\Psi\rangle = 0, \quad (2.1)$$

where $\hat{V}^0$ is the periodic potential of the crystalline lattice. We introduce $\hat{U}$ as a potential introduced by defects in the lattice such as that generated by an impurity. E has energy zero at the lowest conduction band minima.

We understand that, in the absence of defect, the wavefunction resulting from the periodic crystal potential will be a Bloch function. The acceptable energies for free electrons occur in discrete bands that may be separated, resulting in disallowed energies. Each Bloch function is denoted by a wavevector and an index indicating mem-

bership in one of these bands. The space wavefunction of each Bloch state follows the form

$$\langle \mathbf{r} | n\mathbf{k} \rangle = \frac{1}{(2\pi)^{\frac{3}{2}}} e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}\mathbf{r}}, \quad (2.2)$$

where $u_{n\mathbf{k}}$ may be a complicated function, but one which always has the periodicity of the lattice. The Bloch state is the eigenstate of a nearly free electron traveling through a silicon lattice. As such, the Schrödinger equation of the pure system may be written as

$$\left[\frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 - E_{n\mathbf{k}} \right] |n\mathbf{k}\rangle = 0 \quad (2.3)$$

$\hat{V}^0$ is the periodic potential of silicon and $E_{n\mathbf{k}}$ is the energy of a nearly free electron traveling through silicon in the n th conduction band given the wavevector k .

We assume that Bloch functions taken from the first Brillouin zone and from all bands form a complete orthonormal basis to be used to construct the state of an excess electron bound to an impurity or heterostructure. As such, we represent the complete wavefunction as

$$|\Psi\rangle = \sum_n \int_{BZ} d\mathbf{k} A_{n\mathbf{k}} |n\mathbf{k}\rangle \quad (2.4)$$

with the Bloch states such that

$$\langle n\mathbf{k} | n'\mathbf{k}' \rangle = \delta_{nn'} \delta(\mathbf{k} - \mathbf{k}') \quad (2.5)$$

We write the Schrödinger equation in the Bloch basis, letting $\hat{U}$ be the defect potential,

$$(E_{n\mathbf{k}} - E)A_{n\mathbf{k}} + \sum_{n'} \int_{BZ} d\mathbf{k}' A_{n'\mathbf{k}'} \langle n\mathbf{k} | \hat{U} | n'\mathbf{k}' \rangle = 0 \quad (2.6)$$

We can clearly see the role of the defect potential in mixing together different bands, as well as causing the state to spread through each band. However, this equation proves difficult to solve directly, since it relies on the interplay between the Fourier

components of the defect potential and the periodic function $u_{n\mathbf{k}}$.

To resolve these difficulties we employ a variational method. In the variational method one calculates $\mathcal{E}[\Psi]$ from

$$\mathcal{E}[\Psi] = \frac{\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 + \hat{U} | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \quad (2.7)$$

where the state Ψ is judiciously chosen, with variational parameters, to solve the problem approximately. Minimization of the energy with respect to the parameters provides the best possible approximate solution to the problem using the state Ψ .

2.1.2 Shallow one electron states

At this point, we explicitly restrict our concern to localized “shallow” potentials. For our purposes, the “shallowness” of a potential depends on its effect in mixing and occupying the free electron states. Namely, the bound state of a shallow potential only significantly occupies the lowest conduction band centered steeply around band minima. As such, the bound state possesses non-periodic structure that emerges at a scale much longer than the characteristic wavelength of the Bloch function at the conduction band minima. The shallowness assumptions leads to a series of conclusions:

1. The potential does not cause significant band occupation outside the lowest conduction band, and so terms $n \neq 0$ may be ignored.
2. The band occupation will mostly occur in “valleys” – the k -space energy minima of the lowest conduction band, which we will denote as $\mathbf{k}_q$. We write the occupation as $A_{0\mathbf{k}} = \sum_q A_{\mathbf{k}}^{(q)}$. Above, the $A_{\mathbf{k}}^{(q)}$ functions are sharply peaked functions centered at the q th minimum of the conduction band. We expect that the overall form of these occupation peaks will obey symmetry relations and will follow a form corresponding to one of the irreducible representations of the

symmetry group of the composite Hamiltonian including the $\hat{U}$ term. We presume the set containing the $A^{(q)}$ functions is closed under the transformations of the relevant symmetry group.

We wish to see the effect of these assumptions upon Eq. (2.7). Since we are considering only the lowest conduction band, we can drop the explicit band index $n = 0$ from our computations.

The diagonal part is straightforward, and so we write

$$\mathcal{T} \equiv \left\langle \Psi \left| \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 \right| \Psi \right\rangle = \int_{BZ} d\mathbf{k} \tilde{A}_{\mathbf{k}}^{(q)} E_{\mathbf{k}}^{(q')} A_{\mathbf{k}}^{(q')}. \quad (2.8)$$

Turning our concern to the impurity term in Eq. (2.7) yields

$$\mathcal{U} \equiv \langle \Psi | \hat{U} | \Psi \rangle = \sum_{q,q'} \int \int_{BZ} d\mathbf{k} d\mathbf{k}' \tilde{A}_{\mathbf{k}}^{(q)} A_{\mathbf{k}'}^{(q')} \langle \mathbf{k} | \hat{U} | \mathbf{k}' \rangle, \quad (2.9)$$

where the matrix element is computed as as

$$\langle \mathbf{k} | \hat{U} | \mathbf{k}' \rangle = \frac{1}{(2\pi)^3} \int d\mathbf{r} U_{\mathbf{r}} e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} \tilde{u}_{\mathbf{k}\mathbf{r}} u_{\mathbf{k}'\mathbf{r}}. \quad (2.10)$$

We may express the impurity term as a space integral,

$$\mathcal{U} = \int d\mathbf{r} U_{\mathbf{r}} \tilde{\Psi}_{\mathbf{r}}^{(q')} \Psi_{\mathbf{r}}^{(q)}, \quad (2.11)$$

where $\Psi_{\mathbf{r}}^{(q)}$ is

$$\Psi_{\mathbf{r}}^{(q)} = \frac{1}{(2\pi)^{\frac{3}{2}}} \int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(q)} e^{i\mathbf{k} \cdot \mathbf{r}} u_{\mathbf{k}\mathbf{r}}. \quad (2.12)$$

The significance of the function $\Psi_{\mathbf{r}}^{(q)}$ as a single-valley term of the full multi-valley wavefunction is clear when we write the space wavefunction of the bound electron

as a summation over each valley. Utilizing (2.4) and (2.12) we find

$$\Psi_{\mathbf{r}} \equiv \langle \mathbf{r} | \Psi \rangle = \sum_q \Psi_{\mathbf{r}}^{(q)}. \quad (2.13)$$

We expect that the best variational form of function $\Psi^{(q)}$ will be complicated, mixing together both long- and short-range physics.

Because we expect the $A^{(q)}$ to be very sharply peaked at a particular valley in the conduction band, we can expand $\Psi^{(q)}$ to partially separate the long- and short-range scales. In particular, we introduce the "envelope function" as a shifted inverse truncated transform of the occupation function

$$\Phi_{\mathbf{r}}^{(q)} \equiv \frac{1}{(2\pi)^{\frac{3}{2}}} \int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(q)} e^{i(\mathbf{k}-\mathbf{k}_q) \cdot \mathbf{r}} \quad (2.14)$$

This is approximately the Fourier transform of $A^{(q)}$ centered on the valley, and becomes a better approximation the more localized and steeply peaked the $\mathbf{k}$ -space is. Since we are assuming the occupation function is strongly peaked in $\mathbf{k}$ -space, the corresponding "envelope function" decays slowly in real space. In terms of this long-scale envelope function, we can write the real-space valley wavefunction of the bound electron as

$$\Psi_{\mathbf{r}}^{(q)} = \Phi_{\mathbf{r}}^{(q)} (1 + \varphi_{\mathbf{r}}^{(q)}) e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q \mathbf{r}}, \quad (2.15)$$

where $\varphi^{(q)}$ is such that

$$\varphi_{\mathbf{r}}^{(q)} \equiv \frac{\int_{BZ} d\mathbf{k} A_{\mathbf{k}}^{(q)} e^{i(\mathbf{k}-\mathbf{k}_q) \cdot \mathbf{r}} (u_{\mathbf{k} \mathbf{r}} - u_{\mathbf{k}_q \mathbf{r}})}{\Phi_{\mathbf{r}}^{(q)} u_{\mathbf{k}_q \mathbf{r}}}. \quad (2.16)$$

We expect $\varphi^{(q)}$ to be quite small, with no zeroth-order contribution to the integral. While we can expect a first-order contribution, the localization of $A^{(q)}$ around the minima $\mathbf{k}_q$ will keep this contribution very small.

The exact form of the potential $\hat{U}$ resulting from deviations of the pure crystalline structure is still an open question. In this work, we will approach this question of the form of $\hat{U}$ phenomenologically. We structure our impurity term such that

$$\mathcal{U} = \sum_{q,q'} \int d\mathbf{r} e^{i(\mathbf{k}_{q'} - i\mathbf{k}_q) \cdot \mathbf{r}} U_{\mathbf{r}}^{(qq')} u_{\mathbf{k}_q \mathbf{r}}^* u_{\mathbf{k}_{q'} \mathbf{r}} \tilde{\Phi}_{\mathbf{r}}^{(q)} \Phi_{\mathbf{r}}^{(q')} \quad (2.17)$$

where the $\varphi^{(q)}$ contribution to the integral has been folded into the potential factor of the integrand as

$$U_{\mathbf{r}}^{(qq')} \equiv U_{\mathbf{r}} (1 + \varphi_{\mathbf{r}}^{(q)})^* (1 + \varphi_{\mathbf{r}}^{(q')}). \quad (2.18)$$

This form introduces the idea that the form of the potential may differ for different combinations of the minima indexed by q and q' . In this work, however, our approach only considers a single effective potential for all minima combinations. Further work may explore this refinement.

The envelope function also emerges as a convenient way to convert the k -space integral $\mathcal{T}$ to a real-space integral

$$\mathcal{T} = \sum_{q,q'} \int d\mathbf{r} e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} \{\Phi_{\mathbf{r}}^{(q)}\}^* (T\Phi)_{\mathbf{r}}^{(q')}, \quad (2.19)$$

where we can include the effect of the $E_{\mathbf{k}}^{(q)}$ band energy as

$$(T\Phi)_{\mathbf{r}}^{(q)} \equiv \frac{1}{(2\pi)^{\frac{3}{2}}} \int_{BZ} d\mathbf{k} E_{\mathbf{k}}^{(q)} A_{\mathbf{k}}^{(q)} e^{i(\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{r}}. \quad (2.20)$$

As we are presuming that the occupation function $A_{\mathbf{k}}^{(q)}$ is sharply peaked about the minima of $E_{\mathbf{k}}^{(q)}$, we can expand the energy around this minima to lowest order, yielding the following second order expression

$$E_{\mathbf{k}}^{(q)} \approx \frac{\hbar^2}{2} (\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{m}_q^{-1} \cdot (\mathbf{k} - \mathbf{k}_q). \quad (2.21)$$

Inserting this expansion into Eq. (2.20) we find that

$$\begin{aligned}
(T\Phi)_\mathbf{r}^{(q)} &= \frac{1}{(2\pi)^{\frac{3}{2}}} \int_{BZ} d\mathbf{k} \left\{ \frac{\hbar^2}{2} (\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{m}_q^{-1} \cdot (\mathbf{k} - \mathbf{k}_q) \right\} A_\mathbf{k}^{(q)} e^{i(\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{r}} \\
&= \frac{1}{(2\pi)^{\frac{3}{2}}} \int_{BZ} d\mathbf{k} \left\{ -\frac{\hbar^2}{2} \nabla_\mathbf{r} \cdot \mathbf{m}_q^{-1} \cdot \nabla_\mathbf{r} \right\} A_\mathbf{k}^{(q)} e^{i(\mathbf{k} - \mathbf{k}_q) \cdot \mathbf{r}} \\
&= \left\{ -\frac{\hbar^2}{2} \nabla_\mathbf{r} \cdot \mathbf{m}_q^{-1} \cdot \nabla_\mathbf{r} \right\} \Phi_\mathbf{r}^{(q)}
\end{aligned} \tag{2.22}$$

This expression is analogous to a kinetic energy operator acting upon the envelope function, with the quirk that the isotropic inertia mass is replaced with an effective mass tensor. These effective masses are a result of the curvature of the conduction band at the minima. We will further explore this effective mass concept in the context of silicon in the next section.

All together, then, we can restate the above examination succinctly. To wit, our model of a single electron bound to a "shallow" defect in a crystal is encompassed in the following equations for the variational wavefunction Ψ and energy $\mathcal{E}$.

$$\Psi_\mathbf{r} \approx \sum_q \Phi_\mathbf{r}^{(q)} e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q \mathbf{r}}, \tag{2.23}$$

$$\mathcal{E} = \mathcal{T} + \mathcal{U}, \tag{2.24}$$

$$\mathcal{T} \equiv \int d\mathbf{r} \sum_q \sum_{q'} e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} \Phi_\mathbf{r}^{(q)} \hat{T}^{(q)} \{ \Phi_\mathbf{r}^{(q')} \}, \tag{2.25}$$

$$\hat{T}^{(q)} \equiv -\frac{\hbar^2}{2} \nabla_\mathbf{r} \cdot \mathbf{m}_q^{-1} \cdot \nabla_\mathbf{r}, \tag{2.26}$$

$$\mathcal{U} \equiv \int d\mathbf{r} \sum_q \sum_{q'} e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} u_{\mathbf{k}_q \mathbf{r}}^* u_{\mathbf{k}_{q'} \mathbf{r}} U_\mathbf{r} \Phi_\mathbf{r}^{(q)} \Phi_\mathbf{r}^{(q')}. \tag{2.27}$$

In the above, we implicitly assumed that as we are expecting our wavefunction to be square integrable, it is normalizable and so $\langle \Psi | \Psi \rangle = 1$. We also understand that $U_\mathbf{r}$ is not an exact, direct ab-initio expression of the potential introduced by the crystal defect. Instead we consider a phenomenological effective potential informed both

by ab-initio theoretical considerations as well as experimental observations.

2.1.3 The generalized eigenvalue problem emerges.

One of the concerns with the variational method is selection of the wavefunction. The wavefunction must reflect the form of the actual wavefunction, though this form may not be known in any detail. In this work, we consider replacing a single variational wavefunction with a "variational basis". In this section, we will explore the implications of this expanding the envelope function with a set of specially selected states.

Assume there exists a linearly independent countably infinite set of functions such that

$$\alpha_q \Phi_{\mathbf{r}}^{(q)} = \sum_i c_{q,i} \phi_{i\mathbf{r}}. \quad (2.28)$$

We assume the functions of this set can be normalized. For reasons we will explore in a later section, we do not assume this set of functions is orthonormal, but that they do span the function space $L^2(\mathbb{R}^3)$ while remaining linearly independent. We note that q indexes the conduction valley minima of our system.

We can insert this expansion into our variational model of the donors, yielding for the effective kinetic term

$$\mathcal{T} = \sum_{pq} \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \left\{ \sum_i c_{p,i} \phi_{i\mathbf{r}} \right\}^* \hat{T}^{(q)} \left\{ \sum_j c_{q,j} \phi_{j\mathbf{r}} \right\}. \quad (2.29)$$

A little reorganization allows to articulate the following structure for this term,

$$\mathcal{T} = \sum_{pq} \sum_{ij} c_{p,i}^* c_{q,j} \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \phi_{i\mathbf{r}}^* \hat{T}^{(q)} \{ \phi_{j\mathbf{r}} \}. \quad (2.30)$$

With little effort, the above equation can be reinterpreted as a vector-matrix equation,

namely

$$\mathcal{T} = \mathbf{c}^\dagger \mathbb{T} \mathbf{c}, \quad (2.31)$$

where

$$(\mathbb{T})_{pi,qj} \equiv \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \phi_{ir}^* \hat{T}^{(q)} \{\phi_{jr}\}. \quad (2.32)$$

Similarly, the potential term can be reformulated as

$$\mathcal{U} = \mathbf{c}^\dagger \mathbb{U} \mathbf{c}, \quad (2.33)$$

and

$$(\mathbb{U})_{pi,qj} \equiv \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} u_{\mathbf{k}_p \mathbf{r}}^* u_{\mathbf{k}_q \mathbf{r}} U_{\mathbf{r}} \phi_{ir}^* \phi_{jr}. \quad (2.34)$$

An important observation to make here is how the nonorthogonality of the basis affects the normalization condition of the eigenstate.

$$\mathbf{c}^\dagger \mathbb{S} \mathbf{c} = 1 \quad (2.35)$$

$$(\mathbb{S})_{pi,qj} \equiv \int d\mathbf{r} e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \phi_{ir}^* \phi_{jr} \quad (2.36)$$

These pieces together allow us to restate the problem as a generalized eigenvalue problem by noting that 2.24 may be rewritten as

$$\mathbf{c}^\dagger (\mathbb{T} + \mathbb{U} - \mathcal{E} \mathbb{S}) \mathbf{c} = 0. \quad (2.37)$$

The linear independence asserted above means that for 2.37 to be true it must be the case that

$$(\mathbb{T} + \mathbb{U}) \mathbf{c} = \mathcal{E} \mathbb{S} \mathbf{c} \quad (2.38)$$

This restatement of our problem is especially amenable to variational optimization. Solving the eigenvalue problem here automatically optimizes the expansion coeffi-

cients, so no additional optimization beyond the variational parameters we introduce explicitly into the basis states themselves are necessary.

2.1.4 Considering symmetry

The above analysis can be extended to take symmetry into account. Without including an analysis of symmetry, the matrices of the generalized matrix problem will include contributions from many otherwise orthogonal eigenstates. Additionally, there is no clear way to resolve degenerate eigenstates without establishing a symmetry representation.

It is useful to more rigorously clarify what is meant by "symmetry" in this context. We begin by introducing the group of the Hamiltonian. This is a group consisting of all operators that commute with the Hamiltonian. In the sort of systems we are considering, as combinations of crystal structure with an isolated localized structure, we expect the group of these transformations to be one of the 32 crystallographic point groups of 3-D crystals. However, the following analysis can be applied to any Hamiltonian with a finite symmetry group.

Finite groups of quantum mechanical operators are homomorphic to groups of finite square matrices. We consider irreducible representations of the Hamiltonian group using unitary matrices. The specific entries of these matrices are arbitrary as the group structure and properties are left unchanged by a similarity transformation applied to all elements of the group. Groups and matrices linked by these similarity transformations are said to be equivalent. As such, we want to consider a set of inequivalent representations.

Group theory provides useful tools for enumerating, elucidating, and annotating the irreducible representations of finite groups, especially the 32 crystallographic point groups. Once such a collection of group representations has been defined, a projection/transfer operator can be defined to aid in projecting functions onto sym-

metric subspaces.

$$\mathcal{P}_{ij}^{(\kappa)} \equiv \frac{l_\kappa}{L} \sum_{\hat{R}} (\Gamma_{\hat{R}}^{(\kappa)})_{ij}^* \hat{R}, \quad (2.39)$$

with the effect of these operators being denoted as such

$$f_i^{(\kappa)} \equiv \mathcal{P}_{ii}^{(\kappa)} f = \mathcal{P}_{ij}^{(\kappa)} f_j^{(\kappa)}$$

The functions generated by these operators transform according to the symmetry of the system as

$$\hat{R} f_j^{(\kappa)} = \sum_{i=1}^{l_\kappa} f_i^{(\kappa)} (\Gamma_{\hat{R}}^{(\kappa)})_{ij}^*.$$

It can be shown that a function can be broken into symmetric parts, with

$$f = \sum_{\kappa} \sum_i^{l_\kappa} f_i^{(\kappa)}.$$

A prevalent example of this sort of symmetrization can be seen when considering the odd and even parts of a function, namely

$$f(x) = f_{\text{even}} + f_{\text{odd}}$$

where

$$f_{\text{even}} = \frac{f(x) + f(-x)}{2} \quad f_{\text{odd}} = \frac{f(x) - f(-x)}{2}.$$

We can invoke this formulation in our construction of the wavefunction. Recall [2.42](#). We can restate that equation using our symmetrizing machinery, namely

$$\Psi_{\mathbf{r}}^{(\kappa, i)} = \sum_q \alpha_q^{(\kappa, i)} \Psi_{\mathbf{r}}^{(q)} = \mathcal{P}_{ii}^{(\kappa)} \Psi_{\mathbf{r}}^{(q_0)}$$

Much as before we assume there exists a linearly independent countably infinite

set of functions such that

$$\Phi_{\mathbf{r}}^{(q_0)} = \sum_k c_k \Phi_{k\mathbf{r}}. \quad (2.40)$$

Using this form within our wavefunction expression yields

$$\Psi_{\mathbf{r}}^{(\kappa,i)} = \sum_k c_k \mathcal{P}_{ii}^{(\kappa)} \left\{ \Phi_{k\mathbf{r}} \left(1 + \varphi_{\mathbf{r}}^{(q_0)} \right) e^{i\mathbf{k}_{q_0} \cdot \mathbf{r}} u_{\mathbf{k}_{q_0}\mathbf{r}} \right\} \quad (2.41)$$

Since the group of the composite Hamiltonian will be a subgroup of the crystalline Hamiltonian, we expect that the set of valley Bloch functions will be closed under the action of $\mathcal{P}_{ii}^{(\kappa)}$. As such, much as we did before, we can group terms corresponding to the same valley and write our wavefunction expression as

$$\Psi_{\mathbf{r}}^{(\kappa,i)} = \sum_k c_k \sum_q \alpha_q^{(\kappa,i)} \Phi_{kq\mathbf{r}}^{(\kappa,i)} \left(1 + \varphi_{\mathbf{r}}^{(q)} \right) e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q\mathbf{r}}. \quad (2.42)$$

where the coefficients $\alpha_q^{(\kappa,i)}$ have been chosen such that the resulting state $\Psi^{(\kappa,i)}$ belongs to the i -th row of the κ symmetry. This can be expressed operationally

$$\mathcal{P}_{ii}^{(\kappa)} \left\{ \Phi_{kq\mathbf{r}}^{(\kappa,i)} \left(1 + \varphi_{\mathbf{r}}^{(q)} \right) e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q\mathbf{r}} \right\} = \sum_q \alpha_q^{(\kappa,i)} \Phi_{kq\mathbf{r}}^{(\kappa,i)} \left(1 + \varphi_{\mathbf{r}}^{(q)} \right) e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q\mathbf{r}}. \quad (2.43)$$

The exact form of the resulting $\Phi_{kq\mathbf{r}}^{(\kappa,i)}$ functions will depend upon the groups of the crystalline and composite Hamiltonians. This form can be explicitly written as

$$\Phi_{kq\mathbf{r}}^{(\kappa,i)} \propto \sum_{\hat{R}|q_0 \rightarrow q} \left(\Gamma_{\hat{R}}^{(\kappa)} \right)_{ii}^* \hat{R} \Phi_{k\mathbf{r}} \quad (2.44)$$

where the operators $\hat{R}$ are selected such that they transform Bloch functions corresponding to the q_0 -th valley with crystal momentum $\mathbf{k} = \mathbf{k}_{q_0}$ to Bloch functions corresponding to the q -th valley with crystal momentum $\mathbf{k} = \mathbf{k}_q$.

The importance of the form of Eq. (2.42) lies in allowing us to take advantage of symmetry to make analytical observations and simplifications that reduce the amount

of numerical computation necessary. Let's consider the generalized eigenvalue problem using the formalism suggested by Eq. (2.42). We find that the resulting generalized eigenvalue problem in the new formalism has had almost no change in overall structure, stated as

$$(\mathbb{T} + \mathbb{U})\mathbf{c} = \mathcal{E}\mathbb{S}\mathbf{c}. \quad (2.45)$$

However, before the column vector $\mathbf{c}$ consisted of an array of constants, indexed by valley and basis index. By separating valley and basis mixing in this way, the components of the new column vector are indexed only by basis state. The valley physics and symmetry have been offloaded from the purely numerical eigenvalue solver to being onloaded onto the integral evaluation, which has both analytic and numerical features. The new definitions of the matrix quantities under consideration within this formalism are

$$(\mathbb{T})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \{\Phi_{i\mathbf{r}}^{(p)}\}^* \hat{T}^{(q)} \{\Phi_{j\mathbf{r}}^{(q)}\}, \quad (2.46)$$

$$(\mathbb{U})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} u_{\mathbf{k}_p\mathbf{r}}^* u_{\mathbf{k}_q\mathbf{r}} U_{\mathbf{r}} \{\Phi_{i\mathbf{r}}^{(p)}\}^* \Phi_{j\mathbf{r}}^{(q)}, \quad (2.47)$$

$$(\mathbb{S})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \{\Phi_{i\mathbf{r}}^{(p)}\}^* \Phi_{j\mathbf{r}}^{(q)}. \quad (2.48)$$

We may also define matrix elements for the squared Hamiltonian.

$$(\mathbb{TT})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \hat{T}^{(p)} \{\Phi_{i\mathbf{r}}^{(p)}\}^* \hat{T}^{(q)} \{\Phi_{j\mathbf{r}}^{(q)}\}, \quad (2.49)$$

$$(\mathbb{UU})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} u_{\mathbf{k}_p\mathbf{r}}^* u_{\mathbf{k}_q\mathbf{r}} U_{\mathbf{r}}^2 \{\Phi_{i\mathbf{r}}^{(p)}\}^* \Phi_{j\mathbf{r}}^{(q)}, \quad (2.50)$$

$$(\mathbb{UT})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} u_{\mathbf{k}_p\mathbf{r}}^* u_{\mathbf{k}_q\mathbf{r}} U_{\mathbf{r}} \{\Phi_{i\mathbf{r}}^{(p)}\}^* \hat{T}^{(q)} \{\Phi_{j\mathbf{r}}^{(q)}\}, \quad (2.51)$$

$$(\mathbb{TU})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} u_{\mathbf{k}_p\mathbf{r}}^* u_{\mathbf{k}_q\mathbf{r}} U_{\mathbf{r}} \hat{T}^{(p)} \{\Phi_{i\mathbf{r}}^{(p)}\}^* \Phi_{j\mathbf{r}}^{(q)}. \quad (2.52)$$

2.1.5 Introduction of a variational basis

One of the concerns with the variational method is selection of the wavefunction. The wavefunction must reflect the form of the actual wavefunction, though this form may not be known in any detail. In this work, we consider replacing a single variational wavefunction with a "variational basis". In this section, we will introduce, define, and refine this idea.

To start with, we select a set of functions such that we may expand the envelope function of a valley as

$$\Phi_{\mathbf{r}}^{(q)} \equiv \sum_k c_k \Phi_{k\mathbf{r}}. \quad (2.53)$$

We wish to consider the bound states of our system. As such, we can assume that the function $F^{(z)}$ is part of the $L^2_{\mathbb{C}}(\mathbb{R}^3)$ function space. It is also worth noting that this function space is also a separable Hilbert space. We seek a set of orthonormal functions that span this separable Hilbert space. In this work, we explore two possibilities for our complete basis. While a completely rigorous mathematically sound interpretation of the following statement is beyond the scope of this work, I'll state formally that the countable eigenstates of the Hamiltonian of the isotropic harmonic oscillator form a complete basis for the separable Hilbert space. As such, an obvious candidate for our orthonormal basis emerges from the energy eigenstates of the isotropic 3-D harmonic oscillator, namely

$$\varphi_{klm;\eta;\mathbf{r}} \propto L_{k;\rho^2}^{(l+\frac{1}{2})} e^{-\frac{\rho^2}{2}} \rho^l Y_{lm;\chi\phi}. \quad (2.54)$$

Above, $L_k^{(\alpha)}$ denotes the generalized Laguerre polynomials with k being a non-negative integer, and with Y denoting the spherical harmonics such that l is a non-negative integer and m is an integer such that $|m| \leq l$. Our polar spherical coordinate $\chi = z/r = \cos(\theta)$ is an alternate polar spherical coordinate we find more convenient to use in this work than θ directly. As well, $\rho = r/\eta$ is the radial coordinate rescaled to

be dimensionless with η as a unitful constant shared by each state in the basis. We pick this radial scaling η to generate the best representation of our wavefunction. In the isotropic harmonic oscillator, $\eta = \sqrt{\frac{\hbar}{m\omega}}$ with mass m and angular frequency ω . In our system, this constant η may be optimized to generate the lowest expectation energy for our system, thus giving it a variational character.

More generally, we expect the wavefunction of an isolated donor to display some manner of near spherical or ellipsoidal symmetry. As such one of the building blocks of our basis set will be the aforementioned spherical harmonics, complete on $L^2_{\mathbb{C}}(S^2)$ where S^2 is the unit sphere. A further important observation as we design our basis involves the tensor product of Hilbert spaces. To wit, for Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ with orthonormal bases $\{\phi_i\}$ and $\{\eta_j\}$ respectively, it is true that $\{\phi_i \otimes \eta_j\}$ is an orthonormal basis for $\mathcal{H}_1 \otimes \mathcal{H}_2$. Articulating the rigorous mathematical definition of the tensor product of Hilbert spaces is beyond the scope of this document. It is sufficient for our purposes to keep in mind that we can construct a basis for $L^2_{\mathbb{C}}(\mathbb{R}^3)$ by noting that this function space is isomorphic to the tensor product space $L^2_{\mathbb{C}}(S^2) \otimes L^2_{\mathbb{C}}(\mathbb{R}^+, r^2)$.

To find a complete basis on $L^2_{\mathbb{C}}(\mathbb{R}^3)$ then, we only need to find a complete basis on $L^2_{\mathbb{C}}(\mathbb{R}^+, r^2)$. We've already seen such a set of basis functions emerge in the isotropic harmonic oscillator, namely

$$H_{kl;r} \propto L_{k;r^2}^{(l+\frac{1}{2})} e^{-\frac{r^2}{2}} r^l \quad (2.55)$$

However, in addition to this basis, we also want to consider a basis with Slater exponential decay. The formula for the wavefunctions of the isotropic harmonic oscillator includes generalized Laguerre polynomials defined in the conventional way[8]. For our purposes, I note that the Laguerre polynomial $L_{k;x}^{(\alpha)}$ is complete over the Hilbert space $L^2(\mathbb{R}^+, x^\alpha e^{-x})$. For the harmonic oscillator eigenstates with $x = r^2$ and $\alpha = l + \frac{1}{2}$, I note that $x^\alpha e^{-x} dx \propto r^{2l+2} e^{-r^2} dr$. As such, the above statement of completeness for the Laguerre polynomials is equivalent to the completeness of $H_{kl;r}$ on $L^2(\mathbb{R}^+, r^2)$.

To arrive at our Slater-decay basis we will once more introduce the associated Laguerre polynomial as our building block. This time we shall take $x = r$. We introduce then

$$G_{kl;r} \propto L_{k;r}^{(2l+2)} e^{-\frac{r}{2}} r^l \quad (2.56)$$

As L above is complete over $L^2(\mathbb{R}^+, r^\alpha e^{-r})$, then it can be shown that G is complete over $L^2(\mathbb{R}^+, \rho^2)$. As a final aside, we took α above to be such that $\alpha = 2l + 2$. Further discussion on why this choice has been made will be discussed in the [Supplementary Materials](#).

In short, the two forms above for the radial part of our basis can be combined and generalized for all positive integer W , though we are only concerns with $W = 1, 2$. Appropriately normalized, the combined form is

$$F_{kl;\eta;r}^{(W)} = \sqrt{\frac{W}{\eta^3}} L_{k;\rho}^{(\alpha)} e^{-\frac{\rho}{2}} \rho^{\frac{l}{W}}, \quad \rho = \left(\frac{r}{\eta}\right)^W. \quad (2.57)$$

with $\alpha = \frac{2l+3-W}{W}$. With $W = 1, 2$ the above function will be the radial part of our basis. Above, we assume that the Laguerre polynomials are defined conventionally and orthonormalized such that

$$\int d\rho L_{k_0;\rho}^{(\alpha)} L_{k_1;\rho}^{(\alpha)} \rho^\alpha e^{-\rho} = \delta_{k_0 k_1}. \quad (2.58)$$

Thus we may write the complete formula for our naive basis states with positive real η acting as a variational parameter.

$$\Phi_{klm;\eta;r}^{(W)} = F_{kl;\eta;r}^{(W)} Y_{lm;\chi\phi}. \quad (2.59)$$

In the above formula, we assume appropriate normalization factors have been applied to the radial function F and the spherical harmonics Y such that the resulting basis state Φ is normalized.

We note that the envelope function with $W = 1$ yielding Slater-exponential decay bares a strong resemblance to the bound states of the hydrogen atom. As we consider using our basis states to represent the energy eigenstates of our system, this is an important point to consider. Of note, there are two points of divergence between the hydrogenic states and our Slater basis states.

First, we want to be sure that our basis can conveniently represent the hydrogenic eigenstates. The hydrogenic wavefunctions have the form

$$\Psi_{nlk} \propto L_{n-l-1}^{(2l+1)} e^{-\frac{\rho}{2}} \rho^l Y_{lm;\chi\phi} \quad \rho = \frac{r}{\eta_n}. \quad (2.60)$$

It is trivial to reindex the wave function such that $n = k+l+1$ and show that the two set of indices are isomorphic between the hydrogenic states and our Slater basis states. We note that the Laguerre parameter differs between the hydrogenic and Slater basis, being $\alpha_{\text{hydro}} = 2l+1$ and $\alpha_{\text{Slater}} = 2l+2$ respectively. Despite this, it is the case that $L_k^{(\alpha)} = L_k^{(\alpha+1)} - L_{k-1}^{(\alpha+1)}$. In light of the above facts, it is clear that each hydrogenic wavefunction can be represented by, at most, two states from our basis given the correct variational scaling factor η .

Secondly, however, note that the scaling factor η_n for each hydrogen state depends on the quantum numbers of the states. It goes as

$$\eta_n = \frac{n}{2} r_0 \quad (2.61)$$

where

$$r_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} \quad (2.62)$$

We see a conflict between our complete basis with a single parameter η and the bound hydrogenic states. A larger orthonormal basis that tends towards completeness can represent any desired state in principle, of course. In our experience, however, the

computational cost from expanding our orthonormal basis does not see a worthwhile increase in that basis's power to represent more than a single desired state.

To resolve this, we introduce the concepts that we will call in this document the *subbasis* and the *superbasis*. A subbasis is defined to be a truncated orthonormal basis, tending towards completeness, which has been variationally optimized to minimize the energy of one particular state. Therefore, we expect a loose association between a subbasis and an energy eigenstate. The superbasis is a formed from the union of multiple subbases each independently optimized to different eigenstates. The superbasis is no longer orthonormal, as each subbasis is allowed to vary its variational parameters independently. As well, the superbasis tends towards overcompleteness as more states are added. For a numerical method such as ours, we do need to be concerned with numerical instability or intractability introduced by linear dependence and overcompleteness within the superbasis. Despite this, we find that the variational procedure applied to each subbasis helps to ameliorate these problems with numerical instability. As a potential explanation, recall that each subbasis can be loosely associated with each energy eigenstate, and these eigenstates are orthogonal to each other. Serendipitously then, it seems the process of variational optimization mutually orthogonalizes the subbases and reduces numerical instability.

2.1.6 The silicon effective mass.

So far the analysis has been somewhat general, applicable to all “shallow” potentials in the semiconductor bulk. The system of most interest in this document is the phosphorous donor in silicon. As such, we want to consider the meaning of “effective mass” in the context of silicon band structure. In silicon, the six equivalent conduction band minima each lie along one of the three crystallographic axes, $\hat{x}$, $\hat{y}$, or $\hat{z}$.

The crystal momentum of the Bloch function of each valley may be identified as

$$\mathbf{k}_{\pm q} = \pm k_0 \hat{q} \quad (2.63)$$

where $\hat{q} = \hat{x}, \hat{y}, \hat{z}$ and $k_0 \approx .85\left(\frac{2\pi}{a_0}\right)$. We identify a_0 as the lattice constant of silicon.

Significantly, each valley is anisotropic. To the second order in the valley at $\mathbf{k}_q = k_0 \hat{k}_q$, we find that the conduction band energy goes as

$$E_{\mathbf{k}}^{(q)} \approx \frac{\hbar^2}{2m_{\parallel}}(k_q - k_0)^2 + \frac{\hbar^2}{2m_{\perp}}(k_r^2 + k_s^2) \quad (2.64)$$

We take q , r , and s to respectively denote each of the three crystallographic axes in silicon. The quantities $m_{\parallel}$ and $m_{\perp}$ are defined here from the empirical structure of the silicon conduction band, and they take the role of an effective and anisotropic mass in the kinetic expectation energy. As a result of Eq. (2.64) we may define the effective kinetic energy operator $\hat{T}^{(q)}$ as

$$\hat{T}^{(q)} = -\frac{\hbar^2}{2m_{\parallel}} \frac{\partial^2}{\partial r_q^2} - \frac{\hbar^2}{2m_{\perp}} \left(\frac{\partial^2}{\partial r_r^2} + \frac{\partial^2}{\partial r_s^2} \right), \quad (2.65)$$

which more completely specifies the general differential operator $\hat{T}^{(q)}$ introduced in Eq. (2.22) by specifying the inverse effective mass tensor of the system under consideration, namely a silicon donor.

We see that the band structure of silicon influences the motion of excess electrons bound to the shallow impurity, emerging as an ellipsoidal anisotropic effective mass. Since an electron occupying a particular bandstructure valley is effectively heavier along the longitudinal axis of the valley, we expect that the individual components of the complete wavefunction corresponding to each valley will show a deformation, leading to an expected anisotropic ellipsoidal structure for the envelope function, rather than the spherical structure that might be assumed as the solution to a more

hydrogenic problem.

2.1.7 Introducing variational anisotropy.

We previously stated in Eq. (2.59) that our variational subbases will go as

$$\Phi_{klm;\eta;\mathbf{r}} \propto F_{kl;\eta;r}^{(W)} Y_{lm;\chi\phi}. \quad (2.66)$$

To help capture the anisotropic quality of the wavefunctions introduced by the anisotropic effective mass, we introduce a scaling along the particular crystallographic axis. Without loss of generality, we can introduce a scaling $z \rightarrow \lambda z$ with λ playing the role of a second variational parameter for the subbasis alongside η . In spherical coordinates, this scaling takes the form

$$r \rightarrow r\vartheta_{\lambda\chi}, \quad (2.67)$$

$$\chi \rightarrow \frac{\lambda\chi}{\vartheta_{\lambda\chi}}, \quad (2.68)$$

where

$$\vartheta_{\lambda\chi} = \sqrt{1 + (\lambda^2 - 1)\chi^2}, \quad (2.69)$$

with $\chi = z/r = \cos(\theta)$.

This argument can be specialized for the x - and y -axes as well. To accomplish this we can define a set of equivalent spherical coordinates (χ_q, ϕ_q) defined in the

standard way, but taking another axis as the zenith. To wit, we have

$$\chi_x \equiv \frac{x}{r} = \sqrt{1-\chi^2} \cos(\phi) \quad (2.70)$$

$$\chi_y \equiv \frac{y}{r} = \sqrt{1-\chi^2} \sin(\phi) \quad (2.71)$$

$$\chi_z \equiv \frac{z}{r} = \chi \quad (2.72)$$

$$\phi_x \equiv \text{atan2}(\chi_z, \chi_y) \quad (2.73)$$

$$\phi_y \equiv \text{atan2}(\chi_x, \chi_z) \quad (2.74)$$

$$\phi_z \equiv \text{atan2}(\chi_y, \chi_x) = \phi \quad (2.75)$$

where $\text{atan2}(b, a) = \text{Arg}(a + ib)$.

We will need to examine the implications of this introduced anisotropy upon the symmetry of the wavefunctions, particularly in the context of the symmetry of the silicon donor. For now, we can state our basis states with both radial and anisotropic scaling variational parameters as

$$\Phi_{klm;\eta\lambda;\mathbf{r}}^{(Wq)} = F_{lk;\eta\lambda;r\chi_q}^{(W)} Y_{lm;\lambda;\chi_q\phi_q}, \quad (2.76)$$

where

$$F_{lk;\eta\lambda;r\chi_q}^{(W)} = F_{lk;\eta\lambda;\chi_q;r}^{(W)} \quad \eta_{\lambda;\chi_q} = \frac{\eta}{\vartheta_{\lambda;\chi_q}} \quad (2.77)$$

$$Y_{lm;\lambda;\chi_q\phi_q} = \sqrt{\frac{\lambda}{\vartheta_{\lambda;\chi_q}^3}} Y_{lm;\chi_\lambda^{(q)}\phi_q} \quad \chi_\lambda^{(q)} = \frac{\lambda\chi_q}{\vartheta_{\lambda;\chi_q}} \quad (2.78)$$

The additional factors in front of the F and Y functions help to maintain a degree of independent normalization between the parts of the basis state, and makes the scaling transformation of some integrals within this project easier, aiding analytical

simplification and helping numerical stability during computation.

2.1.8 Symmetry concerns within silicon

The operations of the point-group symmetry of our system, a silicon donor, can be conveniently grouped into a few categories. These categories are defined by their transformation effects upon functions with crystalline Cartesian coordinate domain. For convenience, we'll denote the function transformation operators compactly such that $(\bar{y} x \bar{z})$ indicates an operator acting such that $(\bar{y} x \bar{z})f_{xyz} = f_{\bar{y}x\bar{z}}$. For our purposes we recognize

- cyclic permutation e.g. $(z x y)$,
- acyclic swap e.g. $(y x z)$,
- double inversion e.g. $(x \bar{y} \bar{z})$.

An operation in the point group can be part of two categories. An operator in two categories will be a member of the double inversion category, as well as either the cyclic or acyclic category. An example of such is $(\bar{y} x \bar{z})$. These categories form the basis of our particular representation of the symmetries of silicon. Within our formulation of the group, most of the representations of the T_d point symmetry group of silicon can be considered as having definite even or odd symmetry under acyclic swaps and double inversion. For example, functions belonging to the A_1, A_2 representations are invariant under double inversion, but are respectively classified as even or odd under acyclic swap. In comparison, functions belonging to the T_1, T_2 representations are odd under double inversion, but are respectively classified as odd or even under acyclic swap. Note that A_1, T_2 are even and A_2, T_1 are odd under acyclic swap. The final representation, E , is invariant under double inversion, but cannot be said to be simply even or odd under acyclic swaps.

This categorization of the operators of the point-group of our system allows a further useful observation; namely, we may factor the projection operator based upon these categories. To wit,

$$\mathcal{P}_{ii}^{(\kappa)} = \mathcal{V}_{\kappa,i} \mathcal{A}_z^{(\kappa)} \mathcal{D}_z^{(\kappa)} \mathcal{V}_{\kappa,i}^\dagger, \quad (2.79)$$

where the factored operators have been determined to be

$$\mathcal{V}_{\kappa,i} \equiv \alpha_0^{(\kappa,i)}(x y z) + \alpha_1^{(\kappa,i)}(y z x) + \alpha_2^{(\kappa,i)}(z x y), \quad (2.80)$$

$$\mathcal{A}_z^{(\kappa)} \equiv \frac{1 + \delta_{E\kappa}}{2} \{ (x y z) + \sigma^{(\kappa)}(y x z) \} \quad (2.81)$$

$$\mathcal{D}_z^{(\kappa)} \equiv \frac{1}{4} \{ (x y z) + (\bar{x} \bar{y} \bar{z}) + \omega^{(\kappa)}(\bar{x} y \bar{z}) + \omega^{(\kappa)}(x \bar{y} \bar{z}) \}. \quad (2.82)$$

Above, κ denotes the symmetry representation, while i indexes degenerate states. The A_1 and A_2 states are nondegenerate, and do not require indexing. The E states are doubly degenerate and the T_1 and T_2 states are triply degenerate. With our particular irreducible representation for the T symmetries, each of the triply degenerate T states can be indexed by the three valley axes. The values of the ω and σ constants in the expressions above follow from the discussion below. Namely for the states where $\kappa = A_1, A_2, E$ we have $\omega^{(\kappa)} = +1$ and $\kappa = T_1, T_2$ we have $\omega^{(\kappa)} = -1$. For A_1, T_2 states we find $\sigma = +1$ and for A_2, T_1 states we find $\sigma = -1$. The E states have $\sigma = 0$. Finally the valley axis combination vector given by $\boldsymbol{\alpha}$ is determined for A , E , and T states respectively. We find,

$$\boldsymbol{\alpha}^{(A_n)} = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (2.83)$$

$$\boldsymbol{\alpha}^{(E,0)} = \frac{1}{3} \begin{pmatrix} 1 \\ \tau^2 \\ \tau \end{pmatrix}, \quad \boldsymbol{\alpha}^{(E,1)} = \frac{1}{3} \begin{pmatrix} 1 \\ \tau \\ \tau^2 \end{pmatrix}, \quad (2.84)$$

and finally,

$$\boldsymbol{\alpha}^{(T_n, x)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{\alpha}^{(T_n, y)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \boldsymbol{\alpha}^{(T_n, z)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \quad (2.85)$$

Above $\tau = e^{i\frac{2\pi}{3}}$, one of the third roots of unity. We will explore further how these symmetry effects from the silicon media influence the development of our donor electron's basis state.

2.1.9 Symmetry concerns within the silicon wavefunctions.

We expect our full wavefunction to have the form

$$\Psi^{(\kappa, i)} = \sum_p \Phi_p e^{i\mathbf{k}_p \cdot \mathbf{r}} u_{\mathbf{k}_p \mathbf{r}} \quad (2.86)$$

The transformation of the Bloch function terms above is straight-forward, but the symmetry of the full wavefunction involves symmetrization of the product of the Bloch and envelope terms. To understand the symmetry properties of the envelope function requires us to understand the symmetrization of the full wavefunction.

To understand the symmetrization of our full wavefunction, we apply our projection symmetrization operator defined in (2.79) to a general wavefunction. We assert that the full wavefunction is already symmetrized, and as such, the operator leaves our full wavefunction invariant. In essence, we are asserting that Ψ is such that

$$\Psi = \mathcal{P}_{ii}^{(\kappa)} \Psi. \quad (2.87)$$

By expanding Ψ in terms of the form articulated above in (2.86) and matching terms we can assert symmetrization conditions upon the envelope functions.

Our first step involves the following form, namely

$$\mathcal{V}_{\kappa,i}^\dagger \Psi = (\alpha_0^{(\kappa,i)})^* (x y z) \Psi + (\alpha_1^{(\kappa,i)})^* (z x y) \Psi + (\alpha_2^{(\kappa,i)})^* (y z x) \Psi. \quad (2.88)$$

It is useful to examine the operators included in $\mathcal{V}_{\kappa,i}^\dagger$ above. To wit, we can define a set of ancillary functions based off an initial function. We describe these functions as being "valley indexed" by a particular crystallographic axis.

$$f_z \equiv (x y z) f, \quad (2.89)$$

$$f_x \equiv (y z x) f, \quad (2.90)$$

$$f_y \equiv (z x y) f. \quad (2.91)$$

It is useful to note these ancillary functions have the following relationships,

$$f_z = (z x y) f_x = (y z x) f_y, \quad (2.92)$$

$$f_x = (y z x) f_z = (z x y) f_y, \quad (2.93)$$

$$f_y = (z x y) f_z = (y z x) f_x. \quad (2.94)$$

For ease of nomenclature, we'll call these three transformations $(x y z)$, $(y z x)$, and $(z x y)$ the indexing transformations.

Since we've introduced the concept of valley indexed functions, we'll introduce an additional stipulation upon Ψ , namely that

$$\Psi = c_z \psi_z + c_x \psi_x + c_y \psi_y. \quad (2.95)$$

Above, the three functions ψ_x , ψ_y , and ψ_z are arbitrary but linked by valley indexing relationships introduced in (2.89), (2.90), and (2.91). To be consistent with definition (2.95) and equation (2.86) we expect the valley-linked ψ_q terms of the full wavefunc-

tion to themselves be constructed from terms involving the product of the envelope and valley-indexed Bloch functions. We can group those terms by Bloch function axes, and so we expect the following

$$\psi_z = \psi_{\parallel,z} + \psi_{\perp_1,x} + \psi_{\perp_2,y}, \quad (2.96)$$

where

$$\psi_{\mathfrak{K},q} = \Phi_{\mathfrak{K},+q} \theta^{(+q)} + \Phi_{\mathfrak{K},-q} \theta^{(-q)}. \quad (2.97)$$

where the Bloch functions $\theta^{(q)}$ are given as

$$\theta^{(q)} = e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q} \mathbf{r}. \quad (2.98)$$

The indexing relations may be used to construct ψ_x and ψ_y from ψ_z . We may alternatively write the valley-indexed terms of the full wavefunction as

$$\psi_q = \psi_{\mathfrak{K}_{qz},z} + \psi_{\mathfrak{K}_{qx},x} + \psi_{\mathfrak{K}_{qy},y}. \quad (2.96a)$$

with possible values for $\mathfrak{K}_{qq'}$ going as $\mathfrak{K}_{qq} = \parallel$, $\mathfrak{K}_{qr} = \perp_1$ for $qr \in \{yz, zx, xy\}$, and $\mathfrak{K}_{qs} = \perp_2$ for $qs \in \{xz, yx, zy\}$.

When we apply the first factor operator of the symmetrization projection operator to the form in (2.95), we'll find different results depending on the particular symmetry we are projecting upon. For $\kappa = A, E$ we can note the following relation involving the α coefficients, with

$$9\alpha_1\alpha_2 = 1. \quad (2.99)$$

If, in addition to the relationship above, we also note the following relation

$$\gamma_{\kappa,i}^\dagger \psi_z = 3\alpha_1^{(\kappa,i)} \gamma_{\kappa,i}^\dagger \psi_x = 3\alpha_2^{(\kappa,i)} \gamma_{\kappa,i}^\dagger \psi_y, \quad (2.100)$$

we may find for a $\kappa = A, E$ state that

$$\mathcal{V}_{\kappa,i}^\dagger \Psi = (c_z + 3c_x\alpha_2 + 3c_y\alpha_1)(\alpha_0^{(\kappa,i)}\psi_z + \alpha_1^{(\kappa,i)}\psi_x + \alpha_2^{(\kappa,i)}\psi_y). \quad (2.101)$$

This shows a clear projection form, with a resulting symmetrized function form multiplied by a overall factor indicating how much of the original function can be said to "lie along" the symmetrized form. For $\kappa = A, E$ then, we can take the c coefficients to have the following form, with $c_z = C/3$, $c_x = C\alpha_1$, and $c_y = C\alpha_2$, where C is a complex factor useful for normalization. If we define the c_q coefficients as such, we'll find that

$$\mathcal{V}_{\kappa,i}^\dagger \Psi = \Psi = C(\alpha_0^{(\kappa,i)}\psi_z + \alpha_1^{(\kappa,i)}\psi_x + \alpha_2^{(\kappa,i)}\psi_y) \quad (2.102)$$

In the case of $\kappa = T$, the α go as

$$\alpha_0^{(T_n,i)} = \delta_{iz}, \quad \alpha_1^{(T_n,i)} = \delta_{ix}, \quad \alpha_2^{(T_n,i)} = \delta_{iy}. \quad (2.103)$$

This means application of the $\mathcal{V}^\dagger$ operator yields the following effect upon our wavefunction, namely

$$\mathcal{V}_{T_n,i}^\dagger \Psi = d_z\psi_z + d_x\psi_x + d_y\psi_y, \quad (2.104)$$

$$d_z = \delta_{iz}c_z + \delta_{ix}c_x + \delta_{iy}c_y, \quad (2.105)$$

$$d_x = \delta_{iz}c_x + \delta_{ix}c_y + \delta_{iy}c_z, \quad (2.106)$$

$$d_y = \delta_{iz}c_y + \delta_{ix}c_z + \delta_{iy}c_x. \quad (2.107)$$

Alternatively, we can state

$$\mathcal{V}_{T_n,i}^\dagger \Psi = \delta_{iz} \psi_0 + \delta_{ix} \psi_1 + \delta_{iy} \psi_2, \quad (2.108)$$

$$\psi_0 = c_z \psi_z + c_x \psi_x + c_y \psi_y, \quad (2.109)$$

$$\psi_1 = c_x \psi_z + c_y \psi_x + c_z \psi_y, \quad (2.110)$$

$$\psi_2 = c_y \psi_z + c_z \psi_x + c_x \psi_y. \quad (2.111)$$

We see a marked difference between the effect of the $\mathcal{V}_{T_n,i}^\dagger$ factor operator and the $\mathcal{V}_{A_n,i}^\dagger$, $\mathcal{V}_{E,i}^\dagger$ operators. For $\kappa = T$, in essence the effect of the valley mixing/rotation factor operator $\mathcal{V}_{T_n,i}^\dagger$ is to generate a cyclic permutation of the c coefficients without an overall change of function form. Another way to interpret the effect of the $\mathcal{V}_{T_n,i}^\dagger$ factor operator is that it aligns the z -axis of the function with the T -states' axes of symmetry. In the case of $\kappa = A, E$ we find that the factor operator redistributes the weight of each valley.

With appropriate choice of the β coefficients below, we can group results for all symmetries in the following form,

$$\mathcal{V}_{\kappa,i}^\dagger \Psi = \beta_z^{(\kappa,i)} \psi_z + \beta_x^{(\kappa,i)} \psi_x + \beta_y^{(\kappa,i)} \psi_y. \quad (2.112)$$

We can group the terms solely by Bloch axes, which will give us

$$\mathcal{V}_{\kappa,i}^\dagger \Psi = \psi'_z + \psi'_x + \psi'_y, \quad (2.113)$$

$$\psi'_z = \beta_z^{(\kappa,i)} \psi_{\parallel,z} + \beta_y^{(\kappa,i)} \psi_{\perp_1,z} + \beta_x^{(\kappa,i)} \psi_{\perp_2,z}, \quad (2.114)$$

$$\psi'_x = \beta_x^{(\kappa,i)} \psi_{\parallel,x} + \beta_z^{(\kappa,i)} \psi_{\perp_1,x} + \beta_y^{(\kappa,i)} \psi_{\perp_2,x}, \quad (2.115)$$

$$\psi'_y = \beta_y^{(\kappa,i)} \psi_{\parallel,y} + \beta_x^{(\kappa,i)} \psi_{\perp_1,y} + \beta_z^{(\kappa,i)} \psi_{\perp_2,y}. \quad (2.116)$$

We can rewrite this expression to make the underlying envelope-Bloch product form

explicit.

$$\psi'_q = \sum_{v=+,-} \Phi_{vq} \theta^{(vq)}, \quad (2.117)$$

$$\Phi_{vz} = \beta_z^{(\kappa,i)} \Phi_{\parallel,vz} + \beta_y^{(\kappa,i)} \Phi_{\perp_1,vz} + \beta_x^{(\kappa,i)} \Phi_{\perp_2,vz}, \quad (2.118)$$

$$\Phi_{vx} = \beta_x^{(\kappa,i)} \Phi_{\parallel,vx} + \beta_z^{(\kappa,i)} \Phi_{\perp_1,vx} + \beta_y^{(\kappa,i)} \Phi_{\perp_2,vx} \quad (2.119)$$

$$\Phi_{vy} = \beta_y^{(\kappa,i)} \Phi_{\parallel,vy} + \beta_x^{(\kappa,i)} \Phi_{\perp_1,vy} + \beta_z^{(\kappa,i)} \Phi_{\perp_2,vy} \quad (2.120)$$

It's worth noting that the symmetrizing operators can be extended down through the layers of definition introduced in (2.96) and (2.97) to the envelope functions $\Phi_{\kappa,\pm q}$ because the operators are linear. We see the β coefficients show up in our expressions above. The value of the β coefficients depend upon the value of the symmetry index κ and follow directly from (2.102) and (2.104). There's an important distinction between β for $\kappa = T$ and β for $\kappa = A, E$. For the A, E states the coefficients are not arbitrary, with $\beta_q = \alpha_q$. For the T states, the value of $\beta_q = d_q$ and thus depends upon the value of the c arbitrary constants. Since the factor operator for $\kappa = T$ does not remix the valleys, the weight of each valley remains arbitrary. As such we note that the envelope functions for states with $\kappa = T$ are not linked by the indexing expressions introduced in (2.89), (2.90), and (2.91). This is not the case for $\kappa = A, E$. Namely, we find the following valley-indexing relationships.

$$\Phi_{vx} = 3\alpha_1^{(\kappa,i)}(yzx)\Phi_{vz}, \quad (2.121)$$

$$\Phi_{vy} = 3\alpha_2^{(\kappa,i)}(zxy)\Phi_{vz}, \quad (2.122)$$

$$\Phi_{vy} = 3\alpha_1^{(\kappa,i)}(yzx)\Phi_{vx}. \quad (2.123)$$

Over the course of this section, we will determine the necessary properties of the envelope function to generate the desired symmetry. As we apply the rest of the projection operator factors to our full wavefunction, we'll explore how the interplay of

the Bloch and envelope functions influences the process of symmetrization.

Our next step will be to apply $\mathcal{A}_z^{(\kappa)}$ to our wavefunction. We note that for $\kappa = E$, the operator $\mathcal{A}_z^{(E)} = (x y z)$ and so the effect of the operator is trivial. For $\kappa = A, T$, the effect of this operator upon our terms are given below.

Without much trouble we can assert that

$$\mathcal{A}_z^{(\kappa)} \psi'_z = \psi'_z. \quad (2.124)$$

Given the linearity of the symmetry transformations, we can achieve this on a per Bloch function term basis. For the z term it is true that

$$\mathcal{A}_z^{(\kappa)} \Phi_{uz} = \Phi_{uz}, \quad (2.125)$$

if we also assert of our envelope function terms that

$$\Phi_{uz} = \sigma^{(\kappa)}(y x z) \Phi_{uz}. \quad (2.126)$$

However, there are subtleties when applying the acyclic operator $\mathcal{A}_z$ to states with Bloch function factors indexed by x and y . Let's consider the following.

$$\mathcal{A}_z^{(\kappa)} \Phi_{ux} \theta^{(ux)} = \frac{\Phi_{ux} \theta^{(ux)} + [\sigma^{(\kappa)}(y x z) \Phi_{ux}] \theta^{(vy)}}{2}, \quad (2.127)$$

$$\mathcal{A}_z^{(\kappa)} \Phi_{vy} \theta^{(vy)} = \frac{\Phi_{vy} \theta^{(vy)} + [\sigma^{(\kappa)}(y x z) \Phi_{vy}] \theta^{(ux)}}{2} \quad (2.128)$$

Here we note that for $\kappa = T$ the envelope functions are not linked by the indexing relationships. However, we assert that

$$\mathcal{A}_z^{(T)} \{ \Phi_{ux} \theta^{(ux)} + \Phi_{vy} \theta^{(vy)} \} = \Phi_{ux} \theta^{(ux)} + \Phi_{vy} \theta^{(vy)}. \quad (2.129)$$

Combining (2.127) and (2.128) with (2.129) yields

$$\Phi_{vx} = \sigma^{(\kappa)}(y x z) \Phi_{vy} \quad (2.130)$$

$$\Phi_{vy} = \sigma^{(\kappa)}(y x z) \Phi_{vx} \quad (2.130')$$

It should be noted that (2.130) and (2.130') are equivalent. We can clearly state that for $\kappa = T$ states there exists a relationship between the Φ_{vx} and Φ_{vy} envelopes, but no relationship between these envelopes and the Φ_{vz} envelopes can be derived.

However, we saw above in the indexing relations (2.121), (2.122), and (2.123) that relationships exist between the envelope functions of different valley-indices for $\kappa = A, E$. We also observe that

$$(y z x)(y x z)(y z x) = (z x y)(y x z)(z x y) = (y x z). \quad (2.131)$$

In particular, by combining the indexing relations with the observation above and utilizing (2.126) it may be shown that

$$(y x z) \Phi_{vx} = 3\alpha_2^{(\kappa,i)} \sigma^{(\kappa)} \Phi_{vy}. \quad (2.132)$$

or equivalently

$$(y x z) \Phi_{vy} = 3\alpha_1^{(\kappa,i)} \sigma^{(\kappa)} \Phi_{vx}. \quad (2.132')$$

Since the E -states have a trivial acyclic operator, we'll keep our focus on the A -states. In light of this consideration, we find $3\alpha_1^{(A)} = 3\alpha_2^{(A)} = 1$. If this is true then (2.130) is equivalent to (2.132) and (2.130') is equivalent to (2.132'). As such, it is fairly trivial to prove that (2.126) and (2.130) are equivalent relationships for functions linked by the valley-index relations. We should note here that, for $\kappa = T$ states, (2.126) and (2.130) are not equivalent relationships, so while Φ_{vx} and Φ_{vy} are linked they are not linked with Φ_{vz} .

When constructing our envelope functions we find it convenient to use variations of the same underlying naive envelope function for all symmetric states. As such, we require our naive envelope states to satisfy (2.126). However, we note that two such naive envelopes may independently satisfy (2.126) allowing us freedom to construct $\kappa = T$ states where Φ_{vz} is independent of Φ_{vx} and Φ_{vy} . In short, for $\kappa = T$ states we may have Φ_{vz} along with $\Phi_{vx} = (y z x)\Phi'_{vz}$ and $\Phi_{vy} = (z x y)\Phi'_{vz}$ where $\Phi_{vz} \neq \Phi'_{vz}$ and both are such that

$$\sigma^{(\kappa)}(y z x)\Phi_{vz} = \Phi_{vz}, \quad (2.133)$$

$$\sigma^{(\kappa)}(y z x)\Phi'_{vz} = \Phi'_{vz}. \quad (2.134)$$

Now we should apply the double inversion factor operator $\mathcal{D}_z$ to the wavefunction. To ensure symmetrization we wish to allow

$$\mathcal{D}_z^{(\kappa)}\psi'_q = \psi'_q. \quad (2.135)$$

For (2.135) to be true for $q = z$, it must be the case that

$$\Phi_{vz} = \left(\frac{(x y z) + (\bar{x} \bar{y} z)}{4} \right) \Phi_{vz} + \omega^{(\kappa)} \left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{4} \right) \Phi_{-vz}. \quad (2.136)$$

Some further arithmetic work on (2.136) can establish the following relationships. We can show that

$$\Phi_{vz} = (\bar{x} \bar{y} z) \Phi_{vz}, \quad (2.137)$$

$$\Phi_{-vz} = \omega^{(\kappa)} \left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{2} \right) \Phi_{vz}. \quad (2.138)$$

We saw in (2.130) that the presence of x -valley terms implies the presence of y -valley terms in our partially symmetrized wavefunction, and vice versa. Thus, for the

valleys perpendicular to z we must have both

$$\Phi_{vx} = \left(\frac{(xyz) + \omega^{(\kappa)}(x\bar{y}\bar{z})}{4} \right) \Phi_{vx} + \left(\frac{(\bar{x}\bar{y}z) + \omega^{(\kappa)}(\bar{x}y\bar{z})}{4} \right) \Phi_{-vx}, \quad (2.139)$$

and

$$\Phi_{vy} = \left(\frac{(xyz) + \omega^{(\kappa)}(\bar{x}y\bar{z})}{4} \right) \Phi_{vy} + \left(\frac{(\bar{x}\bar{y}z) + \omega^{(\kappa)}(x\bar{y}\bar{z})}{4} \right) \Phi_{-vy}. \quad (2.140)$$

As before we can do some arithmetic work to simplify these expressions. If we assume that (2.139) and (2.140) are both true we find

$$\Phi_{vx} = \omega^{(\kappa)}(x\bar{y}\bar{z})\Phi_{vx}, \quad (2.141)$$

$$\Phi_{-vx} = \left(\frac{(\bar{x}\bar{y}z) + \omega^{(\kappa)}(\bar{x}y\bar{z})}{2} \right) \Phi_{vx}, \quad (2.142)$$

as well as

$$\Phi_{vy} = \omega^{(\kappa)}(\bar{x}y\bar{z})\Phi_{vy}, \quad (2.143)$$

$$\Phi_{-vy} = \left(\frac{(\bar{x}\bar{y}z) + \omega^{(\kappa)}(x\bar{y}\bar{z})}{2} \right) \Phi_{vy}. \quad (2.144)$$

It is worth noting that if we assume (2.130) is true, and if we also observe that

$$(yxz)(\bar{x}y\bar{z})(yxz) = (x\bar{y}\bar{z}). \quad (2.145)$$

then we see that (2.141) is equivalent to (2.143) and (2.142) is equivalent to (2.144).

As before, for the $\kappa = A, E$ states we can use the indexing relationships to try to find additional equivalencies between the above relationships. We once again invoke

(2.121), (2.122), and (2.123). In that spirit, note the following

$$(z x y)(x \bar{y} \bar{z})(y z x) = (\bar{x} \bar{y} z) \quad (2.146)$$

$$(y z x)(\bar{x} \bar{y} \bar{z})(z x y) = (\bar{x} \bar{y} z) \quad (2.147)$$

Applying these relationships to (2.141) and (2.143) yields

$$\Phi_{vz} = \omega^{(\kappa)}(\bar{x} \bar{y} z) \Phi_{vz}. \quad (2.148)$$

For both (2.137) and (2.148) to be true, it must be the case that

$$\omega^{(\kappa)} = +1. \quad (2.149)$$

This condition is satisfied for states with $\kappa = A, E$.

We can also examine the relationships between (2.138), (2.142), and (2.144). Examining the symmetry operators and their products, we assert that

$$(z x y)(\bar{x} \bar{y} z)(y z x) = (\bar{x} \bar{y} z) \quad (2.150)$$

$$(z x y)(\bar{x} \bar{y} \bar{z})(y z x) = (x \bar{y} \bar{z}) \quad (2.151)$$

$$(y z x)(\bar{x} \bar{y} z)(z x y) = (x \bar{y} \bar{z}) \quad (2.152)$$

$$(y z x)(\bar{x} \bar{y} \bar{z})(z x y) = (\bar{x} \bar{y} z) \quad (2.153)$$

Applying these relationships to (2.142), and (2.144) we find

$$\Phi_{-vz} = \left(\frac{(\bar{x} \bar{y} z) + \omega^{(\kappa)}(x \bar{y} \bar{z})}{2} \right) \Phi_{vz}, \quad (2.142a)$$

$$\Phi_{-vz} = \left(\frac{(x \bar{y} \bar{z}) + \omega^{(\kappa)}(\bar{x} \bar{y} z)}{2} \right) \Phi_{vz}, \quad (2.144a)$$

We note that if (2.149) is true, then (2.138), (2.142a), and (2.144a) are equivalent.

So far we have found the form of the partially symmetrized wavefunction to be such that

$$\mathcal{D}_z^{(\kappa)} \mathcal{A}_z^{(\kappa)} \mathcal{V}_{\kappa,i}^\dagger \Psi = \mathcal{V}_{\kappa,i}^\dagger \Psi. \quad (2.95f)$$

In essence, the process by which we have applied the $\mathcal{A}_z$ acyclic operator and the $\mathcal{D}_z$ double inversion operator hasn't altered the overall structure of our wavefunction, but has exerted conditions upon the underlying envelope functions.

It can be shown that the application of the final factor operator $\mathcal{V}$ returns the wavefunction to its initial form. To ensure equality we find the following requirements upon our envelope functions. For the $\kappa = A, E$ states the following requirements are sufficient.

$$\Phi_{vz} = \sigma^{(\kappa)}(y x z) \Phi_{vz}. \quad (2.126)$$

$$\Phi_{vz} = (\bar{x} \bar{y} z) \Phi_{vz}, \quad (2.137)$$

$$\Phi_{-vz} = \left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{2} \right) \Phi_{vz}. \quad (2.138)$$

Due to the valley mixing of the A - and E -states the envelope functions of other valleys may be found from the following indexing relationships.

$$\Phi_{vx} = 3\alpha_1^{(\kappa,i)}(y z x) \Phi_{vz}, \quad (2.121)$$

$$\Phi_{vy} = 3\alpha_2^{(\kappa,i)}(z x y) \Phi_{vz}. \quad (2.122)$$

For the $\kappa = T$ states, the symmetry picks out a longitudinal axis, with the other axes being perpendicular. The longitudinal envelope functions are independent of the transverse envelope functions. We can state the requirements on these envelope

functions as such. For the longitudinal envelopes we see

$$\Phi_{vz} = \sigma^{(\kappa)}(y x z) \Phi_{vz}. \quad (2.126)$$

$$\Phi_{vz} = (\bar{x} \bar{y} z) \Phi_{vz}, \quad (2.137)$$

$$\Phi_{-vz} = -\left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{2}\right) \Phi_{vz}. \quad (2.138)$$

The transverse envelopes are independent of the longitudinal envelopes, but not independent of each other. The relationship between the transverse envelopes follows

$$\Phi_{vx} = \sigma^{(\kappa)}(y x z) \Phi_{vy}. \quad (2.130)$$

The transverse envelopes must also follow

$$\Phi_{vx} = \omega^{(\kappa)}(x \bar{y} \bar{z}) \Phi_{vx}, \quad (2.141)$$

$$\Phi_{-vx} = \left(\frac{(\bar{x} \bar{y} z) - (\bar{x} y \bar{z})}{2}\right) \Phi_{vx}. \quad (2.142)$$

2.1.10 Silicon envelope functions and single axis wavefunctions.

With this discussion on the overall wavefunction and its implications upon the envelope functions in mind, we examine our envelope function basis. In appendix B we explored the transformation properties of the state defined below,

$$\phi_{klm;\eta\lambda;\mathbf{r}}^{(Wq)} = F_{lk;\eta\lambda;r\chi_q}^{(W)} P_{lm;\lambda;\chi_q} Q_{m;\phi_q}, \quad (2.154)$$

where each part may be defined as below. The naive radial part is defined in 2.1.7 in equation (2.77)

$$F_{kl;\eta;r}^{(W)} = \sqrt{\frac{W}{\eta^3}} L_{k;\rho^w}^{(\alpha)} e^{-\frac{\rho^w}{2}} \rho^l, \quad \rho = \frac{r}{\eta}. \quad (2.57)$$

In the equation above, l and k are nonnegative integers and $\eta > 0$ is a variational radial scaling parameter. The function L is a generalized Laguerre polynomial of k -th order and $\alpha = \frac{2l+3-W}{W}$. We expect $W = 1$ or 2 .

When variational anisotropy $\chi_q \rightarrow \lambda\chi_q$ is introduced, we can define a new "radial" function as follows, by introducing a scaling of the radial parameter that goes as $r \rightarrow r\vartheta_{\lambda\chi_q}$. This emerges as a modification of the radial parameters η with angular dependence. We may write it as

$$F_{lk;\eta\lambda;r\chi_q}^{(W)} = F_{lk;\eta\lambda\chi_q;r}^{(W)} \quad \eta_{\lambda;\chi_q} = \frac{\eta}{\vartheta_{\lambda\chi_q}}. \quad (2.77)$$

The "longitudinal" part can be defined as below, with the function P being the associated Legendre polynomials with degree l and order m . m is nonnegative and l is an integer such that $l \geq m$.

$$P_{lm;\lambda;\chi_q} = \sqrt{\frac{\lambda}{\vartheta_{\lambda\chi_q}^3}} P_{lm;\chi_\lambda}^{(q)}. \quad (B.16)$$

The azimuthal part of the envelope basis function are unaffected by the introduction of the variational anisotropy. As discussed in Appendix B the standard definition of the spherical harmonics has been changed to better accommodate the symmetry of our system.

$$Q_{m\zeta;\phi_q} := \delta_{m0}(1 - \delta_{\zeta\bar{1}}) + (1 - \delta_{m0}) \left(\frac{e^{im\phi_q} + \zeta e^{-im\phi_q}}{\sqrt{2}} \right). \quad (B.24)$$

As before m is a nonnegative integer. We consider ζ complex such that $|\zeta| = 1$.

We define the following as our *naive envelope basis function* with anisotropy included as follows,

$$\phi_{klm\zeta;\eta\lambda;r}^{(Wq)} := F_{lk;\eta\lambda;r\chi_q}^{(W)} P_{lm;\lambda;\chi_q} Q_{m\zeta;\phi_q}, \quad (2.155)$$

Earlier, we introduced the concept of valley-indexing. We see of course that naive basis function $\phi^{(z)}$ can be directly indexed by z , as can any arbitrary function. However,

we should also note that the combination of rotated functions $\phi^{(x)} \pm \sigma(yxz)\phi^{(y)}$ can also be indexed by z in the same sense as $\phi^{(z)}$. As such, we shall consider the following generic form for our valley-indexed envelope function.

$$\Phi_{\nu q} := a_z^{(q)} \phi_{\zeta_z^{(q)}}^{(z)} + a_x^{(q)} \phi_{\zeta_x^{(q)}}^{(x)} + a_y^{(q)} \phi_{\zeta_y^{(q)}}^{(y)} \quad (2.156)$$

We can examine the effect of the transformations that form the $\mathcal{A}_z$ and $\mathcal{D}_z^{(\kappa)}$ factor operators upon our naive envelope basis function defined in (2.155). For such a basis function, we previously elucidated in Appendix B the effects of the transformations under consideration. To wit, for the z -index we found

$$\zeta i^{-m} \phi_{(-1)^m \zeta_*}^{(z)} = (y x z) \phi_\zeta^{(z)}, \quad (\text{B.50})$$

$$\zeta (-1)^{l+m} \phi_{\zeta_*}^{(z)} = (x \bar{y} \bar{z}) \phi_\zeta^{(z)}, \quad (\text{B.49})$$

$$(-1)^m \phi_\zeta^{(z)} = (\bar{x} \bar{y} z) \phi_\zeta^{(z)}, \quad (\text{B.47})$$

$$\zeta (-1)^l \phi_{\zeta_*}^{(z)} = (\bar{x} y \bar{z}) \phi_\zeta^{(z)}. \quad (\text{B.48})$$

For the y -index functions we found

$$\zeta i^{-m} \phi_{(-1)^m \zeta_*}^{(x)} = (y x z) \phi_\zeta^{(y)}, \quad (\text{B.54})$$

$$\zeta (-1)^l \phi_{\zeta_*}^{(y)} = (x \bar{y} \bar{z}) \phi_\zeta^{(y)}, \quad (\text{B.53})$$

$$\zeta (-1)^{l+m} \phi_{\zeta_*}^{(y)} = (\bar{x} \bar{y} z) \phi_\zeta^{(y)}, \quad (\text{B.51})$$

$$(-1)^m \phi_\zeta^{(y)} = (\bar{x} y \bar{z}) \phi_\zeta^{(y)}. \quad (\text{B.52})$$

Finally, for the x -index functions we found

$$\zeta i^{-m} \phi_{(-1)^m \zeta_*}^{(y)} = (y x z) \phi_{\zeta}^{(x)}, \quad (\text{B.58})$$

$$(-1)^m \phi_{\zeta}^{(x)} = (x \bar{y} \bar{z}) \phi_{\zeta}^{(x)}, \quad (\text{B.57})$$

$$\zeta (-1)^l \phi_{\zeta_*}^{(x)} = (\bar{x} \bar{y} z) \phi_{\zeta}^{(x)}, \quad (\text{B.55})$$

$$\zeta (-1)^{l+m} \phi_{\zeta_*}^{(x)} = (\bar{x} y \bar{z}) \phi_{\zeta}^{(x)}. \quad (\text{B.56})$$

In the previous section, we determined that our z -indexed envelope functions must obey these conditions,

$$\Phi_{\lambda z} = \sigma^{(\kappa)}(y x z) \Phi_{\lambda z}, \quad (\text{2.126})$$

$$\Phi_{\lambda z} = (\bar{x} \bar{y} z) \Phi_{\lambda z}. \quad (\text{2.137})$$

For $\kappa = A, E$ the envelope functions valley-indexed by x and y may be found from the z -indexed function via (2.121) and (2.122). For $\kappa = T$ we also need to define independent x - and y -indexed envelopes. They will follow these relationships.

$$\Phi_{ux} = \sigma^{(\kappa)}(y x z) \Phi_{uy}, \quad (\text{2.130})$$

$$\Phi_{ux} = -(x \bar{y} \bar{z}) \Phi_{ux}, \quad (\text{2.141})$$

$$\Phi_{uy} = \sigma^{(\kappa)}(y x z) \Phi_{ux}, \quad (\text{2.130}')$$

$$\Phi_{uy} = -(\bar{x} y \bar{z}) \Phi_{uy}. \quad (\text{2.143})$$

It can be shown quite directly that (2.130) and (2.130') are equivalent. Additionally, if (2.130) is true then it can be shown that that (2.141) and (2.143) are also equivalent. As such, we can examine the implications of just (2.130) and (2.141).

Let's start by exploring the implications of (2.126) and (2.137) upon (2.156) for $q =$

z . In particular we can set $\zeta_z^{(z)} = \sigma i^{+m}$. This yields

$$(y x z) \phi_{\sigma i^{+m}}^{(z)} = \sigma \phi_{\sigma i^{+m}}^{(z)}, \quad (2.157)$$

$$(\bar{x} \bar{y} z) \phi_{\sigma i^{+m}}^{(z)} = (-1)^m \phi_{\sigma i^{+m}}^{(z)}. \quad (2.158)$$

For $\kappa = A, T$ we can satisfy equation (2.126) if we set $\sigma = \sigma^{(\kappa)}$. For $\kappa = E$ we recall that $\sigma^{(\kappa)} = 0$. For E -states we can take $\sigma = \pm 1$ as a free index, similar to m, l , and k . We note that (2.137) can only be satisfied for $\Phi^{(z)}$ when m is even. When m is odd, we are unable to construct an appropriate envelope function from $\Phi_{\sigma i^{+m}}^{(z)}$. As such, we'll set

$$a_z^{(z)} \rightarrow \left(\frac{1 + (-1)^m}{2} \right) a_z^{(z)} \quad (2.159)$$

Moving forward, we consider the implications of (2.137) and (2.126) upon the x - and y -indexed functions. To begin, let's examine the implications of (2.137) and (2.126) upon our x, y -indexed naive envelopes, namely

$$(\bar{x} \bar{y} z) \phi_{(-1)^l}^{(x)} = \phi_{(-1)^l}^{(x)}, \quad (2.160)$$

$$(\bar{x} \bar{y} z) \phi_{(-1)^{l+m}}^{(y)} = \phi_{(-1)^{l+m}}^{(y)}, \quad (2.161)$$

$$\sigma(y x z) \phi_{(-1)^l}^{(x)} = \sigma(-1)^l i^{-m} \phi_{(-1)^{l+m}}^{(y)}, \quad (2.162)$$

$$\sigma(y x z) \phi_{(-1)^{l+m}}^{(y)} = \sigma(-1)^{l+m} i^{-m} \phi_{(-1)^l}^{(x)}. \quad (2.163)$$

Above, we set $\zeta_x^{(z)} = (-1)^l$ and $\zeta_y^{(z)} = (-1)^{l+m}$. At this juncture we find it convenient to define the following constant,

$$\Omega := \sigma \omega (-1)^{l+m} i^{+m}. \quad (2.164)$$

Next, consider the following function.

$$\phi_{\sigma}^{(yx)} = \phi_{(-1)^l}^{(x)} + \omega \Omega \phi_{(-1)^{l+m}}^{(y)}. \quad (2.165)$$

This function obeys (2.137) and (2.126) with no caveats.

$$\phi_{\sigma}^{(yx)} = \sigma(yxz) \phi_{\sigma}^{(yx)}, \quad (2.166)$$

$$\phi_{\sigma}^{(yx)} = (\bar{x} \bar{y} z) \phi_{\sigma}^{(yx)}. \quad (2.167)$$

We note that if we set our coefficients in (2.156) for $q = z$ to go as

$$a_x^{(z)} \rightarrow a_{xy}^{(z)}, \quad (2.168)$$

$$a_y^{(z)} \rightarrow \omega \Omega a_{xy}^{(z)}, \quad (2.169)$$

we find that our $+z$ envelope functions goes as

$$\Phi_{+z} = a_z^{(z)} \left(\frac{1 + (-1)^m}{2} \right) \phi_{\sigma i^{+m}}^{(z)} + a_{xy}^{(z)} \phi_{\sigma}^{(yx)} \quad (2.170)$$

Note that the coefficients $a^{(z)}$ in Φ_{+z} above remain arbitrary. As such, for our purposes we find it convenient within our software methodology to treat these terms as different states within our basis. To that end, we can write our z -indexed envelope functions as

$$\Phi_{\parallel 0, +z; \sigma} := \left(\frac{1 + (-1)^m}{2} \right) \phi_{\sigma i^{+m}}^{(z)}, \quad (2.171)$$

$$\Phi_{\parallel 1, +z; \sigma} := \phi_{(-1)^l}^{(x)} + \sigma i^{+m} (-1)^{l+m} \phi_{(-1)^{l+m}}^{(y)}. \quad (2.172)$$

Recall that for $\kappa = A, T$ the index $\sigma = \pm 1$ is set by the symmetry, while for $\kappa = E$ it is a free index.

Moving forward, we can utilize (2.138) as a definition. Namely, we can define the negative valley index envelope functions as

$$\Phi_{-z} := \omega^{(\kappa)} \left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{2} \right) \Phi_{+z}. \quad (2.173)$$

Applying these to our envelope basis functions yields

$$\Phi_{\parallel 0, -z; \sigma} := \omega \left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{2} \right) \Phi_{\parallel 0, +z; \sigma} = \sigma \omega i^{+m} (-1)^{l+m} \Phi_{\parallel 0, +z; \sigma}, \quad (2.174)$$

$$\Phi_{\parallel 1, -z; \sigma} := \omega \left(\frac{(\bar{x} y \bar{z}) + (x \bar{y} \bar{z})}{2} \right) \Phi_{\parallel 1, +z; \sigma} = \omega (-1)^m \Phi_{\parallel 1, +z; \sigma}. \quad (2.175)$$

Having defined our z -indexed envelope function we can introduce *single-axis* wavefunctions, combining the single axis wavefunctions of opposite valleys lying along one axis. To that end, we can introduce the following *axial Bloch function*,

$$\Theta_{\xi}^{(q)} = \theta^{(+q)} + \xi \theta^{(-q)} \quad (2.176)$$

We write our z -axis wavefunction, including the axial Bloch functions.

$$\psi_{\parallel 0, \sigma \omega}^{(z)} := \left(\frac{1 + (-1)^m}{2} \right) \Phi_{\parallel 0, +z; \sigma} \Theta_{\sigma \omega i^{+m} (-1)^{l+m}}^{(z)}, \quad (2.177)$$

$$\psi_{\parallel 1, \sigma \omega}^{(z)} := \Phi_{\parallel 1, +z; \sigma} \Theta_{\omega (-1)^m}^{(z)}. \quad (2.178)$$

This forms the core of the definition of our single axis states for $\kappa = A, E$, as we can find the x - and y -axis single axis states using the cyclic transformations given in (2.121) and (2.122).

However, for the $\kappa = T$ states, the x - and y -indexed perpendicular envelope functions cannot be found from the z -indexed longitudinal envelope function. We invoke (2.130) and (2.141) to specify conditions upon our perpendicular envelopes and again utilize (2.156) as our general form. With this in mind we will explore the qualities of

forms suitable for transverse Φ_{+x} and Φ_{+y} . In particular, let's consider the effect of the operator used in (2.141) upon our naive envelope functions.

$$\omega(x \bar{y} \bar{z}) \phi_{\zeta_x^{(x)}}^{(x)} = \omega(-1)^m \phi_{\zeta_x^{(x)}}^{(x)}, \quad (\text{B.57a})$$

$$\omega(x \bar{y} \bar{z}) \phi_{\zeta_y^{(x)}}^{(y)} = \omega \zeta_y^{(x)} (-1)^l \phi_{\{\zeta_y^{(x)}\}^*}^{(y)}, \quad (\text{B.53a})$$

$$\omega(x \bar{y} \bar{z}) \phi_{\zeta_z^{(x)}}^{(z)} = \omega \zeta_z^{(x)} (-1)^{l+m} \phi_{\{\zeta_z^{(x)}\}^*}^{(z)}. \quad (\text{B.49a})$$

In light of the above information, to ensure we satisfy (2.141) for the terms in Φ_{+x} we set $\zeta_z^{(x)} = \omega(-1)^{l+m}$ and $\zeta_y^{(x)} = \omega(-1)^l$. However, none of the above conditions defining the envelope functions determine $\zeta_x^{(x)}$. Since the ζ indices for the other functions have been determined at this point, we may drop the upper and lower valley indices on $\zeta_x^{(x)}$ without ambiguity. It is convenient for our purposes, with some loss of generality, to have this arbitrary ζ real, so that $\zeta = \pm 1$. We will see how this simplifies the expression of the ψ_{-x} and ψ_{-y} , in particular allowing the Bloch and envelope functions to be factored apart from each other. We do this to simplify the integration expression, helping both analytical and numerical computation of the integrals required to compute the Hamiltonian matrix elements. We can also note that for (B.57a) it must be the case for the x -indexed term that $\omega(-1)^m = 1$ or $\Phi^{(x)} = 0$. This can be accomplished by the same strategy as in (2.159). In light of the above discussion, we can state the $+x$ envelope as

$$\Phi_{+x} = a_x \left(\frac{1 + \omega(-1)^m}{2} \right) \phi_{\zeta}^{(x)} + a_y \phi_{\omega(-1)^l}^{(y)} + a_z \phi_{\omega(-1)^{l+m}}^{(z)}. \quad (2.179)$$

We note that since the coefficients in this expression are shared between the x - and y -valley indexed envelope, the upper valley index on these coefficients are superfluous and can be dropped. In fact, much as we did for the longitudinal envelopes in the

equations (2.171) and (2.172) we can separate out our basis functions.

$$\Phi_{\perp_0,+x;\zeta\omega} := \left(\frac{1 + \omega(-1)^m}{2} \right) \phi_{\zeta}^{(x)}, \quad (2.180)$$

$$\Phi_{\perp_1,+x;\omega} := \phi_{\omega(-1)^l}^{(y)}, \quad (2.181)$$

$$\Phi_{\perp_2,+x;\omega} := \phi_{\omega(-1)^{l+m}}^{(z)} \quad (2.182)$$

Now, to find Φ_{-x} we apply our x -indexed opposite valley operator to the Φ_{+x} envelope basis states. We tabulate the effect of the operator on the naive envelope basis states below.

$$\zeta(-1)^l \left(\frac{1 + \omega(-1)^m}{2} \right) \phi_{\zeta^*}^{(x)} = \left(\frac{(\bar{x}\bar{y}z) + \omega(\bar{x}y\bar{z})}{2} \right) \phi_{\zeta}^{(x)}, \quad (\text{B.55+B.56})$$

$$\omega(-1)^m \phi_{\omega(-1)^l}^{(y)} = \left(\frac{(\bar{x}\bar{y}z) + \omega(\bar{x}y\bar{z})}{2} \right) \phi_{\omega(-1)^l}^{(y)}, \quad (\text{B.51+B.52})$$

$$(-1)^m \phi_{\omega(-1)^{l+m}}^{(z)} = \left(\frac{(\bar{x}\bar{y}z) + \omega(\bar{x}y\bar{z})}{2} \right) \phi_{\omega(-1)^{l+m}}^{(z)}. \quad (\text{B.47+B.48})$$

As such, we find the following for Φ_{-x}

$$\Phi_{\perp_0,-x;\zeta\omega} := \left(\frac{(\bar{x}\bar{y}z) + \omega(\bar{x}y\bar{z})}{2} \right) \Phi_{\perp_0,+x;\zeta\omega} = \zeta(-1)^l \Phi_{\perp_0,+x;\zeta\omega}, \quad (2.183)$$

$$\Phi_{\perp_1,-x;\omega} := \left(\frac{(\bar{x}\bar{y}z) + \omega(\bar{x}y\bar{z})}{2} \right) \Phi_{\perp_1,+x;\omega} = \omega(-1)^m \Phi_{\perp_1,+x;\omega}, \quad (2.184)$$

$$\Phi_{\perp_2,-x;\omega} := \left(\frac{(\bar{x}\bar{y}z) + \omega(\bar{x}y\bar{z})}{2} \right) \Phi_{\perp_2,+x;\omega} = (-1)^m \Phi_{\perp_2,+x;\omega}. \quad (2.185)$$

We may now define our single axial x -axis wavefunctions, going as

$$\psi_{\perp_0,\zeta\omega}^{(x)} = \left(\frac{1 + \omega(-1)^m}{2} \right) \phi_{\zeta}^{(x)} \Theta_{\zeta(-1)^l}^{(x)}, \quad (2.186)$$

$$\psi_{\perp_1,\omega}^{(x)} = \phi_{\omega(-1)^l}^{(y)} \Theta_{\omega(-1)^m}^{(x)}, \quad (2.187)$$

$$\psi_{\perp_2,\omega}^{(x)} = \phi_{\omega(-1)^{l+m}}^{(z)} \Theta_{(-1)^m}^{(x)}. \quad (2.188)$$

We can take (2.130') as definitional for $\psi^{(y)}$, namely

$$\psi_{\mathbf{x},\sigma\omega}^{(y)} := \sigma(yxz)\psi_{\mathbf{x},\omega}^{(x)} \quad (2.189)$$

To further this analysis let's examine the effect of the acyclic swap upon our naive envelope functions. This yields

$$\sigma(yxz)\phi_{\zeta}^{(x)} = \zeta\omega(-1)^l\Omega\phi_{(-1)^m\zeta}^{(y)}, \quad (\text{B.58})$$

$$\sigma(yxz)\phi_{\omega(-1)^l}^{(y)} = \Omega\phi_{\omega(-1)^{l+m}}^{(x)}, \quad (\text{B.54a})$$

$$\sigma(yxz)\phi_{\omega(-1)^{l+m}}^{(z)} = (-1)^m\Omega\phi_{\omega(-1)^l}^{(z)}. \quad (\text{B.50a})$$

Considering the above expressions involving our naive envelope basis functions yield the following expressions for our y -indexed perpendicular single-axis wavefunctions.

$$\psi_{\perp_0,\zeta\omega}^{(y)} = \zeta\omega(-1)^l\Omega\left(\frac{1+\omega(-1)^m}{2}\right)\phi_{\omega\zeta}^{(y)}\Theta_{\zeta(-1)^l}^{(y)}, \quad (2.190)$$

$$\psi_{\perp_1,\omega}^{(y)} = \Omega\phi_{\omega(-1)^{l+m}}^{(x)}\Theta_{\omega(-1)^m}^{(y)}, \quad (2.191)$$

$$\psi_{\perp_2,\omega}^{(y)} = (-1)^m\Omega\phi_{\omega(-1)^l}^{(z)}\Theta_{(-1)^m}^{(y)}. \quad (2.192)$$

2.1.11 Full symmetrized wavefunction, including envelope function anisotropy

This section is a collection of information derived in previous sections, and is collected here to make referencing of the form of our wavefunctions for various symmetries swift and convenient for users and maintainers of the software.

We have a few preliminaries to define. We may start with what we in this document call the *axial Bloch function*. In our system, the Bloch functions we are most concerned are the six energy minima of the conduction band of bulk silicon. Each lies along a particular crystallographic k -space axis, which may be labeled x , y , and

z. We denote the location in k -space where each minima is located as follows

$$\mathbf{k}_{\pm q} = \pm \eta_{\text{Si}} k_0 \hat{q}, \quad (2.193)$$

where

$$k_0 = \frac{2\pi}{a_{\text{Si}}}. \quad (2.194)$$

Above $a_{\text{Si}} = 0.357 \text{ nm}$ is the lattice parameter of the silicon crystal, and $\eta_{\text{Si}} \approx 0.85$.

We use the following notation to describe one of these valley Bloch functions

$$\Theta^{(\pm q)} := e^{i\mathbf{k}_{\pm q} \cdot \mathbf{r}} u_{\mathbf{k}_{\pm q}}(\mathbf{r}), \quad (2.195)$$

$$= e^{i\eta_{\text{Si}} k_0 r \chi_q} u_{\mathbf{k}_{\pm q}}(\mathbf{r}). \quad (2.196)$$

In the previous section we found that our choice of envelope function basis, along with certain parameters specifying functions within that basis, allowed us to factor the envelope function from the Bloch functions of opposite valleys. These factored out Bloch function summations have the following form

$$\Theta_{\xi}^{(q)} = \Theta^{(+q)} + \xi \Theta^{(-q)}. \quad (2.197)$$

We describe this form as the *axial Bloch function*.

There is a feature of notation below we should discuss; previously, we defined the m index of functions in our basis to extend over the nonnegative integers. Because some of the basis functions must be set to zero depending on the parity of m , we introduce new notation to reflect this. Namely, m_e extends over the nonnegative even integers, including zero. Whereas, m_o should be understood to extend over the non-negative odd integers. When invoked without subscript, m retains its original domain of all nonnegative integers.

Now, as we build up our multi-valley wavefunctions we consider what we will describe here as the *z-longitudinal single-axis basis wavefunctions*. These wavefunction terms are longitudinal in the sense that they only include parts of the full wavefunction with only *z*-axial Bloch functions used in their construction. We found that, for $(\kappa, i) = (A_n, 0)$, (E, i) , and (T_n, z) the *z*-longitudinal single-axis basis wavefunctions to go as

$$\psi_{\parallel 0, m_e l k \sigma \omega}^{(z)} := \phi_{klm_e \zeta}^{(z)} \Theta_{\sigma \omega (-1)^{l+m_e/2}}^{(z)}, \quad (2.171+2.177)$$

$$\psi_{\parallel 1, m l k \sigma \omega}^{(z)} := \left\{ \phi_{klm \zeta_0}^{(x)} + \sigma (-1)^{l+m} i^{+m} \phi_{klm \zeta_1}^{(y)} \right\} \Theta_{\omega (-1)^m}^{(z)}. \quad (2.172+2.178)$$

These *z*-longitudinal functions were constructed from the *naive envelope basis function* denoted by lowercase ϕ with suitable azimuthal constants ζ as given below.

$$\zeta = \sigma (-1)^{m_e/2}, \quad (2.198)$$

$$\zeta_0 = (-1)^l \quad \zeta_1 = (-1)^{l+m}. \quad (2.199)$$

We have enough information now to construct the full wavefunctions for the $\kappa = A, E$ symmetries. For those states, we will need *q*-longitudinal single-axis wavefunctions for $q = x, y$ as well as $q = z$. Fortunately, we can invoke (2.121) and (2.122) to find the single-axis wavefunctions for the *x* and *y* axes. The relevant cyclic transformations go simply as $\psi^{(x)} = (y z x) \psi^{(z)}$ and $\psi^{(y)} = (z x y) \psi^{(z)}$. These transformations straightforwardly swap the *z* index in the above expressions for the *x* and *y* expressions respectively. Following this line of reasoning and using the equations

mentioned above we can state the forms of the full A , E wavefunctions to be

$$\Psi_{\varrho,mlk}^{(A_n)} = \frac{1}{3}\psi_{\varrho,mlk\sigma^{(A_n)}(+1)}^{(z)} + \frac{1}{3}\psi_{\varrho,mlk\sigma^{(A_n)}(+1)}^{(x)} + \frac{1}{3}\psi_{\varrho,mlk\sigma^{(A_n)}(+1)}^{(y)}, \quad (2.200)$$

$$\Psi_{\varrho,mlk\sigma}^{(E,0)} = \frac{1}{3}\psi_{\varrho,mlk\sigma(+1)}^{(z)} + \frac{\tau}{3}\psi_{\varrho,mlk\sigma(+1)}^{(x)} + \frac{\tau^2}{3}\psi_{\varrho,mlk\sigma(+1)}^{(y)}, \quad (2.201)$$

$$\Psi_{\varrho,mlk\sigma}^{(E,1)} = \frac{1}{3}\psi_{\varrho,mlk\sigma(+1)}^{(z)} + \frac{\tau^2}{3}\psi_{\varrho,mlk\sigma(+1)}^{(x)} + \frac{\tau}{3}\psi_{\varrho,mlk\sigma(+1)}^{(y)}. \quad (2.202)$$

Above $\tau = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, one of the third roots of unity. The A -type states are nondegenerate, so we drop the degeneracy index above. We retain the degeneracy index for the doubly degenerate E -type states. While σ is a good quantum number for the A and T states, it is not for the E states. As such, since our basis respects this symmetry, we should include contributions from both $\sigma = -1$ and $\sigma = +1$ when considering $\kappa = E$. When necessary we can treat σ as a basis index like m , l , or k . Finally, we introduce a $\varrho = ||_0, ||_1$ index that we will describe here as the single-axis functional form index.

For the A and E symmetries, only q -longitudinal single valley terms are necessary to construct the full multi-valley wavefunction. In essence the idea of a "z-longitudinal" or "z-transverse" single-axis wavefunction is irrelevant for these symmetries – the three axes are all present in the wavefunction, and are in some sense interchangeable. As we will see, however, for T symmetries this distinction between longitudinal and transverse in relation to a given valley is relevant. In the previous section we were able to find an expression for transverse forms for $\kappa = T_n$ states. The T -states are triply degenerate, and each degeneracy may be valley indexed. If we consider the $(\kappa, i) = (T_n, q)$ state, we can define q -longitudinal wavefunction terms to be those that include the q -valley axial Bloch functions, while q -transverse wavefunction terms are defined as those which include axial Bloch functions with valleys perpendicular to q .

To wit, we found these z -*transverse single-axis basis wavefunctions* to be

$$\Psi_{\perp_0, m_0 l k \zeta}^{(T_n, z)} = \phi_{m_0 l k \zeta}^{(x)} \Theta_{\zeta(-1)^l}^{(x)} - \zeta \sigma^{(T_n)} i^{+m} \phi_{m_0 l k \zeta}^{(y)} \Theta_{\zeta(-1)^l}^{(y)}, \quad (2.203)$$

$$\Psi_{\perp_1, m l k}^{(T_n, z)} = \phi_{m l k \zeta_0}^{(y)} \Theta_{(-1)^{m+1}}^{(x)} - \sigma^{(T_n)} (-1)^{l+m} i^{+m} \phi_{m l k \zeta_1}^{(x)} \Theta_{(-1)^{m+1}}^{(y)}, \quad (2.204)$$

$$\Psi_{\perp_2, m l k}^{(T_n, z)} = \phi_{m l k \zeta_1}^{(z)} \Theta_{(-1)^m}^{(x)} - \sigma^{(T_n)} (-1)^l i^{+m} \phi_{m l k \zeta_0}^{(z)} \Theta_{(-1)^m}^{(y)}. \quad (2.205)$$

It should be made clear that in transverse forms, by definition, the valley index of the Bloch functions functions are perpendicular to the valley index of the degeneracy index. As we examine the naive envelope basis functions, it is clear that various valley indices are used above and that which valley index is used for the naive functions depends upon which form index $\perp_0$, $\perp_1$, and $\perp_2$ is used. This is similar to the

We can articulate longitudinal forms for the multi-valley wavefunction with $\kappa = T_n$ built up from the z -*longitudinal single-axis basis wavefunction* we introduced earlier. Namely, for $\rho = \parallel_0, \parallel_1$ we find

$$\Psi_{\rho, m l k}^{(T_n, z)} = \psi_{\rho, m l k \sigma^{(T_n)}(-1)}^{(z)}. \quad (2.206)$$

Above, we articulated wavefunctions for the states of $(\kappa, i) = (T_n, z)$. For the other degenerate states, (T_n, x) and (T_n, y) , we can once again use the appropriate cyclic transformations to swap valley axes.

We will summarize the findings of the preceding chapter here. To that end, we will articulate a condensed form to collate all information required to specify a particular state within a wider basis. We apply the following notational nicety; by writing the symbol I in a fixed-width typewrite font, we are denoting a single arbitrary basis state term of our full multi-valley wavefunction. In particular, we imagine that I may function as a label for the particular state, and as an subscript on a quantity denoting membership in the collection of properties associated with the I state. We expect I to include all primary quantities necessary to define a particular basis state.

Information we might expect to be contained in this structure would include the (κ, i) (symmetry and degeneracy), and ϱ (single-axis function form) indices. It would also necessarily include the full or implicit definitions of the naive envelope basis functions in the form of discrete indices and continuous parameters. This definition consists of the collection of discrete free parameters such as the m , l , and k indices, the η and λ continuous free parameters, and the derived ζ acyclic transformation index. For convenience we can imagine that, in addition to the necessary primary independent parameters, the collection will also include derived secondary quantities used to construct the full wavefunction, such the Ω index used in the axial Bloch functions.

It should be made clear that the exact composition of elements in the collection $\mathbf{I}$ is not static insofar as the number and nature of the values may vary between different possible basis states we wish to explore. The structure of $\mathbf{I}$ and will depend upon the type of symmetry denoted by κ and the envelope function form denoted by ϱ . For example, all functions include the k , l , and m free discrete indices associated with the naive envelope functions. For most states, $0 \leq m \leq l$, and $0 \leq k$ are indices and may take on arbitrary integer values satisfying the given constraints. However, states using the $\parallel_0$ envelope function form do not include states with odd m . Comparatively, states of the $\kappa = T_n$ symmetry using the $\perp_0$ envelope function form only include states with m odd but also include the free index $h = \pm 1$ in addition to (mlk) . Similarly, states with the $\kappa = E$ type of symmetry also require four independent indices, where $\sigma = h$ than having its value determined by the symmetry index κ as it would be for the $\kappa = A_n, T_n$ states.

We introduce this construct $\mathbf{I}$ as useful notation for denoting the source of information and properties used within the analysis and software of Cantankerous Impurity. For example, in this document it will see use in formulae to compute the matrix elements of the Hamiltonian and overlap matrices. However, we feel the primary value of introducing $\mathbf{I}$ lies in the manner in which it will help guide and structure later

software design. More precise and rigorous definition of $\mathbb{I}$, if required, can be found in the relevant software documentation.

We can articulate the following form for the multi-valley basis functions as a combination of the so-called single-axis basis functions defined earlier in this document.

$$\psi_{\mathbb{I}} = \sum_p (\alpha_{\mathbb{I}})_p \psi_{\mathbb{I}}^{(p)}, \quad (2.207)$$

where α above has the same values as defined in subsection 2.1.8. We identify the set of indices $p = (0, 1, 2)$ as given in that section with the valley-indices as defined above going as $p = (z, x, y)$ respectively. Briefly, we may reiterate that

$$\alpha_{A_n} = \frac{1}{3}(1, 1, 1), \quad (2.208)$$

as well as

$$\alpha_{(E,0)} = \frac{1}{3}(1, \tau, \tau^2), \quad \alpha_{(E,1)} = \frac{1}{3}(1, \tau^2, \tau), \quad (2.209)$$

and

$$\alpha_{(T_n,i)} = (\delta_{ix}, \delta_{iy}, \delta_{iz}). \quad (2.210)$$

As articulated previously for all multi-valley states there are multiple independent constructions possible. Following the convention established earlier in the document, we denote which construction is used with the state form index ϱ . For most states, there are only two constructions of the single valley basis possible, which we denote with $\varrho = \parallel_0$, or $\varrho = \parallel_1$. For $\kappa = T_n$ there are five such constructions possible. We denote the additional form indices using $\perp_0$, $\perp_1$, and $\perp_2$. We refer to the $\parallel$ forms as *longitudinal* and the $\perp$ forms as *transverse*, but the distinction only exists and is only relevant for T_n states.

Another way we can articulate the multi-valley wavefunction is directly as the

summation

$$\Psi_{\mathbf{I}} = \sum_q \Phi_{\mathbf{I}}^{(q)} \Theta_{\Omega_{\mathbf{I}}}^{(q)}. \quad (2.211)$$

$$\Phi_{\mathbf{I}}^{(q)} = \sum_p (\boldsymbol{\beta}_{\mathbf{I}})_{pq} \phi_{(mlk)_{\mathbf{I}}(\zeta_{\mathbf{I}})_{pq}}^{(p)}. \quad (2.212)$$

$$\xi = \sigma(-1)^{l+m} i^{+m} \quad (2.213)$$

where

$$(\boldsymbol{\beta}_{(\kappa,i),\parallel_0})_{pq} = (\boldsymbol{\alpha}_{(\kappa,i)})_q \delta_{pq}, \quad \zeta_{\parallel_0} = \xi(-1)^{l+m}, \quad \Omega_{\parallel_0} = \omega \xi(-1)^m, \quad (2.214)$$

$$\boldsymbol{\beta}_{A_n, \parallel_1} = \frac{1}{3} \begin{pmatrix} 0 & \xi & 1 \\ 1 & 0 & \xi \\ \xi & 1 & 0 \end{pmatrix}, \quad (2.215)$$

$$\boldsymbol{\beta}_{(E,0), \parallel_1} = \frac{1}{3} \begin{pmatrix} 0 & \tau^2 \xi & 1 \\ \tau & 0 & \xi \\ \tau \xi & \tau^2 & 0 \end{pmatrix}, \quad (2.216)$$

$$\boldsymbol{\beta}_{(E,0), \parallel_1} = \frac{1}{3} \begin{pmatrix} 0 & \tau \xi & 1 \\ \tau^2 & 0 & \xi \\ \tau^2 \xi & \tau & 0 \end{pmatrix}, \quad (2.217)$$

$$\boldsymbol{\beta}_{(T_n, x), \parallel_1} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ \xi & 0 & 0 \end{pmatrix}, \quad (2.218)$$

$$\boldsymbol{\beta}_{(T_n, y), \parallel_1} = \begin{pmatrix} 0 & \xi & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad (2.219)$$

$$\boldsymbol{\beta}_{(T_n, z), \parallel_1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & \xi \\ 0 & 0 & 0 \end{pmatrix}, \quad (2.220)$$

$$(\zeta_{\parallel_1})_{(q+1)q} = (-1)^l, \quad (\zeta_{\parallel_1})_{(q+2)q} = (-1)^m (\zeta_{\parallel_1})_{q(q+1)}, \quad \Omega_{\parallel_1} = \omega(-1)^m, \quad (2.221)$$

$$(\zeta_{\perp_0})_{(q+1)(q+1)} = h, \quad (\zeta_{\perp_0})_{(q+2)(q+2)} = (-1)^m (\zeta_{\perp_0})_{(q+1)(q+1)}, \quad \Omega_{\perp_0} = h(-1)^m, \quad (2.222)$$

$$(\zeta_{\perp_1})_{(q+1)(q+2)} = -(-1)^l, \quad (\zeta_{\perp_1})_{(q+2)(q+1)} = (-1)^m (\zeta_{\perp_1})_{(q+1)(q+2)} \quad (2.223)$$

$$(\zeta_{\perp_2})_{q(q+1)} = -(-1)^{l+m}, \quad (\zeta_{\perp_2})_{q(q+2)} = (-1)^m (\zeta_{\perp_2})_{q(q+1)} \quad (2.224)$$

$$\boldsymbol{\beta}_{(T_n, z), \perp_0} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\zeta \xi (-1)^{l+m} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{\beta}_{(T_n, z), \perp_1} = \begin{pmatrix} 0 & -\xi & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \boldsymbol{\beta}_{(T_n, z), \perp_2} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -\xi (-1)^m & 0 \end{pmatrix}, \quad (2.225)$$

$$\boldsymbol{\beta}_{(T_n, x), \perp_0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -\zeta \xi (-1)^{l+m} \end{pmatrix}, \quad \boldsymbol{\beta}_{(T_n, x), \perp_1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -\xi \\ 0 & 1 & 0 \end{pmatrix}, \quad \boldsymbol{\beta}_{(T_n, x), \perp_2} = \begin{pmatrix} 0 & 1 & -\xi (-1)^m \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (2.226)$$

$$\boldsymbol{\beta}_{(T_n, y), \perp_0} = \begin{pmatrix} -\zeta \xi (-1)^{l+m} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{\beta}_{(T_n, y), \perp_1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -\zeta & 0 & 0 \end{pmatrix}, \quad \boldsymbol{\beta}_{(T_n, y), \perp_2} = \begin{pmatrix} 0 & 0 & 0 \\ -\xi (-1)^m & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (2.227)$$

In short, we imagine

we can imagine the collection I includes at least 3, and sometimes 4, discrete free parameters. We can denote those parameters by $(mlkh)$ with h representing the value of the fourth discrete free parameters if present and ignored otherwise. To fully define the complete multi-valley

$$\Psi_{\varrho, mlk}^{(A_n)} = \frac{1}{3} \psi_{\varrho, mlk\sigma^{(A_n)}(+1)}^{(z)} + \frac{1}{3} \psi_{\varrho, mlk\sigma^{(A_n)}(+1)}^{(x)} + \frac{1}{3} \psi_{\varrho, mlk\sigma^{(A_n)}(+1)}^{(y)}, \quad (2.228)$$

$$\Psi_{\varrho, mlk\sigma}^{(E,0)} = \frac{1}{3} \psi_{\varrho, mlk\sigma(+1)}^{(z)} + \frac{\tau}{3} \psi_{\varrho, mlk\sigma(+1)}^{(x)} + \frac{\tau^2}{3} \psi_{\varrho, mlk\sigma(+1)}^{(y)}, \quad (2.229)$$

$$\Psi_{\varrho, mlk\sigma}^{(E,1)} = \frac{1}{3} \psi_{\varrho, mlk\sigma(+1)}^{(z)} + \frac{\tau^2}{3} \psi_{\varrho, mlk\sigma(+1)}^{(x)} + \frac{\tau}{3} \psi_{\varrho, mlk\sigma(+1)}^{(y)}. \quad (2.230)$$

$$\Psi_{\varrho, mlk}^{(T_n, q)} = \delta_{qz} \psi_{\varrho, mlk\sigma^{(T_n)}(-1)}^{(z)} + \delta_{qx} \psi_{\varrho, mlk\sigma^{(T_n)}(-1)}^{(x)} + \delta_{qy} \psi_{\varrho, mlk\sigma^{(T_n)}(-1)}^{(y)}. \quad (2.231)$$

$$\psi_{\parallel_0, m_e l k \sigma \omega}^{(z)} := \phi_{klm_e \zeta}^{(z)} \Theta_{\sigma \omega (-1)^{l+m_e/2}}^{(z)}, \quad (2.171+2.177)$$

$$\psi_{\parallel_1, mlk\sigma\omega}^{(z)} := \left\{ \phi_{klm\zeta_0}^{(x)} + \sigma(-1)^{l+m} i^{+m} \phi_{klm\zeta_1}^{(y)} \right\} \Theta_{\omega(-1)^m}^{(z)}. \quad (2.172+2.178)$$

$$\Psi_{\perp_0, m_o l k \zeta}^{(T_n, z)} = \phi_{m_o l k \zeta}^{(x)} \Theta_{\zeta(-1)^l}^{(x)} - \zeta \sigma^{(T_n)} i^{+m} \phi_{m_o l k \zeta}^{(y)} \Theta_{\zeta(-1)^l}^{(y)}, \quad (2.232)$$

$$\Psi_{\perp_1, mlk}^{(T_n, z)} = \phi_{mlk\bar{\zeta}_0}^{(y)} \Theta_{(-1)^{m+1}}^{(x)} - \sigma^{(T_n)} (-1)^{l+m} i^{+m} \phi_{mlk\bar{\zeta}_1}^{(x)} \Theta_{(-1)^{m+1}}^{(y)}, \quad (2.233)$$

$$\Psi_{\perp_2, mlk}^{(T_n, z)} = \phi_{mlk\bar{\zeta}_1}^{(z)} \Theta_{(-1)^m}^{(x)} - \sigma^{(T_n)} (-1)^l i^{+m} \phi_{mlk\bar{\zeta}_0}^{(z)} \Theta_{(-1)^m}^{(y)}. \quad (2.234)$$

$$\mathcal{H} = \int d\mathbf{r} (\Psi_L)^\dagger \hat{U}_r \Psi_R, \quad (2.235)$$

$$= \int d\mathbf{r} \left\{ \sum_{q_L} \Phi_L^{(q_L)} \Theta_{\Omega_L}^{(q_L)} \right\}^\dagger \hat{U}_r \left\{ \sum_{q_R} \Phi_R^{(q_R)} \Theta_{\Omega_R}^{(q_R)} \right\}, \quad (2.236)$$

$$= \int d\mathbf{r} \sum_{q_L} \sum_{q_R} \tilde{\Phi}_L^{(q_L)} \hat{U}_r \Phi_R^{(q_R)} \Theta_{\Omega_L}^{(-q_L)} \Theta_{\Omega_R}^{(+q_R)}, \quad (2.237)$$

$$= \mathcal{H}_\parallel + \mathcal{H}_\perp. \quad (2.238)$$

$$\mathcal{H}_\parallel = \int d\mathbf{r} \sum_{q=x,y,z} \Phi_L^{(q)*} \hat{U}_r \Phi_R^{(q)} \Theta_{\Omega_L}^{(-q)} \Theta_{\Omega_R}^{(q)}, \quad (2.239)$$

$$= \int d\mathbf{r} \varphi_\parallel \Theta_{\Omega_L}^{(-z)} \Theta_{\Omega_R}^{(z)}. \quad (2.240)$$

$$\varphi_\parallel = \Phi_L^{(z)*} \hat{U}_r \Phi_R^{(z)} + (zx y) \Phi_L^{(x)*} \hat{U}_r \Phi_R^{(x)} + (yz x) \Phi_L^{(y)*} \hat{U}_r \Phi_R^{(y)}, \quad (2.241)$$

$$= \sum_{p_L, p_R} \left(\boldsymbol{\beta}_{p_L p_R}^{(z)} \hat{U}_{xyz} \phi_{zz}^{(p_L p_R)} + \boldsymbol{\beta}_{p_L p_R}^{(x)} \hat{U}_{zxy} \phi_{xx}^{(p_L p_R)} + \boldsymbol{\beta}_{p_L p_R}^{(y)} \hat{U}_{yzx} \phi_{yy}^{(p_L p_R)} \right). \quad (2.242)$$

$$\boldsymbol{\beta}_{p_L p_R}^{(z-\Delta)} = (\boldsymbol{\beta}_L^*)_{(p_L-\Delta)(z-\Delta)} (\boldsymbol{\beta}_R)_{(p_R-\Delta)(z-\Delta)}, \quad (2.243)$$

$$\phi_{(z-\Delta)(z-\Delta)}^{(p_L p_R)} = \tilde{\phi}_{L, (\zeta_L)_{(p_L-\Delta)(z-\Delta)}}^{(p_L)} \phi_{R, (\zeta_R)_{(p_R-\Delta)(z-\Delta)}}^{(p_R)}, \quad (2.244)$$

$$\zeta_{(z-\Delta)(z-\Delta)}^{(p_L p_R)} = (\zeta_L^*)_{(p_L-\Delta)(z-\Delta)} (\zeta_R)_{(p_R-\Delta)(z-\Delta)}. \quad (2.245)$$

$$\varphi_{\parallel,zz} = \boldsymbol{\beta}_{zz}^{(z)} \hat{U}_{xyz} \phi_{zz}^{(zz)} + \boldsymbol{\beta}_{zz}^{(x)} \hat{U}_{zxy} \phi_{xx}^{(zz)} + \boldsymbol{\beta}_{zz}^{(y)} \hat{U}_{yzx} \phi_{yy}^{(zz)}, \quad (2.246)$$

$$\varphi_{\parallel,xx} = \boldsymbol{\beta}_{xx}^{(z)} \hat{U}_{xyz} \phi_{zz}^{(xx)} + \boldsymbol{\beta}_{xx}^{(x)} \hat{U}_{zxy} \phi_{xx}^{(xx)} + \boldsymbol{\beta}_{xx}^{(y)} \hat{U}_{yzx} \phi_{yy}^{(xx)}, \quad (2.247)$$

$$\varphi_{\parallel,yy} = \boldsymbol{\beta}_{yy}^{(z)} \phi_{zz}^{(yy)} \hat{U}_{xyz} + \boldsymbol{\beta}_{yy}^{(x)} \hat{U}_{zxy} \phi_{xx}^{(yy)} + \boldsymbol{\beta}_{yy}^{(y)} \hat{U}_{yzx} \phi_{yy}^{(yy)} \quad (2.248)$$

$$(yxz)\varphi_{\parallel,zz} = \boldsymbol{\beta}_{zz}^{(z)} (\zeta_{zz}^{(zz)} \hat{U}_{yxz}) \phi_{zz}^{(zz)} + \left(\boldsymbol{\beta}_{zz}^{(y)} \zeta_{yy}^{(zz)} \hat{U}_{xzy} \right) \phi_{xx}^{(zz)} + \left(\boldsymbol{\beta}_{zz}^{(x)} \zeta_{xx}^{(zz)} \hat{U}_{zyx} \right) \phi_{yy}^{(zz)}, \quad (2.249)$$

$$(yxz)\varphi_{\parallel,xx} = \boldsymbol{\beta}_{xx}^{(z)} \hat{U}_{yxz} \phi_{zz}^{(xx)} + \boldsymbol{\beta}_{xx}^{(x)} \hat{U}_{zyx} \phi_{xx}^{(xx)} + \boldsymbol{\beta}_{xx}^{(y)} \hat{U}_{xzy} \phi_{yy}^{(xx)}, \quad (2.250)$$

$$(yxz)\varphi_{\parallel,yy} = \boldsymbol{\beta}_{yy}^{(z)} \hat{U}_{yxz} \phi_{zz}^{(yy)} + \boldsymbol{\beta}_{yy}^{(x)} \hat{U}_{zyx} \phi_{xx}^{(yy)} + \boldsymbol{\beta}_{yy}^{(y)} \hat{U}_{xzy} \phi_{yy}^{(yy)} \quad (2.251)$$

$$\left(\frac{(xyz) + (yxz)}{2} \right) \varphi_{\parallel,zz} = \boldsymbol{\beta}_{zz}^{(z)} \left(\frac{\hat{U}_{xyz} + \zeta_{zz}^{(zz)} \hat{U}_{yxz}}{2} \right) \phi_{zz}^{(zz)} + \left(\frac{\boldsymbol{\beta}_{zz}^{(x)} \hat{U}_{zxy} + \boldsymbol{\beta}_{zz}^{(y)} \zeta_{yy}^{(zz)} \hat{U}_{xzy}}{2} \right) \phi_{xx}^{(zz)} + \left(\frac{\boldsymbol{\beta}_{zz}^{(y)} \hat{U}_{yzx} + \boldsymbol{\beta}_{zz}^{(x)} \zeta_{xx}^{(zz)} \hat{U}_{zyx}}{2} \right) \phi_{yy}^{(zz)} \quad (2.252)$$

$$\mathcal{H}_{\perp} = \int d\mathbf{r} \sum_{q=x,y,z} \left[\left\{ \Phi_{\text{L}}^{(q)} \right\}^{\dagger} \hat{U}_{\mathbf{r}} \Phi_{\text{R}}^{(q+1)} \Theta_{\Omega_{\text{L}}}^{(-q)} \Theta_{\Omega_{\text{R}}}^{(q+1)} + \left\{ \Phi_{\text{L}}^{(q)} \right\}^{\dagger} \hat{U}_{\mathbf{r}} \Phi_{\text{R}}^{(q-1)} \Theta_{\Omega_{\text{L}}}^{(-q)} \Theta_{\Omega_{\text{R}}}^{(q-1)} \right], \quad (2.253)$$

$$= \int d\mathbf{r} (\varphi_{\perp,0} + (yxz)\varphi_{\perp,0}) \Theta_{\Omega_{\text{L}}}^{(-y)} \Theta_{\Omega_{\text{R}}}^{(x)}. \quad (2.254)$$

$$\varphi_{\perp,0} = \hat{U}_{\mathbf{r}} \Phi_{\text{L}}^{(y)*} \Phi_{\text{R}}^{(x)} + (zxy) \hat{U}_{\mathbf{r}} \Phi_{\text{L}}^{(z)*} \Phi_{\text{R}}^{(y)} + (yzx) \hat{U}_{\mathbf{r}} \Phi_{\text{L}}^{(x)*} \Phi_{\text{R}}^{(z)}, \quad (2.255)$$

$$\varphi_{\perp,1} = \hat{U}_{\mathbf{r}} \Phi_{\text{L}}^{(x)*} \Phi_{\text{R}}^{(y)} + (zxy) \hat{U}_{\mathbf{r}} \Phi_{\text{L}}^{(y)*} \Phi_{\text{R}}^{(z)} + (yzx) \hat{U}_{\mathbf{r}} \Phi_{\text{L}}^{(z)*} \Phi_{\text{R}}^{(x)}. \quad (2.256)$$

Chapter 3

Computational Description

3.1 Overall Program Flow

3.1.1 Overall Program Flow and Structure

This software, **Cantankerous Impurity** has been designed to be easily configurable to the end user. In essence, the user may specify one or more *subbases* as a set of (mlk) tuples along with a radial parameter η and an anisotropic parameter λ . Multiple subbases, with independent radial and anisotropic parameters for each subbases, may be collected together as a *superbasis*. Finally the user designates a radial character W for the whole superbasis, being either $W = 1$ (Slater) or $W = 2$ (Gaussian).

The potential model is very flexible as well, going as

$$\tilde{\mathcal{U}}_{\{r\}} := \mathfrak{E}^s \sum_i \left(\frac{r}{\mathfrak{r}} \right)^{\mathfrak{d}_i} \mathcal{U}_{i\{\Omega\}} e^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\mathfrak{r}}} e^{-\beta_{i\{\Omega\}} \left(\frac{r}{\mathfrak{r}} \right)^2}. \quad (\text{D.229})$$

For each term of the potential, the user specifies $\mathfrak{d} \in \mathbb{R}$ such that $\mathfrak{d} > -2$ as $\mathfrak{d} \leq 2$ leads to a divergent integral and make computation of the matrix elements impossible. The user also specifies, for each term of the potential model, the solid-angle functions $\mathcal{U}_{\{\Omega\}}$, $\tilde{\alpha}_{\{\Omega\}}$, and $\beta_{\{\Omega\}}$. To make sure that symmetry is respected by the user-defined

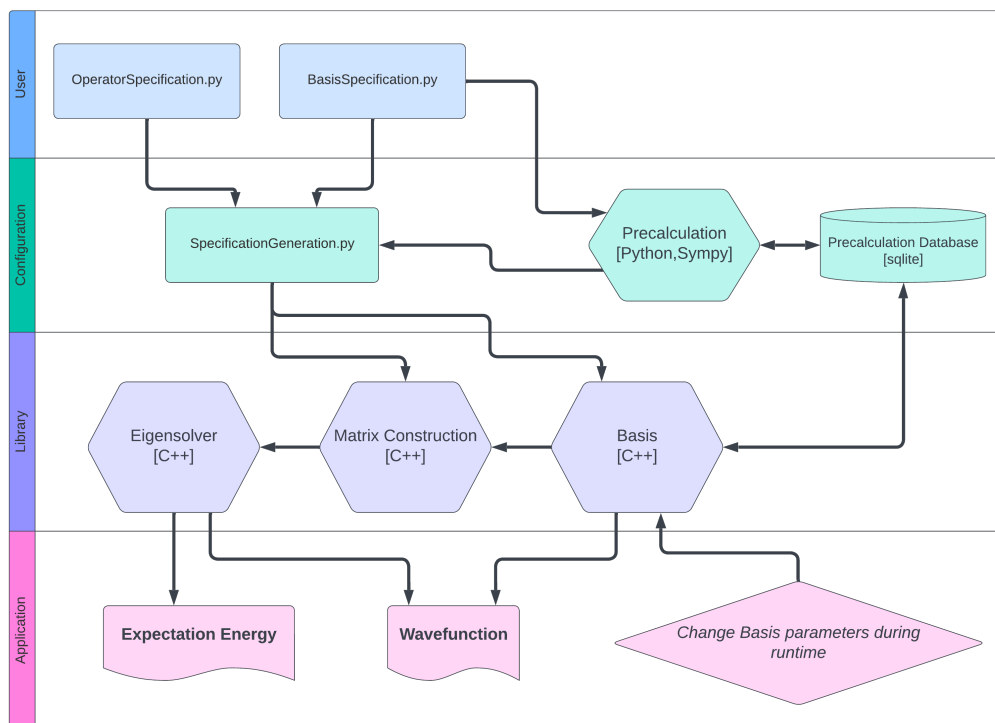


Figure 3.1: Overall Cantankerous Impurity structure.

potential model, we intend to make a set of easy to use function builder tools. At time of writing, these tools are still under development, but as more information becomes available, it will be reported in the [Supplementary Materials](#).

Since we are endeavoring to develop variance as a metric to evaluate our approximate eigenstates, the code must also be able to generate the terms of the potential squared, ideally without extra work on the part of the user. This also features in our generic potential form, which is easy to square.

In figure 3.1 we present a map of the flow and structure of Cantankerous Impurity.

3.2 Precalculation

The most unique feature of Cantankerous Impurity, as far as it is known to the author, is the use of the exact Laguerre polynomials in our basis functions without truncations. As such, we wish to devote some time to discussion of the computations use of

these polynomials entails.

For our purposes, we have defined the *precalculated coefficients* in Appendix D as a set of numbers that must be computed analytically to get accurate results, and are important parts of the integrands used to compute the matrix elements. These computations of analytical numbers are potentially time consume; as such, we have chosen to assemble these coefficients as needed when the program is compiled. The precalculated coefficients are then stored in a database and called into active memory at the start of the program run. Only those coefficients needed to represent the user-designated *sub*- and *superbasis* states are loaded into active memory. As this database persists from run to run, recompiling the program to change various physical or technical parameters does not necessitate adding to the database. This is only required if states with new discrete indices are added to the basis being considered.

We note many of these values we are computing must be stored as exact analytical values. For this reason, we have implemented much of this code within SymPy, storing our values as SymPy objects which represent exact quantities and may be manipulated according to the rules of conventional computer algebra implemented within SymPy. We will also find that the collection of values we are computing and storing have coefficients that don't make conventional rectangular storage easy. As such, much of these arrays of values are implemented as Python dictionaries, allowing easy association between a particular set of coefficients and the desired analytical SymPy object.

As derived in Appendix D, these coefficients go as

$$\mathfrak{A}_{\mathfrak{d}(su)_{01}w} = (-1)^{u_0+u_1+w} \Gamma \left[\frac{\mathfrak{h}_{\mathfrak{d}(s)_{01}}}{W} + w \right] \mathfrak{A}'_{(su)_{01}w}, \quad (\text{D.308})$$

$$\mathfrak{A}'_{(su)_{01}w} = \sum_{v=\text{Max}[0, \mathfrak{K}_0+u_1-w]}^{\text{Min}[\mathfrak{K}_0-u_0, \mathfrak{K}_0+\mathfrak{K}_1-w]} \mathfrak{A}''_{(su)_0v} \mathfrak{A}''_{(su)_1(\mathfrak{K}_0+\mathfrak{K}_1-v-w)}, \quad (\text{D.323}')$$

$$\mathfrak{A}''_{su\nu} = \frac{1}{\sqrt{k!(k+l)!}} \binom{\mathfrak{K}}{u} \binom{\mathfrak{K}-u}{\nu} (k+l)^\nu (1+u\mathfrak{I}_{sv}). \quad (\text{D.319}')$$

The primed equations above and below feature minor modifications to their forms as compared to the versions attested in Appendix D but as the changes are straightforward, we do not anticipate this will generate confusion. We show in Appendix D that

$$\mathfrak{I}_{sv} = \frac{\mathfrak{H}'_{sv}}{(k+l+2-\nu)(k+l+1-\nu)} - 1, \quad (\text{D.320})$$

$$\mathfrak{H}'_{sv} = \sum_{t=0}^{\text{Min}[v, 3+\delta_{s\bar{1}}]} v^t \mathfrak{G}_{st}, \quad (\text{D.321})$$

$$\mathfrak{G}_{st} = g_{kl, s(1+\delta_{s\bar{1}}-t)} \sqrt{\frac{(k+1+\delta_{s\bar{1}}-t)!(k+l+2-t)!}{k!(k+l)!}}. \quad (\text{D.322})$$

As mentioned in Appendix D, the formulae to find the values of $g_{kl, st}$ are cumbersome to write and derive. The formulae for these coefficients may be found in the [Supplementary Materials](#).

When the appropriate values of the coefficients g are used we may write

$$\mathfrak{I}_{\bar{1}v} = -(k+1+v)(k+2+v) + 2v - 1 + \delta_{W,1} \mathfrak{I}'_{\bar{1}v}, \quad (3.1)$$

$$\mathfrak{I}'_{\bar{1}v} = \frac{6v(k+2-v)}{\alpha+1-v} + \frac{12v(v-1)}{\alpha+2-v}, \quad (3.2)$$

$$\mathfrak{I}_{0v} = k+v + \frac{2v(k-\delta_{W,1}(v-1)+\delta_{W,2}v)}{\alpha+1-v} + \delta_{W,1} \frac{4(v-1)v}{\alpha+2-v}, \quad (3.3)$$

$$\mathfrak{I}_{1v} = k-x + \frac{2x(2x(k+\delta_{W,2}-x)+\delta_{W,1})}{\alpha+1-x} + \frac{4x(-k+x-\delta_{W,2})(x-1)}{\alpha+2-x}. \quad (3.4)$$

So let's consider what information is required to know from each state to know which precalculated coefficients to precalculate. To calculate $\mathfrak{J}_{sv}$ we need to specify a given set of $(W; kl)$. Once we know those three pieces of information, we know that the coefficients to calculate for $\mathfrak{J}_{sv}$ have $0 \leq \nu \leq k + 1 + \delta_{s\bar{1}}$ and $s \in \{-1, 0, +1\}$.

We introduce the following notational convention. In this section, *bolded* quantities represent sequences or arrays of objects. These objects themselves depend upon certain quantities we can use to denote to identify a particular entry of the array, which we will consider to be *array indices*. Any remaining quantities necessary to compute the values of a given array will be called *array parameters*. As array indices are conventionally denoted as subscripts, we will denote parameters as superscripts. Of course, the distinction between these two concepts is somewhat arbitrary, and depends upon the requirements to implement the structures in code. In the following discussion, any particular choice to identify as a value as a array index or array parameter should be considered provisional, and ultimately dependent upon the code's final production implementation. Finally, it is possible to imagine multidimensional arrays constructed as an array or arrays, or as a concatenation of arrays. In that case, we can imagine an array parameter in the subarray becoming an array index in the larger superarray.

When we define each bold symbol, we will use set notation to identify and give the limits of the array indices. For $\mathfrak{J}$ we give

$$\mathfrak{J}^{(W;kl)} := \{\mathfrak{J}_{sv} | s = -1, 0, 1 \text{ and } 0 \leq \nu \leq k + 1 + \delta_{s\bar{1}}\} \quad (3.5)$$

So for the basis specified by the user, the precalculation coefficient checker must determine if $\mathfrak{J}^{(W;kl)}$ has been calculated for all $(W; kl)$ present in the specification of the user's basis. We expect that, while it is possible that a user may wish to have a subbasis or superbasis that employs a mix of $W = 1$ and $W = 2$, we expect the likelihood of such an event to be low. Right now our program doesn't implement mixing

states of different W , and we require the user to specify if their overall superbasis uses $W = 1$ (Slater) or $W = 2$ (Gaussian) envelope functions exclusively. A positive integer $W \neq 1, 2$ is possible, but well beyond the scope of the current software.

As we move up the computation chain, our next goal will be to calculate $\mathfrak{A}''$. This coefficient, like before, depends on the $(W; kl)$ quantities from only one state. However, this coefficient will have two forms depending on whether the kinetic toggle $u = 0$ or $u = 1$.

To simplify this, we may consider the following

$$\mathfrak{A}_{su\nu}^{(j)} := \left(\frac{1}{\sqrt{k!}} \binom{k+j}{u} \binom{k+j-u}{v} \right) \frac{(k+l)^\nu}{\sqrt{(k+l)!}}. \quad (3.6)$$

Above, we grouped factors of the formula to point out places where defining and memoizing relevant subfunctions used to compute the coefficients would be very useful. For example we can consider the function

$$\gamma_{x,v} := \frac{(x)^\nu}{\sqrt{x!}}. \quad (3.7)$$

We note that we can use $\gamma_{k+l,v}$ when computing part of the above equation. We can expect integer $x = k+2l+2$ when $W = 1$ and half-integer $x = k+l+\frac{1}{2}$ when $W = 2$. It is clear that multiple particular combinations of $(W; kl)$ could correspond to the same value of $k+l$. Memoizing a function is a computer science term that describes implementing a function with the ability to record and recall some aspect of past computations, such as return values or earlier internal states. This can be quite convenient when optimizing or trying to speed up a computation, such as those of the precalculated coefficients. In this document we are most interested in the broader program structure and flow. While we will not often describe the fine details of specific computations, such as a specific definition and memoizing scheme for useful subfunctions,

we note that some notational and stylistic choices in the following sections are made to make clear opportunities to optimize computation with memoization.

Above, $\mathfrak{A}''^{(j)}$ will have $j = 0, 1, 2$. The value of j determines the limits of s , u , and v . When $j = 0$ it must be that $u = 0$ so $\mathfrak{K} = k$ and $s = 0$. Taking into account the possible values of $\mathcal{A}''$ given a particular value of j , we can compute the following arrays of values.

An important accessory set to define is the set of values (u, v) to run over as indices. We also wish to define a set that includes the possible tuples (s, u) for a given $u = 0, 1$. Let us define

$$\mathfrak{S}^{(u)} := \{s \in \mathbb{Z} | -u \leq s \leq +u\}, \quad (3.8)$$

$$\mathfrak{U}^{(\mathfrak{K})} := \{u \in \mathbb{Z} | 0 \leq u \leq \mathfrak{K}\}, \quad (3.9)$$

$$\mathfrak{V}^{(\mathfrak{K})} := \{(u, v) | u \in \mathfrak{U}^{(\mathfrak{K})} \text{ and } v \in \mathfrak{U}^{(\mathfrak{K}-u)}\}. \quad (3.10)$$

This provides a convenient short-hand we can refer to at points later in this documentation. Note that $\mathfrak{V}^{(\mathfrak{K})}$ is not a conventional rectangular array, as the limits of the trailing dimension v depends on the value of the leading dimension u . This will be true of many of the arrays generated for this section.

Our array of the $\mathfrak{A}''_{W;kl}$ values with indices s , u , v , and u may be define as

$$\mathfrak{A}''_{W;kl} := \left\{ \mathfrak{A}''_{suv}^{(u+\delta_{s\bar{1}})} | (u, v) \in \mathfrak{V}^{(k+u+\delta_{s\bar{1}})} \text{ and } u \in \{0, 1\} \text{ and } s \in \mathfrak{S}^{(u)} \right\}. \quad (3.11)$$

Since we expect every use of the code to include operators with both $u = 0$ and $u = 1$ both will be needed during each program run. As such, we treat it as an array index rather than an array parameter depending upon the user's operator specification.

Up to now, we've been dealing with precalculated coefficients that depend only upon parameters from a single state in the basis. As we move forward to the coefficients that depend upon pairs of states, we can exploit some of the symmetry property

of the coefficients. We note it may be shown that

$$\mathfrak{A}_{(suu)_0(suu)_1 w}^{(W;(lk)_0(lk)_1)} = \mathfrak{A}_{(suu)_1(suu)_0 w}^{(W;(lk)_1(lk)_0)} \quad (3.12)$$

This symmetric behavior allows us to roughly half the number of precalculated coefficients with an efficient implementation. Since two different sets of indices map to the same final value, it's necessary to have a consistent rule to be able map either of the two possibilities to the same final value without repeating the storage of that value unnecessarily. This is equivalent to assigning an ordering scheme to each possible set of (kl) indices. In general, we do not presume a "canonical" ordering to the possible values of (kl) a user may use to construct a basis. Instead, the database itself keeps a list specifying the order of (kl) as they are added. We expect that changes to the order of states within the basis made by the user within the configuration phase of deployment to be invisibly handled by the Precalculation module. More details may be found in the forthcoming Supplementary Materials.

In this document, we won't pay too much attention to the state order. It is sufficient to realize that, for given pair of states, our code implementation will ensure that the appropriate values are calculated and dispatched appropriately as efficiently as necessary. In any case, now that we are considering the two-state precalculated coefficients, it's worthwhile to define accessory sets of indices that go as

$$\mathfrak{W}^{(u)} = \left\{ (s, u) \mid s \in \mathfrak{S}^{(u)} \text{ and } u \in \mathfrak{U}^{(k+u+\delta_{s\bar{1}})} \right\}, \quad (3.13)$$

$$\mathfrak{Y}^{(u)} = \left\{ ((s, u)_{01}, w' + u_0 + u_1) \mid (s, u)_{01} \in \mathfrak{W}^{(u)_{01}} \text{ and } w' \in \mathfrak{U}_{\mathfrak{K}_0 + \mathfrak{K}_1} \right\}. \quad (3.14)$$

where we recall that, for a given basis state, $\mathfrak{K} = k + u + \delta_{s\bar{1}}$.

The final step necessary to calculate the general precalculated coefficients is

$$\mathfrak{A}_{(su)_{01} w}^{(W\mathfrak{D};(lks)_0(lks)_1)} = (-1)^{u_0+u_1+w} \Gamma \left[\frac{\mathfrak{h}_{\mathfrak{D}(s)_{01}}}{W} + w \right] \mathfrak{A}'_{(su)_{01} w} \quad (\text{D.308})$$

We note that $\mathfrak{h}_{\mathfrak{d}(s)_{01}} = l_0 + l_1 - \delta_{w1}(u_0 + u_1) + \mathfrak{d} + 3$. This is straightforward to calculate from $\mathfrak{A}'_{(su)_{01}w}$ with one caveat. Up till now, the precalculated coefficients haven't depended upon the potential model. However, with the introduction of $\mathfrak{d}$ into the formulae, we will need knowledge of the user-made operator specification. Thus, both the basis and the potential model need to be specified at compile time. We make no assumptions about $\mathfrak{d}$ other than $\mathfrak{d} \in \mathbb{R}$ and $\mathfrak{d} > -2$ as $\mathfrak{d} \leq 2$ makes the matrix element integrals divergent.

Finally, the summations found in the integrand of our matrix elements will simplify when certain conditions are met, something we expect will emerge in most use cases of the software. One of the difficulties in using the terms of the Laguerre polynomials emerges in these cases, as we end up needing to compute the summation of sequences of alternating sign. Even using the Kahan summation algorithm, we find these summations are very ill-conditioned when computed using floating point numbers. As such, rather than computing these summations in the matrix element integrand, we compute coefficients that may be used in the simplified sums. To wit, for each $\mathfrak{A}$ we must also compute the following values.

$$\mathfrak{A}_{(s)_{01}uw} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \delta_{u,u_0+u_1} (-1)^{u_0} \mathfrak{A}_{(su)_{01}w}, \quad (3.15)$$

$$\mathfrak{A}'_{(s)_{01}uw} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \delta_{u,u_0+u_1} \mathfrak{A}_{(su)_{01}w}, \quad (3.16)$$

$$\mathfrak{A}_{(s)_{01}w} = \mathfrak{A}_{(s0)_{01}w}, \quad (3.17)$$

$$\mathfrak{a}_{(s)_{01}u} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \delta_{u,u_0+u_1} (-1)^{u_0} \mathfrak{A}_{(su)_{01}w}, \quad (3.18)$$

$$\mathfrak{a}'_{(s)_{01}u} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \delta_{u,u_0+u_1} \mathfrak{A}_{(su)_{01}w}, \quad (3.19)$$

$$\mathfrak{a}_{(s)_{01}} = \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{(s0)_{01}w}. \quad (3.20)$$

Chapter 4

Benchmarking

4.1 Statically screened Coulomb potential

4.1.1 Exploring Single Eigenstates, Single-Valley

The first feature of our program we want to explore is a benchmarking against the single-valley Kohn-Luttinger and Faulkner results. Below we present plots of a single subbasis as we sweep the radial and anisotropic variational parameters of that subbasis. In the single state basis, our program generates results identical to what Kohn-Luttinger found for their ground states. However, as we increase the size of the basis the energy surface flattens. This is to be expected as the basis coefficients themselves are variational parameters that are set by the process of computing the solution to the generalized eigenvalue problem. In short, as the basis size increases, the number of variational parameters set by the eigensolver increases, while the number of free parameters per subbasis stays the same. This means the importance of the free variational parameters decreases.

Benchmarking of this software is on-going, especially as regards to the multi-valley states. In particular, there are still ambiguities in regards to the use of the Bloch functions within Cantankerous Impurity. Further benchmarking results can be found in

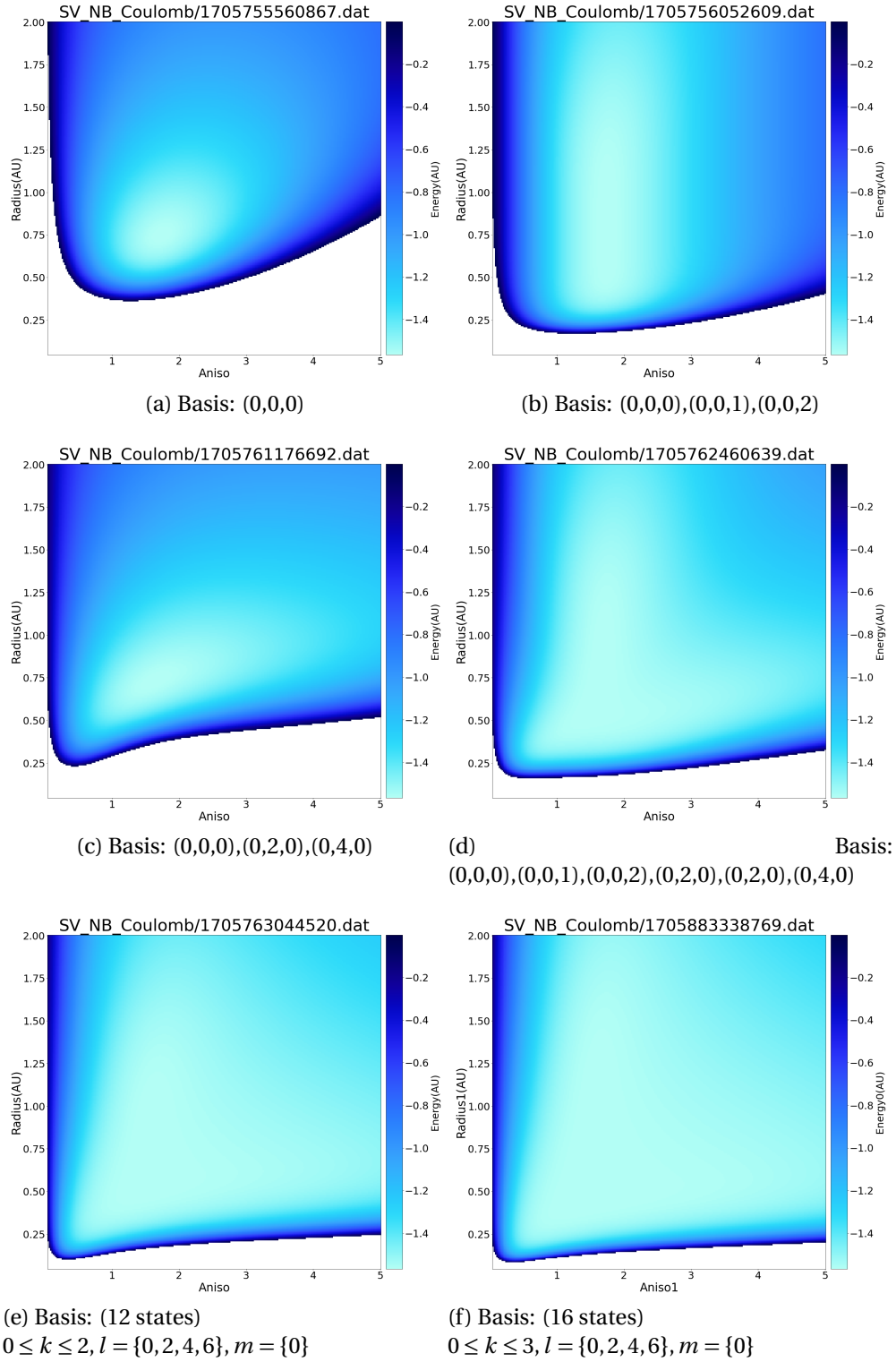


Figure 4.1: This figure shows the effect of basis size on the related variational problem. As the basis size increases, the energy surface becomes less sensitive to changes in the radial and anisotropic parameters within a certain neighborhood of the parameter space.

the [Supplementary Materials](#).

Chapter 5

Potential for future work

Appendix

A System units

For explicit numerical work, it is convenient to use a unitless formulation. In much the same way as is done in the standard treatment of the hydrogen atom problem, we introduce the unitful constants $\mathfrak{E}$ for energy and $\mathfrak{r}$ for length. We define these constants such that

$$\mathfrak{E} \equiv \frac{\hbar^2}{2m_{\perp}\mathfrak{r}^2} = \frac{e^2}{8\pi\epsilon\mathfrak{r}}. \quad (\text{A.1})$$

Using this relationship and solving for the length and energy directly we find

$$\mathfrak{r} = \frac{4\pi\epsilon\hbar^2}{m_{\perp}e^2}, \quad (\text{A.2})$$

$$\mathfrak{E} = \frac{m_{\perp}e^4}{2(4\pi\epsilon\hbar)^2}. \quad (\text{A.3})$$

Please note that above $m_{\perp}$ and ϵ are not fundamental physical constants. $m_{\perp}$ is the transverse effective mass of the bound silicon donor electron, not the full electron mass. As such, $m_{\perp} \approx .1905m_e$. Additionally, it seems as though there is some question as to the best value for the dielectric constant of silicon ϵ . If we take $\epsilon = \eta\epsilon_0$ with η as

a dimensionless parameter we have

$$\tau = .2778\eta \text{ nm}, \quad (\text{A.4})$$

$$\mathfrak{E} = \frac{2592}{\eta^2} \text{ meV}. \quad (\text{A.5})$$

For a commonly quoted value of the silicon dielectric $\epsilon = 11.4\epsilon_0$, we have explicitly

$$\tau = 3.17 \text{ nm}, \quad (\text{A.6})$$

$$\mathfrak{E} = 19.9 \text{ meV}. \quad (\text{A.7})$$

B Transformations

Our underlying scaled basis function was defined in Eq. (2.76) as

$$\phi_{klm;\eta\lambda;r}^{(Wq)} = F_{lk;\eta\lambda;r\chi_q}^{(W)} Y_{lm;\lambda;\chi_q\phi_q}. \quad (\text{B.8})$$

We will examine the transformation qualities of this function, and identify a construction with particular transformation properties that will systematize constructing the full envelope-valley wavefunctions.

$$F_{lk;\eta\lambda;r\chi_q}^{(W)} = \left(\frac{\vartheta_{\lambda;\chi_q}}{\eta} \right)^{\frac{3}{2}} F_{lk;\eta\lambda\chi_q;r}^{(W)} \quad \eta_{\lambda;\chi_q} = \frac{\eta}{\vartheta_{\lambda;\chi_q}} \quad (\text{2.77})$$

$$Y_{lm;\lambda;\chi_q\phi_q} = \sqrt{\frac{\lambda}{\vartheta_{\lambda;\chi_q}^3}} Y_{lm;\chi_\lambda^{(q)}\phi_q} \quad \chi_\lambda^{(q)} = \frac{\lambda\chi_q}{\vartheta_{\lambda;\chi_q}} \quad (\text{2.78})$$

We note that much like the T states, our basis states may be indexed by the crystallographic axes. To clarify, we expect the following relationships to hold for functions indexed by the crystallographic axes,

$$f^{(x)} = (y z x) f^{(z)}, \quad (\text{B.9})$$

$$f^{(y)} = (z x y) f^{(z)}, \quad (\text{B.10})$$

$$f^{(y)} = (y z x) f^{(x)}. \quad (\text{B.11})$$

We note that these three transformations $(x y z)$, $(y z x)$, and $(z x y)$ are part of the factored projection operator defined in (2.80). For ease of nomenclature, we'll call these the indexing transformations. We see of course that $\Phi^{(z)}$ can be indexed by z . However, we should also note that the combination $\Phi^{(x)} \pm \Phi^{(y)}$ can also be indexed by z in the sense established above. We will explore this further below.

After the introduction of the anisotropic scaling, the "radial" part $F^{(q)}$ of the basis

state is no longer isotropic, and gains a dependence upon a valley-indexed coordinate χ_q . As such, it will be expected to transform under the action some of the symmetry transformations. This is true for the indexing transformations. However, χ_q emerges in the function F only to the second order. As such, F is invariant under the action of double inversions. Finally, as F depends only upon one valley-indexed coordinates, it is invariant under acyclic swaps. To present this argument clearly and schematically, we will list the effects of the double inversion and acyclic swap below for F ,

$$F^{(z)} = (\bar{x} \bar{y} z) F^{(z)} = (\bar{x} y \bar{z}) F^{(z)} = (x \bar{y} \bar{z}) F^{(z)} = (y x z) F^{(z)}. \quad (\text{B.12})$$

We see that the symmetry contributions of the "radial" part of the basis state are still simple and mostly invariant, even with the introduction of the anisotropic scaling. Conversely the spherical harmonics Y , as might also have been expected, have the most significant impacts upon symmetry. The spherical harmonics are straightforwardly seperable in longitudinal and azimuthal coordinates as

$$Y_{lm;\chi\phi} \propto P_{lm;\chi} e^{im\phi} \quad (\text{B.13})$$

where the P functions are identified as the associated Legendre polynomials in $\chi = z/r = \cos(\theta)$. When introducing the variational anisotropy, we redefined the "spherical harmonics" as

$$Y_{lm;\lambda;\chi_q\phi_q} = \sqrt{\frac{\lambda}{\vartheta_{\lambda;\chi_q}^3}} Y_{lm;\chi_\lambda^{(q)}\phi_q} \quad \chi_\lambda^{(q)} = \frac{\lambda\chi_q}{\vartheta_{\lambda;\chi_q}} \quad (\text{B.14})$$

We note that this form can still be "seperated" by coordinates, focusing on the valley-indexed coordinates χ_q and ϕ_q . We consider

$$Y_{lm;\lambda;\chi_q\phi_q} = P_{lm;\lambda;\chi_q} Q_{m;\phi_q}. \quad (\text{B.15})$$

where the "longitudinal" part of the stretched spherical harmonics can be identified as

$$P_{lm;\lambda;\chi_q} = \sqrt{\frac{\lambda}{\vartheta_{\lambda;\chi_q}^3}} P_{lm;\chi_\lambda^{(q)}}. \quad (\text{B.16})$$

The "longitudinal" function $P^{(q)}$ defined above in Eq. (B.16) can be valley-indexed much as the "radial" function F was indexed above. Further, we list the effects of the double inversion and acyclic swap below,

$$P^{(z)} = (\bar{x} \bar{y} z) P^{(z)}, \quad (\text{B.17})$$

$$(-1)^{l+m} P^{(z)} = (\bar{x} y \bar{z}) P^{(z)} = (x \bar{y} \bar{z}) P^{(z)}, \quad (\text{B.18})$$

$$P^{(z)} = (y x z) P^{(z)}. \quad (\text{B.19})$$

From this set of equations, we can see that the behavior of P is very similiar to that of F , with both being straightforwardly described by the transformation equations. Another way to state this is to say that F and P are both eigenstates of the transformations in the set $\mathcal{S} = \{(\bar{x} \bar{y} z), (\bar{x} y \bar{z}), (x \bar{y} \bar{z}), (y x z)\}$.

The standard definition of the azimuthal part of the spherical harmonics would have $Q_{m;\phi_q} = e^{im\phi_q}$ where m is an integer. Unfortunately, it can be shown that the azimuthal function $e^{im\phi}$ is not an eigenstate of the transformations in $\mathcal{S}$. To wit,

$$(-1)^m e^{im\phi} = (\bar{x} \bar{y} z) e^{im\phi}, \quad (\text{B.20})$$

$$(-1)^m e^{-im\phi} = (\bar{x} y \bar{z}) e^{im\phi}, \quad (\text{B.21})$$

$$e^{-im\phi} = (x \bar{y} \bar{z}) e^{im\phi}, \quad (\text{B.22})$$

$$i^m e^{-im\phi} = (y x z) e^{im\phi}. \quad (\text{B.23})$$

While $e^{im\phi}$ is an eigenstate of $(\bar{x} \bar{y} z)$, for the other transformations we have $e^{im\phi} \rightarrow e^{-im\phi}$ and so it cannot be an eigenstate of the remaining transformations in $\mathcal{S}$.

To somewhat resolve this, we introduce the following useful construction to find

the highly symmetric states we seek,

$$Q_{m\zeta;\phi_q} \equiv \delta_{m0}(1 - \delta_{\zeta\bar{1}}) + (1 - \delta_{m0}) \left(\frac{e^{im\phi_q} + \zeta e^{-im\phi_q}}{\sqrt{2}} \right), \quad (\text{B.24})$$

where m is a nonnegative integer and $|\zeta| = 1$. Applying the same set of transformations yields the following relationships

$$(-1)^m Q_\zeta^{(z)} = (\bar{x} \bar{y} z) Q_\zeta^{(z)}, \quad (\text{B.25})$$

$$(-1)^m \zeta Q_{\zeta_*}^{(z)} = (\bar{x} y \bar{z}) Q_\zeta^{(z)}, \quad (\text{B.26})$$

$$\zeta Q_{\zeta_*}^{(z)} = (x \bar{y} \bar{z}) Q_\zeta^{(z)}, \quad (\text{B.27})$$

$$\zeta i^{-m} Q_{(-1)^m \zeta_*}^{(z)} = (y x z) Q_\zeta^{(z)}. \quad (\text{B.28})$$

For real ζ and even m this form of Q works well. In particular, we can assert $\sigma = \pm 1$ when $\zeta = \sigma i^m$. We can also identify a double inversion eigenvalue $\omega = \zeta = \sigma i^m$. For odd m , however, we find that the Q functions are not eigenstates under the action of most of the operators of $\mathcal{S}$. Judicious choice of ζ can ameliorate this effect for some of the transformations, not all. For example, asserting that ζ is real allows us to assert that Q is an eigenfunction of $(x \bar{y} \bar{z})$ and $(\bar{x} y \bar{z})$. For Q to be an eigenstate of $(y x z)$, if m is odd then ζ must be purely imaginary. Thus there seems to be no choice of ζ that ensures Q is an eigenstate of all transformations in $\mathcal{S}$, so long as $Q^{(z)} = Q_{m\zeta;\phi_z}$. However, as mentioned above, we must consider combinations of the form $\Phi^{(x)} \pm \Phi^{(y)}$ as candidates.

For this composite form, we would replicate the analysis above. This time we'll examine envelope functions indexed by x and y instead of z directly. The "radial" part $F^{(q)}$ will go as

$$F^{(q)} = (\bar{x} \bar{y} z) F^{(q)} = (\bar{x} y \bar{z}) F^{(q)} = (x \bar{y} \bar{z}) F^{(q)} \quad (\text{B.29})$$

for any choice of $q = x, y, z$. However, for the acyclic swap we find

$$F^{(y)} = (y x z) F^{(x)}, \quad (\text{B.30})$$

$$F^{(x)} = (y x z) F^{(y)}, \quad (\text{B.31})$$

and we see the effect of the acyclic swap is to swap the valley axis indices $y \leftrightarrow x$.

Similiarily, we can examine the "longitudinal" part $P^{(q)}$ and find the following relationships. For the double inversion operators of $\mathcal{S}$ acting on $P^{(x)}$

$$P^{(x)} = (x \bar{y} \bar{z}) P^{(x)}, \quad (\text{B.32})$$

$$(-1)^{l+m} P^{(x)} = (\bar{x} \bar{y} \bar{z}) P^{(x)} = (\bar{x} y z) P^{(x)}, \quad (\text{B.33})$$

and acting on $P^{(y)}$

$$P^{(y)} = (\bar{x} y \bar{z}) P^{(y)}, \quad (\text{B.34})$$

$$(-1)^{l+m} P^{(y)} = (\bar{x} \bar{y} z) P^{(y)} = (x \bar{y} \bar{z}) P^{(y)}. \quad (\text{B.35})$$

The effect of the double inversion operators of $\mathcal{S}$ upon P can be summarized as such; if the double inversion operator includes an inversion of the coordinate q valley-indexed by $P^{(q)}$, that operator acts to multiply $P^{(q)}$ by the factor $(-1)^{m+l}$.

The effects of the acyclic operator on P are the same as on the "radial" F , swapping the valley axis index of the function.

$$P^{(y)} = (y x z) P^{(x)}, \quad (\text{B.36})$$

$$P^{(x)} = (y x z) P^{(y)}. \quad (\text{B.37})$$

Moving onto the "azimuthal" part, we find the following relationships. For the double inversion, we group the equations by effect. Important to understanding the

effects is the idea of cyclic order. For our purposes, in cyclic order x preceeds y which preceeds z which preceeds x to close the cycle. When the double inversion involves the valley index of operand and the coordinate preceeding the valley index in cyclic order, we see the following effect.

$$\zeta Q_{\zeta_*}^{(y)} = (\bar{x} \bar{y} z) Q_{\zeta}^{(y)}, \quad (\text{B.38})$$

$$\zeta Q_{\zeta_*}^{(x)} = (\bar{x} y \bar{z}) Q_{\zeta}^{(x)}. \quad (\text{B.39})$$

The following effect holds for operators inverting the valley index and the coordinate following the valley index in cyclic order.

$$(-1)^m \zeta Q_{\zeta_*}^{(y)} = (x \bar{y} \bar{z}) Q_{\zeta}^{(y)}, \quad (\text{B.40})$$

$$(-1)^m \zeta Q_{\zeta_*}^{(x)} = (\bar{x} \bar{y} z) Q_{\zeta}^{(x)}. \quad (\text{B.41})$$

Finally, when the double inversion operator doesn't include the valley indexed by the operand, we see

$$(-1)^m Q_{\zeta}^{(y)} = (\bar{x} y \bar{z}) Q_{\zeta}^{(y)}, \quad (\text{B.42})$$

$$(-1)^m Q_{\zeta}^{(x)} = (x \bar{y} \bar{z}) Q_{\zeta}^{(x)}. \quad (\text{B.43})$$

We note that some of the double inversion operator transformations involve complex conjugation of ζ .

Considering the effect of the acyclic swap we find

$$\zeta i^{-m} Q_{(-1)^m \zeta_*}^{(x)} = (y x z) Q_{\zeta}^{(y)}, \quad (\text{B.44})$$

$$\zeta i^{-m} Q_{(-1)^m \zeta_*}^{(y)} = (y x z) Q_{\zeta}^{(x)}. \quad (\text{B.45})$$

As such, we can construct an eigenstate of the acyclic swap by combining $Q_{\zeta}^{(y)}$ and

$Q_{(-1)^m \zeta_*}^{(x)}$. This is consistent with our earlier observation that functions valley indexed by x and y can be combined in a way to generate a function that can be described as being indexed by z rather than x and y . However, to reduce computational and numerical complexity, we do not introduce a symmetrization for each part, F , P , and Q , of the basis function. We have chosen to symmetrize the full naive variational basis function. In short, we are defining our envelope basis functions as

$$\phi_\zeta^{(q)} = F^{(q)} P^{(q)} Q_\zeta^{(q)}. \quad (\text{B.46})$$

We can examine the effects of

$$(-1)^m \phi_\zeta^{(z)} = (\bar{x} \bar{y} z) \phi_\zeta^{(z)}, \quad (\text{B.47})$$

$$\zeta (-1)^l \phi_{\zeta_*}^{(z)} = (\bar{x} y \bar{z}) \phi_\zeta^{(z)}, \quad (\text{B.48})$$

$$\zeta (-1)^{l+m} \phi_{\zeta_*}^{(z)} = (x \bar{y} \bar{z}) \phi_\zeta^{(z)}, \quad (\text{B.49})$$

$$\zeta i^{-m} \phi_{(-1)^m \zeta_*}^{(z)} = (y x z) \phi_\zeta^{(z)}. \quad (\text{B.50})$$

$$\zeta (-1)^{l+m} \phi_{\zeta_*}^{(y)} = (\bar{x} \bar{y} z) \phi_\zeta^{(y)}, \quad (\text{B.51})$$

$$(-1)^m \phi_\zeta^{(y)} = (\bar{x} y \bar{z}) \phi_\zeta^{(y)}, \quad (\text{B.52})$$

$$\zeta (-1)^l \phi_{\zeta_*}^{(y)} = (x \bar{y} \bar{z}) \phi_\zeta^{(y)}, \quad (\text{B.53})$$

$$\zeta i^{-m} \phi_{(-1)^m \zeta_*}^{(x)} = (y x z) \phi_\zeta^{(y)}. \quad (\text{B.54})$$

$$\zeta(-1)^l \phi_{\zeta_*}^{(x)} = (\bar{x} \bar{y} z) \phi_{\zeta}^{(x)}, \quad (\text{B.55})$$

$$\zeta(-1)^{l+m} \phi_{\zeta_*}^{(x)} = (\bar{x} y \bar{z}) \phi_{\zeta}^{(x)}, \quad (\text{B.56})$$

$$(-1)^m \phi_{\zeta}^{(x)} = (x \bar{y} \bar{z}) \phi_{\zeta}^{(x)}, \quad (\text{B.57})$$

$$\zeta i^{-m} \phi_{(-1)^m \zeta_*}^{(y)} = (y x z) \phi_{\zeta}^{(x)}. \quad (\text{B.58})$$

C Calculation of matrix elements I: Setup

We will start with a word on notation within this and the following appendix. To start with, it is important to note which numbers are real, and may be represented by a real number data storage format of suitable precision, and which numbers are complex. Complex numbers do not necessarily need to be represented by a general complex number data storage, especially if symmetry properties constrain the possible phase arguments of the complex number. To that end, to make such distinctions more transparent, any quantity which may take on complex values will be denoted by a tilde, for example $\tilde{\varphi}$. As much as possible, we will endeavor to represent purely imaginary numbers as real numbers multiplied by the i , a structure which we can handle programmatically rather than employing unnecessary complex data types and complex algorithms.

Computation of our matrix elements will require integration both analytic and numerical. Because numerical integration may require many evaluations of the integrand, it is important to be clear which quantities may be considered as *direct integration variables* and *indirect integration variables* which must be recalculated for each integrand evaluation, as well as quantities which are *constant over integration* and will only need to be evaluated at the beginning or end of the integration computation. For this analysis, we are doing real space integrals over an approximately spherical system, so our direct integration variables will be real space spherical coordinates (r, χ, ϕ) . r and ϕ are defined in the conventional way, while $\chi = \cos(\theta) = z/r$. Indirect integration variables are quantities which are functions of the direct integration variables and so must be reevaluated with each iteration of the numerical integration algorithm. Indirect integration variables are denoted with Latin script characters in a calligraphic font, or with Greek letters in a conventional font. We denote which direct integration variables affect a given indirect integration variable as subscripts on the indirect variable written within curly brackets. Quantities which are constant

over integration are denoted by letters in a fraktur font. This convention does not extend to the reserved symbols $k, l, m, \xi, \eta, \lambda, q, W$ and κ . These symbols retain their meaning as basis indices and parameters, introduced in the previous sections, and as such would be constant over integration. As well, the symbol δ is reserved for the Kronecker delta symbol or the Dirac delta distribution. We hope that, despite not being in a fraktur font, these symbols' meanings in context make clear that these quantities are fixed over integration.

In this Appendix, we will prepare to calculate the form of matrix elements possible within our software. We begin enumerating the forms of our matrix elements will go as follow. First, the overlap matrix goes as

$$(\mathbb{S})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \Phi_{i\mathbf{r}}^{(p)*} \Phi_{j\mathbf{r}}^{(q)}. \quad (2.48)$$

The Hamiltonian matrix elements $\mathbb{H} = \mathbb{T} + \mathbb{U}$ go as

$$(\mathbb{T})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \Phi_{i\mathbf{r}}^{(p)*} \hat{T}^{(q)} \{ \Phi_{j\mathbf{r}}^{(q)} \}, \quad (2.46)$$

$$(\mathbb{U})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} U_{\mathbf{r}} \{ \Phi_{i\mathbf{r}}^{(p)} u_{\mathbf{k}_p \mathbf{r}} \}^* \Phi_{j\mathbf{r}}^{(q)*} u_{\mathbf{k}_q \mathbf{r}}. \quad (2.47)$$

Finally the Hamiltonian-squared matrix elements $\mathbb{H}\mathbb{H} = \mathbb{T}\mathbb{T} + \mathbb{U}\mathbb{U} + \mathbb{T}\mathbb{U} + \mathbb{U}\mathbb{T}$ go as

$$(\mathbb{T}\mathbb{T})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q e^{i(\mathbf{k}_q - \mathbf{k}_p) \cdot \mathbf{r}} \hat{T}^{(p)} \{ \Phi_{i\mathbf{r}}^{(p)} \}^* \hat{T}^{(q)} \{ \Phi_{j\mathbf{r}}^{(q)} \}, \quad (2.49)$$

$$(\mathbb{U}\mathbb{U})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q \{ e^{i\mathbf{k}_p \cdot \mathbf{r}} u_{\mathbf{k}_p \mathbf{r}} \}^* e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q \mathbf{r}} U_{\mathbf{r}}^2 \{ \Phi_{i\mathbf{r}}^{(p)} u_{\mathbf{k}_p \mathbf{r}} \}^* \Phi_{j\mathbf{r}}^{(q)*} u_{\mathbf{k}_q \mathbf{r}}, \quad (2.50)$$

$$(\mathbb{U}\mathbb{T})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q \{ e^{i\mathbf{k}_p \cdot \mathbf{r}} u_{\mathbf{k}_p \mathbf{r}} \}^* e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q \mathbf{r}} U_{\mathbf{r}} \Phi_{i\mathbf{r}}^{(p)*} \hat{T}^{(q)} \{ \Phi_{j\mathbf{r}}^{(q)} \}, \quad (2.51)$$

$$(\mathbb{T}\mathbb{U})_{ij} \equiv \int d\mathbf{r} \sum_{pq} \alpha_p^* \alpha_q \{ e^{i\mathbf{k}_p \cdot \mathbf{r}} u_{\mathbf{k}_p \mathbf{r}} \}^* e^{i\mathbf{k}_q \cdot \mathbf{r}} u_{\mathbf{k}_q \mathbf{r}} U_{\mathbf{r}} \hat{T}^{(p)} \{ \Phi_{i\mathbf{r}}^{(p)} \}^* \Phi_{j\mathbf{r}}^{(q)*}. \quad (2.52)$$

$$\Psi_{\parallel 0, m_e l k}^{(T_n, z)} = \psi_{\parallel 0, m_e l k \sigma^{(T_n)}}^{(T_n, z)}, \quad (C.59)$$

$$\Psi_{\parallel 1, m l k}^{(T_n, z)} = \psi_{\parallel 1, m l k \sigma^{(T_n)}}^{(T_n, z)}, \quad (C.60)$$

$$\Psi_{\perp 0, m_o l k \zeta}^{(T_n, z)} = \phi_{m_o l k \zeta}^{(T_n, x)} \eta_{\zeta(-1)^l}^{(x)} - \zeta \sigma^{(T_n)} i^{+m} \phi_{m_o l k \bar{\zeta}}^{(T_n, y)} \eta_{\zeta(-1)^l}^{(y)}, \quad (C.61)$$

$$\Psi_{\perp 1, m l k}^{(T_n, z)} = \phi_{m l k(-1)^{l+1}}^{(T_n, y)} \eta_{(-1)^{m+1}}^{(x)} + \Omega_{m l \sigma^{(T_n)}}^{(\kappa)} \phi_{m l k(-1)^{l+m+1}}^{(T_n, x)} \eta_{(-1)^{m+1}}^{(y)}, \quad (C.62)$$

$$\Psi_{\perp 2, m l k}^{(T_n, z)} = \phi_{m l k(-1)^{l+m+1}}^{(T_n, z)} \eta_{(-1)^m}^{(x)} + (-1)^m \Omega_{m l \sigma^{(T_n)}}^{(\kappa)} \phi_{m l k(-1)^{l+1}}^{(T_n, z)} \eta_{(-1)^m}^{(y)}. \quad (C.63)$$

$$\Psi_{\varrho, m_e l k}^{(A_n)} = \frac{1}{3} \psi_{\varrho, m_e l k \sigma^{(A_n)}}^{(A_n, z)} + \frac{1}{3} \psi_{\varrho, m_e l k \sigma^{(A_n)}}^{(A_n, x)} + \frac{1}{3} \psi_{\varrho, m_e l k \sigma^{(A_n)}}^{(A_n, y)}, \quad (C.64)$$

$$\Psi_{\varrho, m_e l k \sigma}^{(E, 0)} = \frac{1}{3} \psi_{\varrho, m_e l k \sigma}^{(E, z)} + \frac{\tau}{3} \psi_{\varrho, m_e l k \sigma}^{(E, x)} + \frac{\tau^2}{3} \psi_{\varrho, m_e l k \sigma}^{(E, y)}, \quad (C.65)$$

$$\Psi_{\varrho, m_e l k \sigma}^{(E, 1)} = \frac{1}{3} \psi_{\varrho, m_e l k \sigma}^{(E, z)} + \frac{\tau^2}{3} \psi_{\varrho, m_e l k \sigma}^{(E, x)} + \frac{\tau}{3} \psi_{\varrho, m_e l k \sigma}^{(E, y)}. \quad (C.66)$$

We also have the possibility of applying an effective kinetic operator to the envelopes of either of the states forming our integrand, and as such we define the following notational nicety.

$$\phi_{k l m \zeta; \eta \lambda \{ \mathbf{r} \}}^{(Wq)(u)} := \left\{ (1-u) + u \hat{T}^{(q)} \right\} \phi_{k l m \zeta; \eta \lambda \{ \mathbf{r} \}}^{(Wq)} \quad (C.67)$$

Above $u = \pm 1$ is a "switch" allowing us to selectively turn on or off the effect of the effective valley kinetic operator in the formulae below. The form of the above expression may be written as

$$\phi_{k l m \zeta; \eta \lambda \{ \mathbf{r} \}}^{(Wq)(u)} = \sum_{s=-u}^{+u} \mathfrak{E}^u \left(\frac{\mathbf{r}}{\eta} \right)^{2u} \mathfrak{H}_{m l; \lambda, s}^{(u)} (F'_s)^{(Wu)}_{l k; \eta \lambda \{ r \chi_q \}} P_{(l+2s)m; \lambda \{ \chi_q \}} Q_{m \zeta \{ \phi_q \}}, \quad (C.68)$$

where the "radial" part of the expression can be given as

$$(F'_s)^{(Wu)}_{lk;\eta\lambda\{r\chi_q\}} := \left(\frac{\eta}{\vartheta}\right)^{-\frac{3}{2}} \sqrt{W} \rho^{\frac{l}{W} - u\delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \sum_{t=-2u}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} L_{(k+t)\{\rho\}}^{(l+u-\delta_{s\bar{1}})}, \quad (C.69)$$

$$\vartheta := \sqrt{1 + (\lambda^2 - 1)\chi_q^2}, \quad (C.70)$$

$$\rho := \left(r \frac{\vartheta}{\eta}\right)^W, \quad (C.71)$$

$$\mathfrak{l} = \frac{2l + 3 - W}{W}. \quad (C.72)$$

We have introduced the following coefficients that go as

$$\mathfrak{H}_{ml;\lambda,s}^{(u)} = \delta_{s0} - u(1 - \gamma\lambda^2)\mathfrak{Y}_{ml;s}, \quad (C.73)$$

$$h_{kl,st}^{(u)} = (1 - u)\delta_{s0}\delta_{t0} + u g_{kl,st}. \quad (C.74)$$

More information on the derivation of the above expression, as well as the specific values of the $\mathfrak{Y}$ and g constants introduced above are cumbersome to reproduce here. Their full definition, formulae, and derivation can be found in the [Supplementary Materials](#).

We described the possible symmetrized wavefunctions we wish to consider in section 2.13. The most general form of wavefunction we found may be expressed as

$$\tilde{\Psi}_{\varrho,mlk\xi}^{(\kappa,i)} = \alpha_z^{(\kappa,i)} \tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,z)} + \alpha_x^{(\kappa,i)} \tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,x)} + \alpha_y^{(\kappa,i)} \tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,y)} \quad (C.75)$$

The above valley-indexed single-axis wavefunctions $\tilde{\psi}$ are linked by the cyclic indexing transformations defined in (2.89), (2.90), and (2.91).

However, the full wavefunction is not used in all matrix elements. Overlap and purely kinetic elements do not include the periodic part of the Bloch functions. Taking all of the above into account our overlap, Hamiltonian, and Hamiltonian-squared

matrix elements are computed with the following formula

$$(\mathcal{M})_{01} = \sum_{\alpha_{q_0}^{(\kappa_0, i_0)}} \sum_{\alpha_{q_1}^{(\kappa_1, i_1)}} \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^\infty r^2 dr \left(\phi_{\varrho, mlk\zeta; \eta\lambda\{\mathbf{r}\}}^{(Wq)(u)} \Theta^{(q)} \right)_0^* \tilde{\mathcal{U}}_{\{\mathbf{r}\}} \left(\phi_{\varrho, mlk\zeta; \eta\lambda\{\mathbf{r}\}}^{(Wq)(u)} \Theta^{(q)} \right)_1, \quad (\text{C.76})$$

$$= \sum_{\alpha_{q_0}^{(\kappa_0, i_0)}} \sum_{\alpha_{q_1}^{(\kappa_1, i_1)}} \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^\infty r^2 dr \left[\phi_{\varrho, mlk\zeta; \eta\lambda\{\mathbf{r}\}}^{(Wq)(u)} \Theta^{(q)} \right]_{01} \left[\Theta^{(q)} \right]_{01} \tilde{\mathcal{U}}_{\{\mathbf{r}\}}. \quad (\text{C.77})$$

where we may define a *generalized axial Bloch function* that goes as

$$\Theta^{(q)} = e^{i\mathbf{k}_q \cdot \mathbf{r}} \left[(1 - \mathfrak{s}) + \mathfrak{s} \eta_\xi^{(q)} \right] \quad (\text{C.78})$$

Much like u , we have $\mathfrak{s} = 0, 1, 2$ as a "switch" that communicates information about the potential operator being used in the matrix elements. 0 is for the overlap $\mathbb{S}$ and purely kinetic elements $\mathbb{K}$ and $\mathbb{KK}$. $\mathfrak{s} = 1$ is for the $\mathbb{U}$, $\mathbb{TU}$, and the $\mathbb{UT}$ elements. Finally 2 denotes the $\mathbb{UU}$ element.

$$\Psi_{\parallel 0, m_e l k}^{(T_n, z)} = \psi_{\parallel 0, m_e l k \sigma^{(T_n)}}^{(T_n, z)} \quad (\text{C.79})$$

$$\Psi_{\parallel 1, mlk}^{(T_n, z)} = \psi_{\parallel 1, mlk \sigma^{(T_n)}}^{(T_n, z)} \quad (\text{C.80})$$

$$\Psi_{\perp 0, m_o l k \zeta}^{(T_n, z)} = \phi_{m_o l k \zeta}^{(T_n, x)} \eta_{\zeta(-1)^l}^{(x)} - \zeta \sigma^{(T_n)} i^{+m} \phi_{m_o l k \bar{\zeta}}^{(T_n, y)} \eta_{\zeta(-1)^l}^{(y)}, \quad (\text{C.81})$$

$$\Psi_{\perp 1, mlk}^{(T_n, z)} = \phi_{mlk(-1)^{l+1}}^{(T_n, y)} \eta_{(-1)^{m+1}}^{(x)} + \Omega_{ml\sigma^{(T_n)}}^{(\kappa)} \phi_{mlk(-1)^{l+m+1}}^{(T_n, x)} \eta_{(-1)^{m+1}}^{(y)}, \quad (\text{C.82})$$

$$\Psi_{\perp 2, mlk}^{(T_n, z)} = \phi_{mlk(-1)^{l+m+1}}^{(T_n, z)} \eta_{(-1)^m}^{(x)} + (-1)^m \Omega_{ml\sigma^{(T_n)}}^{(\kappa)} \phi_{mlk(-1)^{l+1}}^{(T_n, z)} \eta_{(-1)^m}^{(y)}. \quad (\text{C.83})$$

$$\psi_{\varrho, m_e l k}^{(A_n)} = \frac{1}{3} \psi_{\varrho, m_e l k \sigma^{(A_n)}}^{(A_n, z)} + \frac{1}{3} \psi_{\varrho, m_e l k \sigma^{(A_n)}}^{(A_n, x)} + \frac{1}{3} \psi_{\varrho, m_e l k \sigma^{(A_n)}}^{(A_n, y)}, \quad (\text{C.84})$$

$$\psi_{\varrho, m_e l k \sigma}^{(E, 0)} = \frac{1}{3} \psi_{\varrho, m_e l k \sigma}^{(E, z)} + \frac{\tau}{3} \psi_{\varrho, m_e l k \sigma}^{(E, x)} + \frac{\tau^2}{3} \psi_{\varrho, m_e l k \sigma}^{(E, y)}, \quad (\text{C.85})$$

$$\psi_{\varrho, m_e l k \sigma}^{(E, 1)} = \frac{1}{3} \psi_{\varrho, m_e l k \sigma}^{(E, z)} + \frac{\tau^2}{3} \psi_{\varrho, m_e l k \sigma}^{(E, x)} + \frac{\tau}{3} \psi_{\varrho, m_e l k \sigma}^{(E, y)}. \quad (\text{C.86})$$

Now, as we build up our multi-valley wavefunctions we consider what we will describe here as the *longitudinal single-axis basis wavefunction*. We found that, for $(\kappa, i) = (A_n, 0)$, (E, i) , and (T_n, z) the part of the wavefunction lying along the z -axial Bloch function have the following forms

$$\psi_{\parallel 0, m_e l k \sigma}^{(\kappa, z)} := \Phi_{\parallel 0, m_e l k \sigma}^{(\kappa, z)} \eta_{\Omega_{m_e l \sigma}^{(\kappa)}}^{(z)}, \quad (\text{2.177a})$$

$$\psi_{\parallel 1, m l k \sigma}^{(\kappa, z)} := \Phi_{\parallel 1, m l k \sigma}^{(\kappa, z)} \eta_{\omega^{(\kappa)}(-1)^m}^{(z)}. \quad (\text{2.178a})$$

The envelope basis functions associated with the basis wavefunctions above are constructed from our naive envelope basis functions and go as follows

$$\Phi_{\parallel 0, m_e l k \sigma}^{(\kappa, z)} = \phi_{k l m_e \sigma; (-1)^{m_e/2}}^{(z)}, \quad (\text{C.87})$$

$$\Phi_{\parallel 1, m l k \sigma}^{(\kappa, z)} = \phi_{k l m \sigma; (-1)^l}^{(x)} + \omega^{(\kappa)} \Omega_{m l \sigma}^{(\kappa)} \phi_{k l m \sigma; (-1)^{l+m}}^{(y)}. \quad (\text{2.165a})$$

Above we introduced the following constant.

$$\Omega_{m l \sigma}^{(\kappa)} := \sigma \omega^{(\kappa)} (-1)^{l+m} i^{+m}. \quad (\text{C.88})$$

A quick word on a notational brevity introduced above. We denote the two basis states above by the subscripts 0 and 1. Each index on the wavefunction belongs to that wavefunction, and should properly be subscripted by this label. Since the author suspects that will introduce an untenable amount of cognitive load within the reader,

we use the convention that terms or tuples surrounded by parentheses subscripted by 0 and 1 individually or 01 together are understood to have indices that are likewise subscripted by that index. As an example, consider below.

$$\left(\tilde{\Psi}_{\varrho, mlk\xi}^{(\kappa, q)(u)}\right)_0 = \tilde{\Psi}_{\varrho_0, m_0 l_0 k_0 \xi_0}^{(\kappa_0, q_0)(u_0)} \quad (\text{C.89})$$

We also extend this convention to superscripted or subscripted indices of functions, namely

$$\mathcal{V}_{\|\{\mathbf{r}\}}^{(\kappa, i)_{01}} = \mathcal{V}_{\|\{\mathbf{r}\}}^{(\kappa_0, i_0, \kappa_1, i_1)} \quad (\text{C.90})$$

This also serves as an example to show how parenthesis subscripted by 01 should be interpreted. Finally, terms surrounded by brackets subscripted by 01 are understood to be repeated, first with the subscript 0 and then with subscript 1. Often, this means the repeated symbols inside the bracket are to be multiplied together, for example

$$\left[\tilde{\Psi}_{\varrho, mlk\xi}^{(\kappa, q)(u)}\right]_{01} = \tilde{\Psi}_{\varrho_0, m_0 l_0 k_0 \xi_0}^{(\kappa_0, q_0)(u_0)} \tilde{\Psi}_{\varrho_1, m_1 l_1 k_1 \xi_1}^{(\kappa_1, q_1)(u_1)}. \quad (\text{C.91})$$

In the interests of making use of this software as flexible as possible, we employ a potential of the form

$$\mathcal{U}_{\{\mathbf{r}\}} := \mathfrak{E}^{\mathfrak{s}} \sum_i \left(\frac{r}{\mathfrak{r}}\right)^{\mathfrak{d}_i} \mathcal{U}_{i\{\Omega\}} e^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\mathfrak{r}}} e^{-\beta_{i\{\Omega\}} \left(\frac{r}{\mathfrak{r}}\right)^2} \quad (\text{C.92})$$

Above we introduce the integration "variable" $\{\Omega\}$. This symbol denotes a solid angle functional dependence and we consider it interchangeable with an expression using our angular spherical coordinates, namely that $\{\Omega\} = \{\chi \phi\}$. As well, $\mathfrak{E}$ represents the energy scale of our system and $\mathfrak{r}$ represents the length scale. Both have been defined in Appendix A. While we described the expression above as a "potential", we see that by judicious choice of the constants (such as $\mathfrak{s}$, $\mathfrak{d}$ and $\mathfrak{U}$) and functions (such as $\mathcal{U}$, $\tilde{\alpha}$ and β) above this expression can be used for both the overlap and Hamiltonian

squared matrix elements in addition to the regular Hamiltonian elements. We consider the potential above to be "complex" and apply a tilde because while we expect the overall potential to be nondispersive and real it is possible for individual terms to have complex coefficients and functional dependence, depending upon how that potential is formulated.

We can start to simplify our integration by taking advantage of the cyclic nature of the single-axis wavefunction. To wit, we may expand the full wavefunctions in the matrix element formula and group by similar pairs of valley-indices. Further we may exploit the transformation properties of the single-axis wavefunction by using the cyclic transformations as a change of integration variables to show that several terms of the integrand may be "folded back over" each other. To that end, we may write

$$(\mathcal{M})_{01} = (\mathcal{M})_{\parallel,01} + (\mathcal{M})_{\perp,01} \quad (\text{C.93})$$

where

$$(\mathcal{M})_{\parallel,01} = \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^{\infty} r^2 dr \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,z)(u)} \right)_0^* \tilde{\mathcal{V}}_{\parallel\{\mathbf{r}\}}^{(\kappa,i)01} \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,z)(u)} \right)_1, \quad (\text{C.94})$$

and

$$(\mathcal{M})_{\perp,01} = \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^{\infty} r^2 dr \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,y)(u)} \right)_0^* \tilde{\mathcal{V}}_{\perp\{\mathbf{r}\}}^{(\kappa,i)01} \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,x)(u)} \right)_1. \quad (\text{C.95})$$

The expressions denoted by $\tilde{\mathcal{V}}$ are the original general potential form "folded over" together after change of integration variables using the T_d symmetry operations. These

new forms may be compactly expressed as below.

$$\mathcal{V}_{\parallel\{\mathbf{r}\}}^{(\kappa,i)_{01}} := \{\boldsymbol{\alpha}^{(\kappa,i)_0}\}^\dagger \begin{pmatrix} \tilde{\mathcal{U}}_{\{xyz\}} & 0 & 0 \\ 0 & \tilde{\mathcal{U}}_{\{zxy\}} & 0 \\ 0 & 0 & \tilde{\mathcal{U}}_{\{yxz\}} \end{pmatrix} \boldsymbol{\alpha}^{(\kappa,i)_1}, \quad (\text{C.96})$$

and

$$\mathcal{V}_{\perp\{\mathbf{r}\}}^{(\kappa,i)_{01}} := \{\boldsymbol{\alpha}^{(\kappa,i)_0}\}^\dagger \begin{pmatrix} 0 & \xi_0 \xi_1 \tilde{\mathcal{U}}_{\{xzy\}} & \tilde{\mathcal{U}}_{\{zxy\}} \\ \tilde{\mathcal{U}}_{\{yzx\}} & 0 & \xi_0 \xi_1 \tilde{\mathcal{U}}_{\{yxz\}} \\ \xi_0 \xi_1 \tilde{\mathcal{U}}_{\{zyx\}} & \tilde{\mathcal{U}}_{\{xyz\}} & 0 \end{pmatrix} \boldsymbol{\alpha}^{(\kappa,i)_1}, \quad (\text{C.97})$$

We note that we haven't a priori set $\kappa_0 = \kappa_1$, because we wish to examine and include, in addition to the potential of the system, other external potentials such as applied electric fields that might break the symmetry and allowing mixing of states with different κ . For our purposes here, we note that while a general potential of the form we gave for $\tilde{\mathcal{U}}$ does not have any particular sense of symmetry, it is well known that general functions can be written as a sum of functions of definite symmetry. To wit,

$$\tilde{\mathcal{U}} = \tilde{\mathcal{U}}^{(A_1)} + \tilde{\mathcal{U}}^{(A_2)} + \tilde{\mathcal{U}}^{(E,0)} + \tilde{\mathcal{U}}^{(E,1)} + \sum_{q=x,y,z} \left(\tilde{\mathcal{U}}^{(T_1,q)} + \tilde{\mathcal{U}}^{(T_2,q)} \right) \quad (\text{C.98})$$

We examine such selection rules and how they might simplify the above expressions for $\tilde{\mathcal{V}}$ in the [Supplementary Materials](#). In any case, for this derivation we only consider potentials of a given symmetry. The process of resolving a general potential into symmetric components doesn't significantly alter the underlying summation structure of the potential we defined in (D.229) and so we will continue to use it. To that end, we write

$$\mathcal{V}_{\mathbf{r}\{\mathbf{r}\}}^{(\kappa',j)(\kappa,i)_{01}} = \mathfrak{e}^5 \sum_i \left(\frac{r}{\mathfrak{r}} \right)^{\partial_i} \mathcal{V}_{i\{\chi,\phi\}} e^{-\tilde{\alpha}_{i\{\chi,\phi\}} \frac{r}{\mathfrak{r}}} e^{-\beta_{i\{\chi,\phi\}} \left(\frac{r}{\mathfrak{r}} \right)^2}. \quad (\text{C.99})$$

We may employ this form within our integration formula. One of the computational benefits of the potential form we have chosen is that we may complete the radial integration analytically, which we have seen will simplify the numerical computational complexity of the integration over the solid angle degrees of freedom.

$$(\mathcal{M})_{\mathfrak{K},01}^{(z,j)} = \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^{\infty} r^2 dr \xi_0 \left[\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,q_{\mathfrak{K}})(u)} \right]_{01} \tilde{\mathcal{V}}_{\mathfrak{K}\{r\}}^{(\kappa',j)(\kappa,i)_{01}}, \quad (\text{C.100})$$

$$= \int_{\mathcal{S}_0} d\Omega \sum_i \sum_{s_0=-u_0}^{s_0=+u_0} \sum_{s_1=-u_1}^{s_1=+u_1} \mathfrak{E}_{i(ml;\eta\lambda,s)_{01}}^{(u)_{01}} \mathcal{J}_{i(s)_{01}\{\Omega\}}, \quad (\text{C.101})$$

$$\mathfrak{E}_{i(ml;\eta\lambda,s)_{01}}^{(u)_{01}} := \xi_0 \mathfrak{E}^{u_0+u_1+\mathfrak{s}} \mathfrak{r}^{-\mathfrak{d}_i} \left[\frac{k! \mathfrak{H}_{ml;\lambda,s}^{(u)}}{\mathfrak{K}! \left(\frac{\eta}{\tau} \right)^{2u}} \right]_{01}. \quad (\text{C.102})$$

We will see $\mathfrak{K}$ emerge later in this appendix. For now, we merely remark that

$$\mathfrak{K}_n = (k + u + \delta_{s\bar{1}})_n. \quad (\text{C.103})$$

Because we expect that, in general, the integration over solid angle will be conducted numerically, we do not exploit the linear nature of integration to pull out the summation symbols. We find that for our integrals it is computationally much simpler and faster to compute one numerical integration for a summed-over integrand rather than complete many numerical integrations that are then summed. We have introduced a few sequences which we will define below,

$$\mathcal{J}_{i(s)_{01}\{\chi,\phi\}} := \left[P_{(l+2s)m;\lambda\{\chi_q\}} Q_{m\xi\{\phi_q\}} \right]_{01} \mathcal{V}_{i\{\Omega\}} \mathcal{R}_{i(s)_{01}\{\Omega\}}, \quad (\text{C.104})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} := \int_0^{\infty} dr r^{\mathfrak{d}_i+2} \left[\frac{\mathfrak{K}!}{k!} (F_s')^{(Wu)}_{lk;\eta\lambda\{r\chi_q\}} \Theta^{(q)} \right]_{01} e^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\tau} - \beta_{i\{\Omega\}} \left(\frac{r}{\tau} \right)^2}. \quad (\text{C.105})$$

We might note here that we have not explicitly included the Bloch functions in our formulae above. However, with this form, incorporation of the Bloch functions is fair straightforward. More details on how the full Bloch functions may be incorporated

into these formulae without much additional complexity may be found in the [Supplementary Materials](#).

We recall that

$$\left[(F'_s)^{(Wu)}_{lk;\eta\lambda\{\chi_q\}} \right]_{01} = \left[\left(\sqrt{\frac{\vartheta}{\eta}} \right)^3 \sqrt{W} \rho^{\frac{l}{W} - u \delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \right]_{01} \mathcal{L}_{(s\{\rho\})_{01}}, \quad (\text{C.106})$$

$$= r^{-3} W \left[\rho^{\frac{l+\frac{3}{2}}{W} - u \delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \right]_{01} \mathcal{L}_{(s\{\rho\})_{01}}, \quad (\text{C.107})$$

$$\mathcal{L}_{(s\{\rho\})_{01}} := \left[\frac{\Re!}{k!} \sum_{t=-2u}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} L_{(k+t)\{\rho\}}^{(l+u-\delta_{s\bar{1}})} \right]_{01}. \quad (\text{C.108})$$

We may make our notation even further compact. If we introduce

$$G_{\{r\}(s\{\chi_q\})_{01}} := W r^{\mathfrak{d}_i-1} \left[\rho^{\frac{l+\frac{3}{2}}{W} - u \delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \right]_{01}, \quad (\text{C.109})$$

than we may write our analytic integral as

$$\mathcal{R}_{i\{\Omega\}} = \int_0^\infty dr G_{\{r\}(s\{\chi_q\})_{01}} \mathcal{L}_{(s\{\rho\})_{01}} e^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\tau} - \beta_{i\{\Omega\}} \left(\frac{r}{\tau} \right)^2}. \quad (\text{C.110})$$

We do not assume that the anisotropy of the two states are the same in general, and so $\rho_0 \neq \rho_1$ over the integration domain. We introduce a new "averaged" variable. We define

$$\hat{\rho}_{\{\Omega\}} := \frac{\rho_0 + \rho_1}{2} \quad (\text{C.111})$$

which yields

$$\left[e^{-\frac{\rho}{2}} \right]_{01} = e^{-\hat{\rho}}. \quad (\text{C.112})$$

We find it convenient to introduce a new radial scaling $\check{r}$ with solid angle dependence so that we may write our "averaged" $\hat{\rho}$ as

$$\hat{\rho}_{\{\Omega\}} = \left(\frac{r}{\check{r}_{\{\Omega\}}} \right)^W. \quad (\text{C.113})$$

We will define $\check{r}$ below. If we employ this form of $\hat{\rho}$ we can define our original ρ_n in the following terms below

$$\Gamma_{n\{\Omega\}} := \sqrt{\check{r}_{\{\Omega\}} \left(\frac{\vartheta_{\{\Omega\}}}{\eta} \right)_n}, \quad (\text{C.114})$$

$$\rho_n = \Gamma_{n\{\Omega\}}^{2W} \hat{\rho}. \quad (\text{C.115})$$

Using these new expressions we can simplify and rewrite part of our integrand using $\hat{\rho}$ as

$$\left[\rho^{\frac{l+\frac{3}{2}}{W} - u\delta_{W1} - \delta_{s1}} \right]_{01} = \left[\Gamma_{\{\Omega\}}^{\mathfrak{L} - 2W\delta_{s1}} \right]_{01} \hat{\rho}^{\frac{\mathfrak{L}_0 + \mathfrak{L}_1}{2W} - \delta_{s_01} - \delta_{s_11}}, \quad (\text{C.116})$$

$$\mathfrak{L}_n := 2l_n + 3 - 2u_n \delta_{W1}. \quad (\text{C.117})$$

Through experience, we have found it convenient in the development of our software to rewrite ρ in terms of some accessory constants. We have an effective radial functions that goes as

$$\check{r}_{\{\Omega\}} := \frac{\hat{\eta}}{\hat{\vartheta}_{\{\Omega\}} \Xi_{\{\Omega\}}} \quad (\text{C.118})$$

where we defined an additional parameter

$$\hat{\eta}^W := \frac{2(\eta_0 \eta_1)^W}{\eta_0^W + \eta_1^W} \quad (\text{C.119})$$

and some additional functions

$$\hat{\vartheta}_{\{\Omega\}}^W := \frac{\vartheta_{0\{\Omega\}}^W + \vartheta_{1\{\Omega\}}^W}{2} \quad (\text{C.120})$$

$$\Xi_{\{\Omega\}}^W := 1 + \mathfrak{f} \epsilon_{\{\Omega\}} \quad (\text{C.121})$$

The Ξ function depends on the mismatch between the parameters of the states, which

go as

$$\mathfrak{f} := \frac{\eta_1^W - \eta_0^W}{\eta_0^W + \eta_1^W} \quad (\text{C.122})$$

$$\epsilon_{\{\Omega\}} := \frac{\vartheta_0^W - \vartheta_1^W}{\vartheta_0^W + \vartheta_1^W} \quad (\text{C.123})$$

Within a subbasis $\eta_0 = \eta_1$ and $\lambda_0 = \lambda_1$ so many of the above expressions simplify. However, $\epsilon \neq 0$ can emerge when integrating over matrix elements involving single-valley states of different axes.

With these accessor functions and parameters, we can rewrite one of the factors of our integrand as such

$$G_{\{r\}(s\{\chi_q\})_{01}} = W r^{\mathfrak{d}_i-1} \left[\rho^{\frac{l+\frac{3}{2}}{W} - u\delta_{W1} - \delta_{s1}} e^{-\frac{\rho}{2}} \right]_{01}, \quad (\text{C.124})$$

$$= W \left(\hat{\rho}^{\frac{1}{W}} \check{r}_{\{\Omega\}} \right)^{\mathfrak{d}_i-1} \left[\Gamma_{\{\Omega\}}^{\mathfrak{L}-2W\delta_{s1}} \right]_{01} \hat{\rho}^{\frac{\mathfrak{L}_0+\mathfrak{L}_1}{2W} - \delta_{s_01} - \delta_{s_11}} e^{-\hat{\rho}}, \quad (\text{C.125})$$

$$= \frac{W}{\check{r}_{\{\Omega\}}} \hat{\rho}^{1-\frac{1}{W}} H_{\{\hat{\rho}\}(s\{\chi_q\})_{01}} e^{-\hat{\rho}}. \quad (\text{C.126})$$

Above we have introduced a new function and some associated constants, which go as

$$H_{\{\hat{\rho}\}(s\{\chi_q\})_{01}} := \check{r}_{\{\Omega\}}^{\mathfrak{d}_i} \left[\Gamma_{\{\Omega\}}^{\mathfrak{L}-2W\delta_{s1}} \right]_{01} \hat{\rho}^{\frac{\mathfrak{L}\mathfrak{d}_i}{W} - \delta_{s_01} - \delta_{s_11} - 1}, \quad (\text{C.127})$$

$$\hat{\mathfrak{L}}_{\mathfrak{d}} := \frac{\mathfrak{L}_0 + \mathfrak{L}_1}{2} + \mathfrak{d} = l_0 + l_1 - \delta_{W1}(u_0 + u_1) + \mathfrak{d} + 3. \quad (\text{C.128})$$

We note that

$$d\hat{\rho} = \frac{W}{\hat{r}} \hat{\rho}^{1-\frac{1}{W}} dr \quad (\text{C.129})$$

as such we may write

$$dr G_{\{r\}(s\{\chi_q\})_{01}} = d\hat{\rho} H_{\{\hat{\rho}\}(s\{\chi_q\})_{01}} \quad (\text{C.130})$$

The expressions above suggests we should redefine $\hat{\rho}_{\{\Omega\}}$ as our integration variable.

Over the radial integration we can consider $\{\Omega\}$ to be constant, and so we will drop this subscript from expressions that form the radial integrand. We will restore this subscripts as needed, and once the analytical radial integration is completed we will ensure these subscripts are included in the numerical solid-angle integrand. With this in mind we may write our analytic integral over $\hat{\rho}$ as

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} = \int_0^\infty d\hat{\rho} H_{\{\hat{\rho}\}(s)_{01}} \mathcal{L}_{(s\{\rho\})_{01}} e^{-\mathcal{F}_{i\{\hat{\rho}\}}}. \quad (\text{C.131})$$

Expanding H as previously defined yields

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} = \check{r}_{\{\Omega\}}^{\mathfrak{d}_i} \left[\Gamma_{\{\Omega\}}^{\mathcal{L}-2W\delta_{s\bar{1}}} \right]_{01} \mathcal{R}_{i(s)_{01}\{\Omega\}}, \quad (\text{C.132})$$

with the radial integral partitioned as $\mathcal{R}$. We'll focus on the radial integral, namely

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} := \int_0^\infty d\hat{\rho} \hat{\rho}^{\frac{\mathfrak{L}_{\mathfrak{d}_i}-\delta_{s_0\bar{1}}-\delta_{s_1\bar{1}}-1}{W}} \mathcal{L}_{(s\{\rho\})_{01}} e^{-\mathcal{F}_{i\{\hat{\rho}\}}}, \quad (\text{C.133})$$

$$\mathcal{L}_{(s\{\rho\})_{01}} := \left[\frac{\mathfrak{K}!}{k!} \sum_{t=-2u}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} L_{(k+t)\{\rho\}}^{(l+u-\delta_{s\bar{1}})} \right]_{01}. \quad (\text{C.134})$$

In the analytical radial integral above we introduced a new function,

$$\mathcal{F}_{i\{\hat{\rho}\}} := \hat{\rho} + \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) \hat{\rho}^{\frac{1}{W}} + \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2 \hat{\rho}^{\frac{2}{W}}, \quad (\text{C.135})$$

$$= (1 + \mathcal{A}_i) \hat{\rho} + \mathcal{B}_i \hat{\rho}^{\frac{2}{W^2}}, \quad (\text{C.136})$$

where

$$\mathcal{A}_i := \delta_{W1} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) + \delta_{W2} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2, \quad (\text{C.137})$$

$$\mathcal{B}_i := \delta_{W1} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2 + \delta_{W2} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) \quad (\text{C.138})$$

We can rewrite $\mathcal{R}$ by pulling the summation over $(t)_{01}$ outside the analytical radial

integration.

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{t=-2u}^{u+\delta_{s1}} \frac{\mathfrak{R}!}{k!} h_{(kl, st)}^{(u)} \right]_{01} \mathcal{R}'_{i(st)_{01}}, \quad (\text{C.139})$$

$$\mathcal{R}'_{i(st)_{01}} := \int_0^\infty d\hat{\rho} \hat{\rho}^{\frac{\hat{\mathfrak{z}}_{\delta_i}}{W} - \delta_{s_0 1} - \delta_{s_1 1} - 1} \left[L_{(k+v)\{\rho\}}^{(+u-\delta_{s1})} \right]_{01} e^{-\mathcal{F}_i\{\rho\}}. \quad (\text{C.140})$$

In analogy to our rules for subscripted brackets, we introduce a related rule for summation symbols when they are the only symbols included in the brackets. We want to be clear that this rule might be counterintuitive when first seen because it represents an abuse of commonly understood rules of parenthesis and bracket in arithmetic. However, we think the formal meaning of repeated symbols is clear and it can be understood by the reader that in certain contexts repeated symbols do necessarily mean "product" exclusively. As such, we want to make it clear that symbols outside and to the right of subscripted brackets are understood to be part of the summation, not outside it as might be expected. Namely, we define the following.

$$\left[\sum_{n=a}^b \right]_{01} (f_n)_{01} := \sum_{n_0=a_0}^{b_0} \sum_{n_1=a_1}^{b_1} f_{n_0 n_1} \quad (\text{C.141})$$

For example, equation (D.334) above can be considered and understood in light of the definition above to be such that

$$\left[\sum_{s=-u}^{s=+u} \right]_{01} \left(\mathfrak{e}_{i(ml; \eta \lambda, s)}^{(u)} \mathcal{G}'_{i(s)\{\Omega\}} \right)_{01} = \sum_{s_0=-u_0}^{s_0=+u_0} \sum_{s_1=-u_1}^{s_1=+u_1} \mathfrak{e}_{i(ml; \eta \lambda, s)}^{(u)_{01}} \mathcal{G}'_{i(s)_{01}\{\Omega\}}. \quad (\text{C.142})$$

While it represents a slight abuse of standard bracket notation, we hope that it is clear that this notation has the benefit of compacting otherwise long and repetitive expressions involving double or greater summation. To prevent confusion, we note that forms with brackets containing summation symbol AND summations terms, for example as in equation (D.271) should continue to be understood as a product of terms subscripted with 0 and 1 respectively.

To complete the radial integration, we consider an integral of the form given below.

$$\mathcal{R}' := \int_0^\infty d\rho \rho^{\mathfrak{g}-1} L_{\mathfrak{T}_0\{\rho(1+\Delta)\}}^{(\mathfrak{a}_0)} L_{\mathfrak{T}_1\{\rho(1-\Delta)\}}^{(\mathfrak{a}_1)} e^{-(1+\mathcal{A})\rho - \mathcal{B}\rho^{\frac{2}{W^2}}}, \quad (\text{C.143})$$

$$= \left[\sum_{u=0}^{\mathfrak{T}} \sum_{v=u}^{\mathfrak{T}} \right]_{01} \mathfrak{F}_{\mathfrak{g}(uv)_{01}} \mathcal{G}_{(W\mathfrak{g})\{u\}_{01}\{\Delta\mathcal{Z}_{\mathcal{G}}\}}^{(W)} \mathcal{H}_{(W\mathfrak{g}+Ww)\{\mathcal{Z}_{\mathcal{H}}\}}^{(W)}. \quad (\text{C.144})$$

The L functions used above are the associated Laguerre functions. The significance of the above expression is that it includes the coefficients of the Laguerre polynomials. It should be noted that the summations above can involve sequences of terms of near magnitude but alternating sign. This is a computation ill-posed to be completed numerically, so we will endeavor to simplify the expressions as much as possible. Some of the expressions used in the formula above are defined below

$$\mathfrak{F}_{\mathfrak{g}(uv)_{01}} := \left[\frac{(-1)^{u+v} \sqrt{\mathfrak{T}!(\mathfrak{T}+\mathfrak{a})!}}{(\mathfrak{T}-v)!(\mathfrak{a}+v)!u!(v-u)!} \right]_{01} \Gamma(\mathfrak{g} + v_0 + v_1), \quad (\text{C.145})$$

$$\mathcal{G}_{L;\{u\}_{01}\{\Delta\mathcal{Z}\}}^{(W)} := (1 - \mathcal{Z}(1 + \Delta))^{u_0} (1 - \mathcal{Z}(1 - \Delta))^{u_1} \mathcal{Z}^{\frac{L}{W}}, \quad (\text{C.146})$$

$$\mathcal{H}_{M\{\mathcal{Z}\}}^{(W)} := \left(\delta_{W1} \frac{\sqrt{\pi}}{\Gamma[\frac{M+1}{2}]} \mathcal{Z}^M + \delta_{W2} \right) h_{M\{\mathcal{Z}\}}, \quad (\text{C.147})$$

$$\mathcal{Z}_{\mathcal{G}} := \frac{1}{1 + \mathcal{A}}, \quad (\text{C.148})$$

$$\mathcal{Z}_{\mathcal{H}} := \delta_{W1} \left(\frac{1 + \mathcal{A}}{2\sqrt{\mathcal{B}}} \right) + \delta_{W2} \left(\frac{\mathcal{B}}{2\sqrt{1 + \mathcal{A}}} \right). \quad (\text{C.149})$$

The expression h_{WL} refers to the hypergeometric function defined in the [Supplementary Materials](#).

We can use the above expressions if we set the constants to appropriate values. As we are defining new quantities and functions, we've taken the liberty of explicitly including the dependence over solid-angle in the final expression of indirect integra-

tion variables. We define or set the following

$$\mathfrak{h}_{\mathfrak{d}(s)_{01}} := W \mathfrak{g} = \hat{\mathfrak{L}}_{\mathfrak{d}} - W(\delta_{s_0 \bar{1}} + \delta_{s_1 \bar{1}}), \quad (\text{C.150})$$

$$\mathfrak{a}_n := (\mathfrak{l} + \mathfrak{u} - \delta_{s \bar{1}})_n, \quad (\text{C.151})$$

$$\mathfrak{T}_n := (k + t)_n, \quad (\text{C.152})$$

$$\Delta := \frac{\rho_0 - \rho_1}{2\hat{\rho}}. \quad (\text{C.153})$$

We can further refine Δ .

$$\Delta = \frac{\Gamma_0^{2W} - \Gamma_1^{2W}}{2} = \frac{\check{r}^W}{2} \left[\left(\frac{\vartheta}{\eta} \right)_0^W - \left(\frac{\vartheta}{\eta} \right)_1^W \right], \quad (\text{C.154})$$

$$= \frac{1}{2} \left(\frac{\hat{\eta}}{\hat{\vartheta} \Xi} \right)^W \left(\frac{(\eta_1 \vartheta_0)^W - (\eta_0 \vartheta_1)^W}{[\eta]_{01}^W} \right), \quad (\text{C.155})$$

$$= \frac{1}{2} \left(\frac{4[\eta]_{01}^W}{(\eta_0^W + \eta_1^W)(\vartheta_0^W + \vartheta_1^W) \Xi^W} \right) \left(\frac{(\eta_1 \vartheta_0)^W - (\eta_0 \vartheta_1)^W}{[\eta]_{01}^W} \right), \quad (\text{C.156})$$

$$= \frac{(\eta_1^W - \eta_0^W)(\vartheta_0^W + \vartheta_1^W) + (\eta_0^W + \eta_1^W)(\vartheta_0^W - \vartheta_1^W)}{(\vartheta_0^W + \vartheta_1^W)(\eta_0^W + \eta_1^W) \Xi^W}. \quad (\text{C.157})$$

Distributing the denominator through the sum in the numerator, and identifying existing quantities, we may state our final expression for Δ to be

$$\Delta_{\{\Omega\}} = \frac{\mathfrak{f} + \epsilon_{\{\Omega\}}}{1 + \mathfrak{f} \epsilon_{\{\Omega\}}} \quad (\text{C.158})$$

We must also consider $\mathcal{L}_{\mathcal{G}}$ and $\mathcal{L}_{\mathcal{H}}$. We had defined these quantities in terms of $\mathcal{A}$ and $\mathcal{B}$. Here we recall that we had previously identified those quantities as indirect integration variables, namely

$$\tilde{\mathcal{A}}_i := \delta_{W1} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) + \delta_{W2} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2, \quad (\text{D.274a})$$

$$\tilde{\mathcal{B}}_i := \delta_{W1} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2 + \delta_{W2} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) \quad (\text{D.275a})$$

Using these expressions for $\mathcal{A}$ and $\mathcal{B}$, we can write the particular $\mathcal{Z}_{\mathcal{G}}$ and $\mathcal{Z}_{\mathcal{H}}$ we wish to use as functions over solid-angle. To wit,

$$\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}} = \frac{1}{1 + \delta_{W1} \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right) + \delta_{W2} \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right)^2}, \quad (\text{C.159})$$

$$\tilde{\mathcal{Z}}_{i\mathcal{H}\{\Omega\}} = \frac{\delta_{W1} + \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right)}{\sqrt{\delta_{W2} + \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right)^2}}. \quad (\text{C.160})$$

While we wanted to ensure the solid-angle dependence of Δ and the $\mathcal{Z}$ were clear in the expressions above, we will resume dropping this dependence in expressions while examining the analytic radial integration. In light of the arguments above, we can write

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{t=-2u}^{u+\delta_{s\bar{1}}} \frac{\mathfrak{K}!}{k!} h_{(kl, st)}^{(u)} \right]_{01} \mathcal{R}'_{i(st)_{01}}, \quad (\text{C.161})$$

$$\mathcal{R}'_{i(st)_{01}} = \left[\sum_{u=0}^{k+t} \sum_{v=u}^{k+t} \right]_{01} \mathfrak{F} \left(\frac{\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}}{W} \right)_{(uv)_{01}} \mathcal{G}_{i(su)_{01}} \mathcal{H}_{i(s)_{01}(v_0+v_1)}. \quad (\text{C.162})$$

where we found it convenient to introduce the following sequences

$$\mathcal{G}_{i(su)_{01}} := \mathcal{G}_{\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}(u)_{01} \{ \Delta \tilde{\mathcal{Z}}_{i\mathcal{G}} \}}^{(W)}, \quad (\text{C.163})$$

$$\mathcal{H}_{i(s)_{01}w} := \mathcal{H}_{(\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}+Ww) \{ \tilde{\mathcal{Z}}_{i\mathcal{H}} \}}^{(W)}. \quad (\text{C.164})$$

Earlier, we had pulled out the summation over t to complete the radial integration analytically. We can reorganize the terms of the double summation in $\mathcal{R}$ to allow the summation over t to be completed. We introduce a new limit for summations that goes as

$$\mathfrak{K}_n = (k + u + \delta_{s\bar{1}})_n. \quad (\text{C.165})$$

We recognize that u can range from 0 to $\mathfrak{K}$, v can range from u to $\mathfrak{K}$, and t must always be larger than $v - k$. We reorganize the order of our summations, and we pull out to

the front summations over u_0 , and u_1 . We may write

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{u=0}^{\mathfrak{K}} \sum_{v=u}^{\mathfrak{K}} \right]_{01} \mathfrak{A}_{\mathfrak{d}_i(suv)_{01}} \mathcal{G}_{i(su)_{01}} \mathcal{H}_{i(s)_{01}}(v_0+v_1) \quad (\text{C.166})$$

where we introduced

$$\mathfrak{A}_{\mathfrak{d}(suv)_{01}} := \left[\sum_{t=\text{Max}[v-k, -2u]}^{u+\delta_{s\bar{1}}} \frac{\mathfrak{K}!}{k!} h_{(kl, st)}^{(u)} \right]_{01} \mathfrak{F}_{\left(\frac{\mathfrak{h}_{\mathfrak{d}}}{W}\right)}(uv)_{01} \quad (\text{C.167})$$

However, we can further simplify this from a four-index summation to a three-index summation by introducing $w = v_0 + v_1$. We do this by introducing a summation over a Kronecker delta, going as

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{u=0}^{\mathfrak{K}} \sum_{v=u}^{\mathfrak{K}} \right]_{01} \mathfrak{A}_{\mathfrak{d}_i(suv)_{01}} \mathcal{G}_{i(su)_{01}} \sum_{w=u_0+u_1}^{\mathfrak{K}_0+\mathfrak{K}_1} \delta_{w, v_0+v_1} \mathcal{H}_{i(s)_{01}} w, \quad (\text{C.168})$$

$$= \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \sum_{w=u_0+u_1}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(su)_{01} w} \mathcal{G}_{i(su)_{01}} \mathcal{H}_{i(s)_{01}} w. \quad (\text{C.169})$$

where we introduce a new constant

$$\mathfrak{A}_{\mathfrak{d}(su)_{01} w} := \left[\sum_{v=u}^{\mathfrak{K}} \right]_{01} \delta_{w, v_0+v_1} \mathfrak{A}_{\mathfrak{d}(suv)_{01}}. \quad (\text{C.170})$$

$\mathfrak{A}$ is constant over integration, and in particular its value does not depend on any runtime parameters, as long as the basis and operator/potential specification occurs at or prior to compile time. It is a potentially complicated expression, and we take some effort to simplify it and attempt to find a form of the expression that may be computed in reusable pieces. We can consider

$$\mathfrak{A}_{\mathfrak{d}(su)_{01} w} = (-1)^{u_0+u_1+w} \Gamma \left[\frac{\mathfrak{h}_{\mathfrak{d}(s)_{01}}}{W} + w \right] \mathfrak{A}'_{(su)_{01} w}, \quad (\text{C.171})$$

$$\mathfrak{A}'_{(su)_{01} w} = \left[\sum_{v=u}^{\mathfrak{K}} \right]_{01} \delta_{w, v_0+v_1} [\mathfrak{A}''_{su v}]_{01}. \quad (\text{C.172})$$

The product of coefficients above has factors that depend only on information from a single state. We can write this single-state coefficient as so,

$$\mathfrak{A}_{suv}'' = \frac{1}{\sqrt{k!(k+l)!}} \frac{\mathfrak{K}!}{u!(v-u)!(\mathfrak{K}-v)!} \frac{(k+l)!}{(\mathfrak{a}-2u+v)!} \mathfrak{G}_{sv}, \quad (\text{C.173})$$

$$= \frac{1}{\sqrt{k!(k+l)!}} \binom{\mathfrak{K}}{u} \binom{\mathfrak{K}-u}{\mathfrak{K}-v} (k+l)^{\mathfrak{K}-v} \mathfrak{G}_{sv}, \quad (\text{C.174})$$

$$\mathfrak{G}_{sv} = \frac{(\mathfrak{a}-2u+v)!}{(\mathfrak{a}+v)!} \mathfrak{H}_{sv}^{(u)}, \quad (\text{C.175})$$

$$= (1-u) \mathfrak{H}_{sv}^{(0)} + u \frac{\mathfrak{H}_{sv}^{(1)}}{(\mathfrak{l}-\delta_{s\bar{1}}+v+1)(\mathfrak{l}-\delta_{s\bar{1}}+v)}. \quad (\text{C.176})$$

We have

$$\mathfrak{H}_{sv}^{(u)} = \sum_{t=\text{Max}[v-k, -2u]}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} \frac{(\mathfrak{K}-v)! \sqrt{(k+t)!(k+t+\mathfrak{a})!}}{(k+t-v)! \sqrt{k!(k+l)!}}. \quad (\text{C.177})$$

We also find it convenient to explicitly the separate the $u = 0$ (kinetic-off) and the $u = 1$ (kinetic-on) forms somewhat. I note that when the kinetic operator is "off" that $s = 0$ and that $v - k \leq 0$. This means

$$\mathfrak{H}_{sv}^{(0)} = 1. \quad (\text{C.178})$$

We also have

$$\mathfrak{H}_{sv}^{(1)} = \sum_{t=\text{Max}[v-k, -2]}^{1+\delta_{s\bar{1}}} g_{kl,st} \frac{(\mathfrak{K}-v)! \sqrt{(k+t)!(k+t+\mathfrak{l}+1-\delta_{s\bar{1}})!}}{(k+t-v)! \sqrt{k!(k+l)!}}. \quad (\text{C.179})$$

We note that we can treat this summation as a polynomial in $\mathfrak{K} - v$. Let us define

$$\mathfrak{H}_{sv}' = \sum_{t=0}^{\text{Min}[\mathfrak{N}, 3+\delta_{s\bar{1}}]} v^t \mathfrak{G}_{st}, \quad (\text{C.180})$$

$$\mathfrak{G}_{st} := g_{kl,s(1+\delta_{s\bar{1}}-t)} \frac{\sqrt{(k+1+\delta_{s\bar{1}}-t)!(k+\mathfrak{l}+2-t)!}}{\sqrt{k!(k+l)!}}. \quad (\text{C.181})$$

We may pull together these realizations and write

$$\mathfrak{A}''_{su(\mathfrak{K}-v)} = \frac{1}{\sqrt{k!(k+l)!}} \binom{\mathfrak{K}}{u} \binom{\mathfrak{K}-u}{v} (k+l)^v (1+u\mathfrak{J}_{sv}), \quad (\text{C.182})$$

$$\mathfrak{J}_{sv} = \frac{\mathfrak{H}'_{sv}}{(k+l+2-v)(k+l+1-v)} - 1, \quad (\text{C.183})$$

$$\mathfrak{H}'_{sv} = \sum_{t=0}^{\text{Min}[v, 3+\delta_{s\bar{1}}]} v^t \mathfrak{G}_{st}, \quad (\text{C.184})$$

$$\mathfrak{G}_{st} = g_{kl,s(1+\delta_{s\bar{1}}-t)} \frac{\sqrt{(k+1+\delta_{s\bar{1}}-t)!(k+l+2-t)!}}{\sqrt{k!(k+l)!}}. \quad (\text{C.185})$$

Please note that, in the interests of space and clarity, we did not make explicit the fact that $\mathfrak{A}''$ and the various basis indices and parameters within the expression should be indexed by the state identifiers 0 or 1.

It is generally less computationally intense to use the Kronecker delta to reduce the above double summation to a single summation to implement, rather than implementing a double summation with many zero terms. To accomplish this we let $v_0 \rightarrow \mathfrak{K}_0 - v$ and $v_1 \rightarrow w - \mathfrak{K}_0 + v$. Our expression becomes

$$\mathfrak{A}'_{(su)_{01}w} = \sum_{v=\text{Max}[0, \mathfrak{K}_0+u_1-w]}^{\text{Min}[\mathfrak{K}_0-u_0, \mathfrak{K}_0+\mathfrak{K}_1-w]} \mathfrak{A}''_{(su)_0(\mathfrak{K}_0-v)} \mathfrak{A}''_{(su)_1(w-\mathfrak{K}_0+v)}. \quad (\text{C.186})$$

Earlier we found some sequences of values we needed to compute to generate our matrix elements.

$$\tilde{\mathcal{G}}_{i(su)_{01}} := \mathcal{G}_{\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}; (u)_{01}}^{(W)} \{\Delta \tilde{\mathcal{Z}}_{i\mathcal{G}}\}, \quad (\text{D.300})$$

$$\tilde{\mathcal{H}}_{i(s)_{01}w} := \mathcal{H}_{(\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}+W)w}^{(W)} \{\tilde{\mathcal{Z}}_{i\mathcal{H}}\}. \quad (\text{D.301})$$

We note that $\mathcal{G}$ and $\mathcal{H}$ above will simplify to less complicated forms depending on the values of $\tilde{\mathcal{Z}}$ which themselves depend on the functions $\tilde{\alpha}$, β and constant W . If both $\delta_{w1}\tilde{\alpha} \equiv 0$ and $\delta_{w2}\beta \equiv 0$ identically, then $\mathcal{Z}_{\mathcal{G}} \equiv 1$. We also see another simplification

when $\Delta \equiv 0$ identically as well. There is a third form when both conditions apply. To wit, we have

$$\mathcal{G}_{L;(u)_{01}\{\Delta(1)\}}^{(W)} = (-1)^{u_0} \Delta^{u_0+u_1}, \quad (\text{C.187})$$

$$\mathcal{G}_{L;(u)_{01}\{(0)\tilde{\mathcal{Z}}_{\mathcal{G}}\}}^{(W)} = (1 - \tilde{\mathcal{Z}}_{\mathcal{G}})^{u_0+u_1} \tilde{\mathcal{Z}}_{\mathcal{G}}^{\frac{1}{W}}, \quad (\text{C.188})$$

$$\mathcal{G}_{L;(u)_{01}\{(0)(1)\}}^{(W)} = \delta_{0,u_0+u_1}. \quad (\text{C.189})$$

We can also simplify $\tilde{\mathcal{H}}$. In the case when $W = 1$ and $\beta = 0$ we see that $\tilde{\mathcal{Z}}_{\mathcal{H}}$ becomes ill-defined. Returning to the original analytical integration we can see that in such a case, $\tilde{\mathcal{H}} \equiv 1$. This can also be shown as a limit to complex infinity, which we will denote as $\infty_{\mathbb{C}}$. Namely, we see $\lim_{\tilde{\mathcal{Z}}_{\mathcal{H}} \rightarrow \infty_{\mathbb{C}}} \tilde{\mathcal{H}} = 1$. In essence, as β tends to become identically zero, it is the case that $\tilde{\mathcal{H}}$ tends to become identically unity. A similar effect emerges when $W = 2$ and $\tilde{\alpha} \equiv 0$, but in that case $\tilde{\mathcal{Z}}_{\mathcal{H}} \equiv 0$. However, $\tilde{\mathcal{H}}$ becomes undefined for that value of $\tilde{\mathcal{Z}}_{\mathcal{G}}$. As before, examining the analytic continuation versus direct recalculation with $\tilde{\alpha} = 0$ yields the same result – as before, under these conditions $\tilde{\mathcal{H}} \equiv 1$.

These changes to the terms of the summation over u_0 , u_1 , and w significantly alters the structure of the summation. We present the following contractions of the main precalculated coefficients, namely

$$\mathfrak{A}_{\mathfrak{d}(s)_{01}uw} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \delta_{u,u_0+u_1} (-1)^{u_0} \mathfrak{A}_{\mathfrak{d}(s)_{01}w}, \quad (\text{C.190})$$

$$\mathfrak{A}'_{\mathfrak{d}(s)_{01}uw} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \delta_{u,u_0+u_1} \mathfrak{A}_{\mathfrak{d}(s)_{01}w}, \quad (\text{C.191})$$

$$\mathfrak{A}_{\mathfrak{d}(s)_{01}w} = \mathfrak{A}_{\mathfrak{d}(s0)_{01}w}. \quad (\text{C.192})$$

In instances when $\tilde{\mathcal{H}} \equiv 1$ we find an additional contraction along the w index, which

provides the following

$$\alpha_{\mathfrak{d}(s u)_{01}} = \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \mathfrak{A}_{\mathfrak{d}(s u)_{01} w}, \quad (\text{C.193})$$

$$\alpha_{\mathfrak{d}(s)_{01} u} = \left[\sum_{u=0}^{\mathfrak{R}} \right]_{01} \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \delta_{u, u_0+u_1} (-1)^{u_0} \mathfrak{A}_{\mathfrak{d}(s u)_{01} w}, \quad (\text{C.194})$$

$$\alpha'_{(s)_{01} u} = \left[\sum_{u=0}^{\mathfrak{R}} \right]_{01} \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \delta_{u, u_0+u_1} \mathfrak{A}_{\mathfrak{d}(s u)_{01} w}, \quad (\text{C.195})$$

$$\alpha_{(s)_{01}} = \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \mathfrak{A}_{\mathfrak{d}(s 0)_{01} w}. \quad (\text{C.196})$$

As before, we note that the summations to generate these coefficients feature alternating signs, making this computations ill-posed to be done using standard floating-point computations. Additionally, the coefficients themselves can become quite large. These summations have problems with both overflow and underflow error when floating point numbers are used.

To work with this form within our numerical integrator, which will use standard floating-point numerical computation, we opted to compute the coefficients analytically and store the results as a floating point number. The analytic computation can be achieved and automated with a computer algebra system, and for this work we used the SymPy Python package. We can then store the results as floating-point numbers in a database that can be recalled as needed. More information is available in [3.2](#).

Such a database can be constructed at compile time, or may be referenced from previous runs of the software. As such, we call this set of coefficients the *precalculated coefficients*. Unfortunately, the database does prove to be quite large, making portability between systems an issue. Additionally, we find that pulling the needed coefficients from the database into runtime does add a non-negligible amount of time to add computation. However, this is anticipated to be a one-time expenditure of memory access time during a computation, one which is anticipated to scale linearly with

problem size.

In light of these simplifications and considerations we can attempt to rewrite our integral by assessing and partitioning certain expressions. Our goal with this formulation of the matrix elements is based on minimizing repeated computations as well as exploiting certain features of our subbases/superbasis structure. In particular, within a subbasis, we will have $\eta_0 = \eta_1 = \hat{\eta}$ and so we will also have $\mathfrak{f} = 0$, $\Xi_{\{\Omega\}} \equiv 1$, and quantities such as $\hat{\eta}/\eta_n = 1$. Additionally, when considering matrix elements with $q_0 = q_1 = z$ within the same subbasis, we find that $\vartheta_0 \equiv \vartheta_1$ which also permits further simplifications in the expressions we formulate below.

$$(\mathcal{M})_{\mathfrak{K},01}^{(\kappa',j)} := \mathfrak{E}^{u_0+u_1+s} \mathfrak{C} \mathcal{I}, \quad (\text{C.197})$$

$$\mathfrak{C} := \frac{\xi_0}{\left[\left(\frac{\eta}{\hat{\eta}} \right)^{\frac{s}{2}+2u} \right]_{01} \left(\frac{\hat{\eta}}{\mathfrak{r}} \right)^{2u_0+2u_1}}, \quad (\text{C.198})$$

$$\mathcal{I} := \int_{\mathcal{S}_0} d\Omega \left[\left(\frac{\vartheta_{\{\Omega\}}}{\hat{\vartheta}_{\{\Omega\}}} \right)^{\frac{s}{2}} Q_{m\zeta\{\phi_q\}} \right]_{01} \mathcal{S}_{\{\Omega\}}, \quad (\text{C.199})$$

$$\mathcal{S}_{\{\Omega\}} := \left[\sum_{s=-u}^{s=+u} \right]_{01} \sum_i \mathcal{P}_{(s)_{01}\{\Omega\}} \mathcal{X}_{\mathfrak{d}_i\{\Omega\}} \mathcal{V}_{i\{\Omega\}} \mathcal{R}_{i(s)_{01}\{\Omega\}}, \quad (\text{C.200})$$

$$\mathcal{P}_{(s)_{01}\{\Omega\}} := \left[\frac{k! \mathfrak{H}_{ml;\lambda,s}^{(\kappa)}}{\mathfrak{K}! \left(\frac{\vartheta_{\{\Omega\}} \hat{\mathfrak{r}}_{\{\Omega\}}}{\eta} \right)^{W \delta_{s1}}} P_{(l+2s)m;\lambda\{\chi_q\}} \right]_{01}, \quad (\text{C.201})$$

$$\mathcal{X}_{\mathfrak{d}\{\Omega\}} := \left(\frac{\hat{\eta}}{\mathfrak{r} \hat{\vartheta}} \right)^{\mathfrak{d}} (\Xi_{\{\Omega\}})^{-\hat{\mathcal{L}}_{\mathfrak{d}}}. \quad (\text{C.202})$$

Above we have defined the following constants,

$$\mathfrak{K} := k + u + \delta_{s1}, \quad (\text{C.203})$$

$$\hat{\mathcal{L}}_{\mathfrak{d}} := l_0 + l_1 - \delta_{W1}(u_0 + u_1) + \mathfrak{d} + 3. \quad (\text{C.204})$$

We can articulate the various conditions that affect the summation structure of $\mathcal{R}$.

There are three conditions that determine which form should be used. They are

$$\delta_{W1}\tilde{\alpha}_i + \delta_{W2}\beta_i = 0 \quad \Rightarrow \quad \tilde{\mathcal{E}}_i \equiv 1, \quad (\text{C1a})$$

$$\eta_0 = \eta_1 \text{ and } \lambda_0 = \lambda_1 \text{ and } q_0 = q_1 \quad \Rightarrow \quad \Delta \equiv 0, \quad (\text{C1b})$$

$$\delta_{W1}\beta_i + \delta_{W2}\tilde{\alpha}_i = 0 \quad \Rightarrow \quad \tilde{\mathcal{H}} \equiv 1. \quad (\text{C2})$$

If condition (C2) is not true, then the radially integrated part of the integrand will go as

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(su)_{01}w} \mathcal{G}_{i(u)_{01}\{\Omega\}} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}, \quad (\text{C.205})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C1a})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(s)_{01}uw} \mathcal{G}_{u\{\Omega\}} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}, \quad (\text{C.206})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C1b})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}'_{\mathfrak{d}_i(s)_{01}uw} \tilde{\mathcal{G}}_{i(s)_{01}u\{\Omega\}} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}, \quad (\text{C.207})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C1a+C1b})} = \sum_{w=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(s)_{01}w} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}. \quad (\text{C.208})$$

However, if condition (C2) is true, then the radially integrated part of the integrand will go as

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C2})} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \mathfrak{a}_{\mathfrak{d}_i(su)_{01}} \tilde{\mathcal{G}}_{(u)_{01}\{\Omega\}}, \quad (\text{C.209})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C2;C1a})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{a}_{\mathfrak{d}_i(s)_{01}u} \mathcal{G}_{u\{\Omega\}}, \quad (\text{C.210})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C2;C1b})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{a}'_{\mathfrak{d}_i(s)_{01}u} \tilde{\mathcal{G}}_{i(s)_{01}u\{\Omega\}}, \quad (\text{C.211})$$

$$\mathcal{R}_{i(s)_{01}}^{(\text{C2;C1a+C1b})} = \mathfrak{a}_{\mathfrak{d}_i(s)_{01}}. \quad (\text{C.212})$$

where we employ sequences of the form

$$\tilde{\mathcal{G}}_{i(s)u_{01}\{\Omega\}} := \tilde{\mathcal{G}}_{\mathfrak{h}_{\mathfrak{d}_i(s)01};(u)_{01}\{\Delta_{\{\Omega\}}\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}}\}}^{(W)}, \quad (\text{C.213})$$

$$\mathcal{G}_{u\{\Omega\}} := \Delta_{\{\Omega\}}^u, \quad (\text{C.214})$$

$$\tilde{\mathcal{G}}_{i(s)01u\{\Omega\}} := \tilde{\mathcal{G}}_{\mathfrak{h}_{\mathfrak{d}_i(s)01};u\{\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}}\}}^{(W)}, \quad (\text{C.215})$$

$$\tilde{\mathcal{H}}_{i(s)01w\{\Omega\}} := \tilde{\mathcal{H}}_{(\mathfrak{h}_{\mathfrak{d}_i(s)01}+Ww);\{\tilde{\mathcal{Z}}_{i\mathcal{H}\{\Omega\}}\}}^{(W)}. \quad (\text{C.216})$$

The sequence values may be calculated with the following constant

$$\mathfrak{h}_{\mathfrak{d}(s)01} := \hat{\mathcal{L}}_0 - W(\delta_{s_0\bar{1}} + \delta_{s_1\bar{1}}), \quad (\text{C.217})$$

and with the following accessory functions.

$$\tilde{\mathcal{G}}_{L;(u)_{01}\{\Delta\tilde{\mathcal{Z}}\}}^{(W)} := (1 - \tilde{\mathcal{Z}}(1 + \Delta))^{u_0} (1 - \tilde{\mathcal{Z}}(1 - \Delta))^{u_1} \tilde{\mathcal{Z}}^{\frac{L}{W}}, \quad (\text{C.218})$$

$$\tilde{\mathcal{G}}_{L;u\{\tilde{\mathcal{Z}}\}}^{(W)} := (1 - \tilde{\mathcal{Z}})^u \tilde{\mathcal{Z}}^{\frac{L}{W}}, \quad (\text{C.219})$$

$$\tilde{\mathcal{H}}_{M\{\tilde{\mathcal{Z}}\}}^{(W)} := \left(\delta_{W1} \frac{\sqrt{\pi}}{\Gamma\left[\frac{M+1}{2}\right]} \tilde{\mathcal{Z}}^M + \delta_{W2} \right) h_{M\{\tilde{\mathcal{Z}}\}}, \quad (\text{C.220})$$

$$\Delta_{\{\Omega\}} = \frac{\mathfrak{f} + \epsilon_{\{\Omega\}}}{1 + \mathfrak{f}\epsilon_{\{\Omega\}}}, \quad (\text{C.221})$$

$$\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}} = \frac{1}{1 + \delta_{W1} \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right) + \delta_{W2} \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right)^2}, \quad (\text{C.222})$$

$$\tilde{\mathcal{Z}}_{i\mathcal{H}\{\Omega\}} = \frac{\delta_{W1} + \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right)}{\sqrt{\delta_{W2} + \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right)^2}}. \quad (\text{C.223})$$

D Calculation of matrix elements II: Analytical results

Now, as we build up our multi-valley wavefunctions we consider what we will describe here as the *longitudinal single-axis basis wavefunction*. We found that, for $(\kappa, i) = (A_n, 0)$, (E, i) , and (T_n, z) the part of the wavefunction lying along the z -axial Bloch function have the following forms

$$\psi_{\parallel 0, m_e l k \sigma}^{(\kappa, z)} := \Phi_{\parallel 0, m_e l k \sigma}^{(\kappa, z)} \eta_{\Omega_{m_e l \sigma}^{(\kappa)}}^{(z)}, \quad (2.177a)$$

$$\psi_{\parallel 1, m l k \sigma}^{(\kappa, z)} := \Phi_{\parallel 1, m l k \sigma}^{(\kappa, z)} \eta_{\omega^{(\kappa)}(-1)^m}^{(z)}. \quad (2.178a)$$

The envelope basis functions associated with the basis wavefunctions above are constructed from our naive envelope basis functions and go as follows

$$\Phi_{\parallel 0, m_e l k \sigma}^{(\kappa, z)} = \phi_{k l m_e \sigma; (-1)^{m_e/2}}^{(z)}, \quad (D.224)$$

$$\Phi_{\parallel 1, m l k \sigma}^{(\kappa, z)} = \phi_{k l m \sigma; (-1)^l}^{(x)} + \omega^{(\kappa)} \Omega_{m l \sigma}^{(\kappa)} \phi_{k l m \sigma; (-1)^{l+m}}^{(y)}. \quad (2.165a)$$

Above we introduced the following constant.

$$\Omega_{m l \sigma}^{(\kappa)} := \sigma \omega^{(\kappa)} (-1)^{l+m} i^{+m}. \quad (D.225)$$

A quick word on a notational brevity introduced above. We denote the two basis states above by the subscripts 0 and 1. Each index on the wavefunction belongs to that wavefunction, and should properly be subscripted by this label. Since the author suspects that will introduce an untenable amount of cognitive load within the reader, we use the convention that terms or tuples surrounded by parentheses subscripted by 0 and 1 individually or 01 together are understood to have indices that are likewise

subscripted by that index. As an example, consider below.

$$\left(\tilde{\Psi}_{\varrho, mlk\xi}^{(\kappa, q)(u)}\right)_0 = \tilde{\Psi}_{\varrho_0, m_0 l_0 k_0 \xi_0}^{(\kappa_0, q_0)(u_0)} \quad (\text{D.226})$$

We also extend this convention to superscripted or subscripted indices of functions, namely

$$\gamma_{\|\{\mathbf{r}\}}^{\tilde{\gamma}(\kappa, i)_{01}} = \gamma_{\|\{\mathbf{r}\}}^{\tilde{\gamma}(\kappa_0, i_0, \kappa_1, i_1)} \quad (\text{D.227})$$

This also serves as an example to show how parenthesis subscripted by 01 should be interpreted. Finally, terms surrounded by brackets subscripted by 01 are understood to be repeated, first with the subscript 0 and then with subscript 1. Often, this means the repeated symbols inside the bracket are to be multiplied together, for example

$$\left[\tilde{\Psi}_{\varrho, mlk\xi}^{(\kappa, q)(u)}\right]_{01} = \tilde{\Psi}_{\varrho_0, m_0 l_0 k_0 \xi_0}^{(\kappa_0, q_0)(u_0)} \tilde{\Psi}_{\varrho_1, m_1 l_1 k_1 \xi_1}^{(\kappa_1, q_1)(u_1)}. \quad (\text{D.228})$$

In the interests of making use of this software as flexible as possible, we employ a potential of the form

$$\tilde{\mathcal{U}}_{\{\mathbf{r}\}} := \mathfrak{E}^{\mathfrak{s}} \sum_i \left(\frac{r}{\mathfrak{r}}\right)^{\mathfrak{d}_i} \mathcal{U}_{i\{\Omega\}} \mathbf{e}^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\mathfrak{r}}} \mathbf{e}^{-\beta_{i\{\Omega\}} \left(\frac{r}{\mathfrak{r}}\right)^2} \quad (\text{D.229})$$

Above we introduce the integration "variable" $\{\Omega\}$. This symbol denotes a solid angle functional dependence and we consider it interchangeable with an expression using our angular spherical coordinates, namely that $\{\Omega\} = \{\chi \phi\}$. As well, $\mathfrak{E}$ represents the energy scale of our system and $\mathfrak{r}$ represents the length scale. Both have been defined in [Appendix A](#). While we described the expression above as a "potential", we see that by judicious choice of the constants (such as $\mathfrak{s}$, $\mathfrak{d}$ and $\mathfrak{U}$) and functions (such as $\mathcal{U}$, $\tilde{\alpha}$ and β) above this expression can be used for both the overlap and Hamiltonian squared matrix elements in addition to the regular Hamiltonian elements. We consider the potential above to be "complex" and apply a tilde because while we expect

the overall potential to be nondispersive and real it is possible for individual terms to have complex coefficients and functional dependence, depending upon how that potential is formulated.

We can start to simplify our integration by taking advantage of the cyclic nature of the single-axis wavefunction. To wit, we may expand the full wavefunctions in the matrix element formula and group by similar pairs of valley-indices. Further we may exploit the transformation properties of the single-axis wavefunction by using the cyclic transformations as a change of integration variables to show that several terms of the integrand may be "folded back over" each other. To that end, we may write

$$(\mathcal{M})_{01} = (\mathcal{M})_{\parallel,01} + (\mathcal{M})_{\perp,01} \quad (\text{D.230})$$

where

$$(\mathcal{M})_{\parallel,01} = \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^{\infty} r^2 dr \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,z)(u)} \right)_0^* \tilde{\mathcal{V}}_{\parallel\{\mathbf{r}\}}^{(\kappa,i)01} \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,z)(u)} \right)_1, \quad (\text{D.231})$$

and

$$(\mathcal{M})_{\perp,01} = \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^{\infty} r^2 dr \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,y)(u)} \right)_0^* \tilde{\mathcal{V}}_{\perp\{\mathbf{r}\}}^{(\kappa,i)01} \left(\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,x)(u)} \right)_1. \quad (\text{D.232})$$

The expressions denoted by $\tilde{\mathcal{V}}$ are the original general potential form "folded over" together after change of integration variables using the T_d symmetry operations. These new forms may be compactly expressed as below.

$$\tilde{\mathcal{V}}_{\parallel\{\mathbf{r}\}}^{(\kappa,i)01} := \left\{ \boldsymbol{\alpha}^{(\kappa,i)0} \right\}^\dagger \begin{pmatrix} \tilde{\mathcal{U}}_{\{xyz\}} & 0 & 0 \\ 0 & \tilde{\mathcal{U}}_{\{zxy\}} & 0 \\ 0 & 0 & \tilde{\mathcal{U}}_{\{yzx\}} \end{pmatrix} \boldsymbol{\alpha}^{(\kappa,i)1}, \quad (\text{D.233})$$

and

$$\mathcal{V}_{\perp\{\mathbf{r}\}}^{(\kappa,i)_{01}} := \{\boldsymbol{\alpha}^{(\kappa,i)_0}\}^\dagger \begin{pmatrix} 0 & \xi_0 \xi_1 \tilde{\mathcal{U}}_{\{xzy\}} & \tilde{\mathcal{U}}_{\{zxy\}} \\ \tilde{\mathcal{U}}_{\{yzx\}} & 0 & \xi_0 \xi_1 \tilde{\mathcal{U}}_{\{yxz\}} \\ \xi_0 \xi_1 \tilde{\mathcal{U}}_{\{zyx\}} & \tilde{\mathcal{U}}_{\{xyz\}} & 0 \end{pmatrix} \boldsymbol{\alpha}^{(\kappa,i)_1}, \quad (\text{D.234})$$

We note that we haven't a priori set $\kappa_0 = \kappa_1$, because we wish to examine and include, in addition to the potential of the system, other external potentials such as applied electric fields that might break the symmetry and allowing mixing of states with different κ . For our purposes here, we note that while a general potential of the form we gave for $\tilde{\mathcal{U}}$ does not have any particular sense of symmetry, it is well known that general functions can be written as a sum of functions of definite symmetry. To wit,

$$\tilde{\mathcal{U}} = \tilde{\mathcal{U}}^{(A_1)} + \tilde{\mathcal{U}}^{(A_2)} + \tilde{\mathcal{U}}^{(E,0)} + \tilde{\mathcal{U}}^{(E,1)} + \sum_{q=x,y,z} \left(\tilde{\mathcal{U}}^{(T_1,q)} + \tilde{\mathcal{U}}^{(T_2,q)} \right) \quad (\text{D.235})$$

We examine such selection rules and how they might simplify the above expressions for $\tilde{\mathcal{V}}$ in the [Supplementary Materials](#). In any case, for this derivation we only consider potentials of a given symmetry. The process of resolving a general potential into symmetric components doesn't significantly alter the underlying summation structure of the potential we defined in (D.229) and so we will continue to use it. To that end, we write

$$\mathcal{V}_{\mathbf{r}\{\mathbf{r}\}}^{(\kappa',j)(\kappa,i)_{01}} = \mathfrak{E}^5 \sum_i \left(\frac{r}{\mathfrak{r}} \right)^{\mathfrak{d}_i} \mathcal{V}_{i\{\chi,\phi\}} e^{-\tilde{\alpha}_{i\{\chi,\phi\}} \frac{r}{\mathfrak{r}}} e^{-\beta_{i\{\chi,\phi\}} \left(\frac{r}{\mathfrak{r}} \right)^2}. \quad (\text{D.236})$$

We may employ this form within our integration formula. One of the computational benefits of the potential form we have chosen is that we may complete the radial integration analytically, which we have seen will simplify the numerical com-

putational complexity of the integration over the solid angle degrees of freedom.

$$(\mathcal{M})_{\mathbf{K},01}^{(z,j)} = \int_{-\pi}^{+\pi} d\phi \int_{-1}^{+1} d\chi \int_0^{\infty} r^2 dr \xi_0 \left[\tilde{\psi}_{\varrho,mlk\xi}^{(\kappa,q_{\mathbf{K}})(u)} \right]_{01} \tilde{\mathcal{V}}_{\mathbf{K}\{\mathbf{r}\}}^{(\kappa',j)(\kappa,i)_{01}}, \quad (\text{D.237})$$

$$= \int_{\mathcal{S}_0} d\Omega \sum_i \sum_{s_0=-u_0}^{s_0=+u_0} \sum_{s_1=-u_1}^{s_1=+u_1} \mathfrak{C}_{i(ml;\eta\lambda,s)_{01}}^{(u)_{01}} \mathcal{J}_{i(s)_{01}\{\Omega\}}, \quad (\text{D.238})$$

$$\mathfrak{C}_{i(ml;\eta\lambda,s)_{01}}^{(u)_{01}} := \xi_0 \mathfrak{E}^{u_0+u_1+s} \mathfrak{r}^{-\mathfrak{d}_i} \left[\frac{k! \mathfrak{H}_{ml;\lambda,s}^{(u)}}{\mathfrak{K}! \left(\frac{\eta}{\mathfrak{r}} \right)^{2u}} \right]_{01}. \quad (\text{D.239})$$

We will see $\mathfrak{K}$ emerge later in this appendix. For now, we merely remark that

$$\mathfrak{K}_n = (k + u + \delta_{s\bar{1}})_n. \quad (\text{D.240})$$

Because we expect that, in general, the integration over solid angle will be conducted numerically, we do not exploit the linear nature of integration to pull out the summation symbols. We find that for our integrals it is computationally much simpler and faster to compute one numerical integration for a summed-over integrand rather than complete many numerical integrations that are then summed. We have introduced a few sequences which we will define below,

$$\mathcal{J}_{i(s)_{01}\{\chi,\phi\}} := \left[P_{(l+2s)m;\lambda\{\chi_q\}} Q_{m\xi\{\phi_q\}} \right]_{01} \mathcal{V}_{i\{\Omega\}} \mathcal{R}_{i(s)_{01}\{\Omega\}}, \quad (\text{D.241})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} := \int_0^{\infty} dr r^{\mathfrak{d}_i+2} \left[\frac{\mathfrak{K}!}{k!} (F'_s)^{(Wu)}_{lk;\eta\lambda\{r\chi_q\}} \Theta^{(q)} \right]_{01} e^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\mathfrak{r}} - \beta_{i\{\Omega\}} \left(\frac{r}{\mathfrak{r}} \right)^2}. \quad (\text{D.242})$$

We might note here that we have not explicitly included the Bloch functions in our formulae above. However, with this form, incorporation of the Bloch functions is fair straightforward. More details on how the full Bloch functions may be incorporated into these formulae without much additional complexity may be found in the [Supplementary Materials](#).

We recall that

$$\left[(F'_s)^{(Wu)}_{lk;\eta\lambda\{r\chi_q\}} \right]_{01} = \left[\left(\sqrt{\frac{\vartheta}{\eta}} \right)^3 \sqrt{W} \rho^{\frac{l}{W} - u \delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \right]_{01} \mathcal{L}_{(s\{\rho\})_{01}}, \quad (\text{D.243})$$

$$= r^{-3} W \left[\rho^{\frac{l+\frac{3}{2}}{W} - u \delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \right]_{01} \mathcal{L}_{(s\{\rho\})_{01}}, \quad (\text{D.244})$$

$$\mathcal{L}_{(s\{\rho\})_{01}} := \left[\frac{\mathfrak{K}!}{k!} \sum_{t=-2u}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} L_{(k+t)\{\rho\}}^{(l+u-\delta_{s\bar{1}})} \right]_{01}. \quad (\text{D.245})$$

We may make our notation even further compact. If we introduce

$$G_{\{r\}(s\{\chi_q\})_{01}} := W r^{\mathfrak{d}_i-1} \left[\rho^{\frac{l+\frac{3}{2}}{W} - u \delta_{W1} - \delta_{s\bar{1}}} e^{-\frac{\rho}{2}} \right]_{01}, \quad (\text{D.246})$$

than we may write our analytic integral as

$$\mathcal{R}_{i\{\Omega\}} = \int_0^\infty dr G_{\{r\}(s\{\chi_q\})_{01}} \mathcal{L}_{(s\{\rho\})_{01}} e^{-\tilde{\alpha}_{i\{\Omega\}} \frac{r}{\check{r}} - \beta_{i\{\Omega\}} \left(\frac{r}{\check{r}} \right)^2}. \quad (\text{D.247})$$

We do not assume that the anisotropy of the two states are the same in general, and so $\rho_0 \neq \rho_1$ over the integration domain. We introduce a new "averaged" variable. We define

$$\hat{\rho}_{\{\Omega\}} := \frac{\rho_0 + \rho_1}{2} \quad (\text{D.248})$$

which yields

$$\left[e^{-\frac{\rho}{2}} \right]_{01} = e^{-\hat{\rho}}. \quad (\text{D.249})$$

We find it convenient to introduce a new radial scaling $\check{r}$ with solid angle dependence so that we may write our "averaged" $\hat{\rho}$ as

$$\hat{\rho}_{\{\Omega\}} = \left(\frac{r}{\check{r}_{\{\Omega\}}} \right)^W. \quad (\text{D.250})$$

We will define $\check{r}$ below. If we employ this form of $\hat{\rho}$ we can define our original ρ_n in

the following terms below

$$\Gamma_{n\{\Omega\}} := \sqrt{\check{\eta}_{\{\Omega\}} \left(\frac{\vartheta_{\{\Omega\}}}{\eta} \right)_n}, \quad (\text{D.251})$$

$$\rho_n = \Gamma_{n\{\Omega\}}^{2W} \hat{\rho}. \quad (\text{D.252})$$

Using these new expressions we can simplify and rewrite part of our integrand using $\hat{\rho}$ as

$$\left[\rho^{\frac{l+\frac{3}{2}}{W} - u\delta_{W1} - \delta_{s1}} \right]_{01} = \left[\Gamma_{\{\Omega\}}^{\mathfrak{L} - 2W\delta_{s1}} \right]_{01} \hat{\rho}^{\frac{\mathfrak{L}_0 + \mathfrak{L}_1}{2W} - \delta_{s_01} - \delta_{s_11}}, \quad (\text{D.253})$$

$$\mathfrak{L}_n := 2l_n + 3 - 2u_n \delta_{W1}. \quad (\text{D.254})$$

Through experience, we have found it convenient in the development of our software to rewrite ρ in terms of some accessory constants. We have an effective radial functions that goes as

$$\check{\eta}_{\{\Omega\}} := \frac{\hat{\eta}}{\hat{\vartheta}_{\{\Omega\}} \Xi_{\{\Omega\}}} \quad (\text{D.255})$$

where we defined an additional parameter

$$\hat{\eta}^W := \frac{2(\eta_0 \eta_1)^W}{\eta_0^W + \eta_1^W} \quad (\text{D.256})$$

and some additional functions

$$\hat{\vartheta}_{\{\Omega\}}^W := \frac{\vartheta_{0\{\Omega\}}^W + \vartheta_{1\{\Omega\}}^W}{2} \quad (\text{D.257})$$

$$\Xi_{\{\Omega\}}^W := 1 + \mathfrak{f} \epsilon_{\{\Omega\}} \quad (\text{D.258})$$

The Ξ function depends on the mismatch between the parameters of the states, which

go as

$$\mathfrak{f} := \frac{\eta_1^W - \eta_0^W}{\eta_0^W + \eta_1^W} \quad (\text{D.259})$$

$$\epsilon_{\{\Omega\}} := \frac{\vartheta_0^W - \vartheta_1^W}{\vartheta_0^W + \vartheta_1^W} \quad (\text{D.260})$$

Within a subbasis $\eta_0 = \eta_1$ and $\lambda_0 = \lambda_1$ so many of the above expressions simplify. However, $\epsilon \neq 0$ can emerge when integrating over matrix elements involving single-valley states of different axes.

With these accessor functions and parameters, we can rewrite one of the factors of our integrand as such

$$G_{\{r\}(s\{\chi_q\})_{01}} = W r^{\mathfrak{d}_i-1} \left[\rho^{\frac{l+\frac{3}{2}}{W} - u\delta_{W1} - \delta_{s1}} e^{-\frac{\rho}{2}} \right]_{01}, \quad (\text{D.261})$$

$$= W \left(\hat{\rho}^{\frac{1}{W}} \check{r}_{\{\Omega\}} \right)^{\mathfrak{d}_i-1} \left[\Gamma_{\{\Omega\}}^{\mathfrak{L}-2W\delta_{s1}} \right]_{01} \hat{\rho}^{\frac{\mathfrak{L}_0+\mathfrak{L}_1}{2W} - \delta_{s_01} - \delta_{s_11}} e^{-\hat{\rho}}, \quad (\text{D.262})$$

$$= \frac{W}{\check{r}_{\{\Omega\}}} \hat{\rho}^{1-\frac{1}{W}} H_{\{\hat{\rho}\}(s\{\chi_q\})_{01}} e^{-\hat{\rho}}. \quad (\text{D.263})$$

Above we have introduced a new function and some associated constants, which go as

$$H_{\{\hat{\rho}\}(s\{\chi_q\})_{01}} := \check{r}_{\{\Omega\}}^{\mathfrak{d}_i} \left[\Gamma_{\{\Omega\}}^{\mathfrak{L}-2W\delta_{s1}} \right]_{01} \hat{\rho}^{\frac{\mathfrak{L}_0}{W} - \delta_{s_01} - \delta_{s_11}}, \quad (\text{D.264})$$

$$\hat{\mathfrak{L}}_0 := \frac{\mathfrak{L}_0 + \mathfrak{L}_1}{2} + \mathfrak{d} = l_0 + l_1 - \delta_{W1}(u_0 + u_1) + \mathfrak{d} + 3. \quad (\text{D.265})$$

We note that

$$d\hat{\rho} = \frac{W}{\hat{r}} \hat{\rho}^{1-\frac{1}{W}} dr \quad (\text{D.266})$$

as such we may write

$$dr G_{\{r\}(s\{\chi_q\})_{01}} = d\hat{\rho} H_{\{\hat{\rho}\}(s\{\chi_q\})_{01}} \quad (\text{D.267})$$

The expressions above suggests we should redefine $\hat{\rho}_{\{\Omega\}}$ as our integration variable.

Over the radial integration we can consider $\{\Omega\}$ to be constant, and so we will drop this subscript from expressions that form the radial integrand. We will restore this subscripts as needed, and once the analytical radial integration is completed we will ensure these subscripts are included in the numerical solid-angle integrand. With this in mind we may write our analytic integral over $\hat{\rho}$ as

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} = \int_0^\infty d\hat{\rho} H_{\{\hat{\rho}\}(s)_{01}} \mathcal{L}_{(s\{\rho\})_{01}} e^{-\mathcal{F}_{i\{\hat{\rho}\}}}. \quad (\text{D.268})$$

Expanding H as previously defined yields

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} = \check{r}_{\{\Omega\}}^{\mathfrak{d}_i} \left[\Gamma_{\{\Omega\}}^{\mathcal{L}-2W\delta_{s\bar{1}}} \right]_{01} \mathcal{R}_{i(s)_{01}\{\Omega\}}, \quad (\text{D.269})$$

with the radial integral partitioned as $\mathcal{R}$. We'll focus on the radial integral, namely

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} := \int_0^\infty d\hat{\rho} \hat{\rho}^{\frac{\mathfrak{L}_{\mathfrak{d}_i}-\delta_{s_0\bar{1}}-\delta_{s_1\bar{1}}-1}{W}} \mathcal{L}_{(s\{\rho\})_{01}} e^{-\mathcal{F}_{i\{\hat{\rho}\}}}, \quad (\text{D.270})$$

$$\mathcal{L}_{(s\{\rho\})_{01}} := \left[\frac{\mathfrak{K}!}{k!} \sum_{t=-2u}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} L_{(k+t)\{\rho\}}^{(l+u-\delta_{s\bar{1}})} \right]_{01}. \quad (\text{D.271})$$

In the analytical radial integral above we introduced a new function,

$$\mathcal{F}_{i\{\hat{\rho}\}} := \hat{\rho} + \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) \hat{\rho}^{\frac{1}{W}} + \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2 \hat{\rho}^{\frac{2}{W}}, \quad (\text{D.272})$$

$$= (1 + \mathcal{A}_i) \hat{\rho} + \mathcal{B}_i \hat{\rho}^{\frac{2}{W^2}}, \quad (\text{D.273})$$

where

$$\mathcal{A}_i := \delta_{W1} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) + \delta_{W2} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2, \quad (\text{D.274})$$

$$\mathcal{B}_i := \delta_{W1} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2 + \delta_{W2} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) \quad (\text{D.275})$$

We can rewrite $\mathcal{R}$ by pulling the summation over $(t)_{01}$ outside the analytical radial

integration.

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{t=-2u}^{u+\delta_{s1}} \frac{\mathfrak{R}!}{k!} h_{(kl, st)}^{(u)} \right]_{01} \mathcal{R}'_{i(st)_{01}}, \quad (\text{D.276})$$

$$\mathcal{R}'_{i(st)_{01}} := \int_0^\infty d\hat{\rho} \hat{\rho}^{\frac{\hat{\mathfrak{z}}_{\delta_i}}{W} - \delta_{s_0 1} - \delta_{s_1 1} - 1} \left[L_{(k+v)\{\rho\}}^{(+u-\delta_{s1})} \right]_{01} e^{-\mathcal{F}_{i\{\rho\}}}. \quad (\text{D.277})$$

In analogy to our rules for subscripted brackets, we introduce a related rule for summation symbols when they are the only symbols included in the brackets. We want to be clear that this rule might be counterintuitive when first seen because it represents an abuse of commonly understood rules of parenthesis and bracket in arithmetic. However, we think the formal meaning of repeated symbols is clear and it can be understood by the reader that in certain contexts repeated symbols do necessarily mean "product" exclusively. As such, we want to make it clear that symbols outside and to the right of subscripted brackets are understood to be part of the summation, not outside it as might be expected. Namely, we define the following.

$$\left[\sum_{n=a}^b \right]_{01} (f_n)_{01} := \sum_{n_0=a_0}^{b_0} \sum_{n_1=a_1}^{b_1} f_{n_0 n_1} \quad (\text{D.278})$$

For example, equation (D.334) above can be considered and understood in light of the definition above to be such that

$$\left[\sum_{s=-u}^{s=+u} \right]_{01} \left(\mathfrak{e}_{i(ml; \eta \lambda, s)}^{(u)} \mathcal{G}'_{i(s)\{\Omega\}} \right)_{01} = \sum_{s_0=-u_0}^{s_0=+u_0} \sum_{s_1=-u_1}^{s_1=+u_1} \mathfrak{e}_{i(ml; \eta \lambda, s)}^{(u)_{01}} \mathcal{G}'_{i(s)_{01}\{\Omega\}}. \quad (\text{D.279})$$

While it represents a slight abuse of standard bracket notation, we hope that it is clear that this notation has the benefit of compacting otherwise long and repetitive expressions involving double or greater summation. To prevent confusion, we note that forms with brackets containing summation symbol AND summations terms, for example as in equation (D.271) should continue to be understood as a product of terms subscripted with 0 and 1 respectively.

To complete the radial integration, we consider an integral of the form given below.

$$\mathcal{R}' := \int_0^\infty d\rho \rho^{\mathfrak{g}-1} L_{\mathfrak{T}_0\{\rho(1+\Delta)\}}^{(\mathfrak{a}_0)} L_{\mathfrak{T}_1\{\rho(1-\Delta)\}}^{(\mathfrak{a}_1)} e^{-(1+\mathcal{A})\rho - \mathcal{B}\rho^{\frac{2}{W^2}}}, \quad (\text{D.280})$$

$$= \left[\sum_{u=0}^{\mathfrak{T}} \sum_{v=u}^{\mathfrak{T}} \right]_{01} \mathfrak{F}_{\mathfrak{g}(uv)_{01}} \mathcal{G}_{(W\mathfrak{g});(u)_{01}\{\Delta\mathcal{Z}_{\mathcal{G}}\}}^{(W)} \mathcal{H}_{(W\mathfrak{g}+Wu)\{\mathcal{Z}_{\mathcal{H}}\}}^{(W)}. \quad (\text{D.281})$$

The L functions used above are the associated Laguerre functions. The significance of the above expression is that it includes the coefficients of the Laguerre polynomials. It should be noted that the summations above can involve sequences of terms of near magnitude but alternating sign. This is a computation ill-posed to be completed numerically, so we will endeavor to simplify the expressions as much as possible. Some of the expressions used in the formula above are defined below

$$\mathfrak{F}_{\mathfrak{g}(uv)_{01}} := \left[\frac{(-1)^{u+v} \sqrt{\mathfrak{T}!(\mathfrak{T}+\mathfrak{a})!}}{(\mathfrak{T}-v)!(\mathfrak{a}+v)!u!(v-u)!} \right]_{01} \Gamma(\mathfrak{g} + v_0 + v_1), \quad (\text{D.282})$$

$$\mathcal{G}_{L;(u)_{01}\{\Delta\mathcal{Z}\}}^{(W)} := (1 - \mathcal{Z}(1 + \Delta))^{u_0} (1 - \mathcal{Z}(1 - \Delta))^{u_1} \mathcal{Z}^{\frac{L}{W}}, \quad (\text{D.283})$$

$$\mathcal{H}_{M\{\mathcal{Z}\}}^{(W)} := \left(\delta_{W1} \frac{\sqrt{\pi}}{\Gamma[\frac{M+1}{2}]} \mathcal{Z}^M + \delta_{W2} \right) h_{M\{\mathcal{Z}\}}, \quad (\text{D.284})$$

$$\mathcal{Z}_{\mathcal{G}} := \frac{1}{1 + \mathcal{A}}, \quad (\text{D.285})$$

$$\mathcal{Z}_{\mathcal{H}} := \delta_{W1} \left(\frac{1 + \mathcal{A}}{2\sqrt{\mathcal{B}}} \right) + \delta_{W2} \left(\frac{\mathcal{B}}{2\sqrt{1 + \mathcal{A}}} \right). \quad (\text{D.286})$$

The expression h_{WL} refers to the hypergeometric function defined in the [Supplementary Materials](#).

We can use the above expressions if we set the constants to appropriate values. As we are defining new quantities and functions, we've taken the liberty of explicitly including the dependence over solid-angle in the final expression of indirect integra-

tion variables. We define or set the following

$$\mathfrak{h}_{\mathfrak{d}(s)_{01}} := W \mathfrak{g} = \hat{\mathfrak{L}}_{\mathfrak{d}} - W(\delta_{s_0 \bar{1}} + \delta_{s_1 \bar{1}}), \quad (\text{D.287})$$

$$\mathfrak{a}_n := (\mathfrak{l} + \mathfrak{u} - \delta_{s \bar{1}})_n, \quad (\text{D.288})$$

$$\mathfrak{T}_n := (k + t)_n, \quad (\text{D.289})$$

$$\Delta := \frac{\rho_0 - \rho_1}{2\hat{\rho}}. \quad (\text{D.290})$$

We can further refine Δ .

$$\Delta = \frac{\Gamma_0^{2W} - \Gamma_1^{2W}}{2} = \frac{\check{r}^W}{2} \left[\left(\frac{\vartheta}{\eta} \right)_0^W - \left(\frac{\vartheta}{\eta} \right)_1^W \right], \quad (\text{D.291})$$

$$= \frac{1}{2} \left(\frac{\hat{\eta}}{\hat{\vartheta} \Xi} \right)^W \left(\frac{(\eta_1 \vartheta_0)^W - (\eta_0 \vartheta_1)^W}{[\eta]_{01}^W} \right), \quad (\text{D.292})$$

$$= \frac{1}{2} \left(\frac{4[\eta]_{01}^W}{(\eta_0^W + \eta_1^W)(\vartheta_0^W + \vartheta_1^W) \Xi^W} \right) \left(\frac{(\eta_1 \vartheta_0)^W - (\eta_0 \vartheta_1)^W}{[\eta]_{01}^W} \right), \quad (\text{D.293})$$

$$= \frac{(\eta_1^W - \eta_0^W)(\vartheta_0^W + \vartheta_1^W) + (\eta_0^W + \eta_1^W)(\vartheta_0^W - \vartheta_1^W)}{(\vartheta_0^W + \vartheta_1^W)(\eta_0^W + \eta_1^W) \Xi^W}. \quad (\text{D.294})$$

Distributing the denominator through the sum in the numerator, and identifying existing quantities, we may state our final expression for Δ to be

$$\Delta_{\{\Omega\}} = \frac{\mathfrak{f} + \epsilon_{\{\Omega\}}}{1 + \mathfrak{f} \epsilon_{\{\Omega\}}} \quad (\text{D.295})$$

We must also consider $\mathcal{L}_{\mathcal{G}}$ and $\mathcal{L}_{\mathcal{H}}$. We had defined these quantities in terms of $\mathcal{A}$ and $\mathcal{B}$. Here we recall that we had previously identified those quantities as indirect integration variables, namely

$$\tilde{\mathcal{A}}_i := \delta_{W1} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) + \delta_{W2} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2, \quad (\text{D.274a})$$

$$\tilde{\mathcal{B}}_i := \delta_{W1} \beta_i \left(\frac{\check{r}}{\mathfrak{r}} \right)^2 + \delta_{W2} \tilde{\alpha}_i \left(\frac{\check{r}}{\mathfrak{r}} \right) \quad (\text{D.275a})$$

Using these expressions for $\mathcal{A}$ and $\mathcal{B}$, we can write the particular $\mathcal{Z}_{\mathcal{G}}$ and $\mathcal{Z}_{\mathcal{H}}$ we wish to use as functions over solid-angle. To wit,

$$\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}} = \frac{1}{1 + \delta_{W1} \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right) + \delta_{W2} \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right)^2}, \quad (\text{D.296})$$

$$\tilde{\mathcal{Z}}_{i\mathcal{H}\{\Omega\}} = \frac{\delta_{W1} + \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right)}{\sqrt{\delta_{W2} + \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\tau} \right)^2}}. \quad (\text{D.297})$$

While we wanted to ensure the solid-angle dependence of Δ and the $\mathcal{Z}$ were clear in the expressions above, we will resume dropping this dependence in expressions while examining the analytic radial integration. In light of the arguments above, we can write

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{t=-2u}^{u+\delta_{s\bar{1}}} \frac{\mathfrak{K}!}{k!} h_{(kl,st)}^{(u)} \right]_{01} \mathcal{R}'_{i(st)_{01}}, \quad (\text{D.298})$$

$$\mathcal{R}'_{i(st)_{01}} = \left[\sum_{u=0}^{k+t} \sum_{v=u}^{k+t} \right]_{01} \mathfrak{F} \left(\frac{\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}}{W} \right)_{(uv)_{01}} \mathcal{G}_{i(su)_{01}} \mathcal{H}_{i(s)_{01}(v_0+v_1)}. \quad (\text{D.299})$$

where we found it convenient to introduce the following sequences

$$\mathcal{G}_{i(su)_{01}} := \mathcal{G}_{\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}(u)_{01} \{ \Delta \tilde{\mathcal{Z}}_{i\mathcal{G}} \}}^{(W)}, \quad (\text{D.300})$$

$$\mathcal{H}_{i(s)_{01}w} := \mathcal{H}_{(\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}+Ww) \{ \tilde{\mathcal{Z}}_{i\mathcal{H}} \}}^{(W)}. \quad (\text{D.301})$$

Earlier, we had pulled out the summation over t to complete the radial integration analytically. We can reorganize the terms of the double summation in $\mathcal{R}$ to allow the summation over t to be completed. We introduce a new limit for summations that goes as

$$\mathfrak{K}_n = (k + u + \delta_{s\bar{1}})_n. \quad (\text{D.302})$$

We recognize that u can range from 0 to $\mathfrak{K}$, v can range from u to $\mathfrak{K}$, and t must always be larger than $v - k$. We reorganize the order of our summations, and we pull out to

the front summations over u_0 , and u_1 . We may write

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{u=0}^{\mathfrak{K}} \sum_{v=u}^{\mathfrak{K}} \right]_{01} \mathfrak{A}_{\mathfrak{d}_i(suv)_{01}} \mathcal{G}_{i(su)_{01}} \mathcal{H}_{i(s)_{01}}(v_0+v_1) \quad (\text{D.303})$$

where we introduced

$$\mathfrak{A}_{\mathfrak{d}(suv)_{01}} := \left[\sum_{t=\text{Max}[v-k, -2u]}^{u+\delta_{s\bar{1}}} \frac{\mathfrak{K}!}{k!} h_{(kl, st)}^{(u)} \right]_{01} \mathfrak{F}_{\left(\frac{\mathfrak{h}_{\mathfrak{d}}}{W}\right)}(uv)_{01} \quad (\text{D.304})$$

However, we can further simplify this from a four-index summation to a three-index summation by introducing $w = v_0 + v_1$. We do this by introducing a summation over a Kronecker delta, going as

$$\mathcal{R}_{i(s)_{01}} = \left[\sum_{u=0}^{\mathfrak{K}} \sum_{v=u}^{\mathfrak{K}} \right]_{01} \mathfrak{A}_{\mathfrak{d}_i(suv)_{01}} \mathcal{G}_{i(su)_{01}} \sum_{w=u_0+u_1}^{\mathfrak{K}_0+\mathfrak{K}_1} \delta_{w, v_0+v_1} \mathcal{H}_{i(s)_{01}} w, \quad (\text{D.305})$$

$$= \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \sum_{w=u_0+u_1}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(su)_{01}w} \mathcal{G}_{i(su)_{01}} \mathcal{H}_{i(s)_{01}} w. \quad (\text{D.306})$$

where we introduce a new constant

$$\mathfrak{A}_{\mathfrak{d}(su)_{01}w} := \left[\sum_{v=u}^{\mathfrak{K}} \right]_{01} \delta_{w, v_0+v_1} \mathfrak{A}_{\mathfrak{d}(suv)_{01}}. \quad (\text{D.307})$$

$\mathfrak{A}$ is constant over integration, and in particular its value does not depend on any runtime parameters, as long as the basis and operator/potential specification occurs at or prior to compile time. It is a potentially complicated expression, and we take some effort to simplify it and attempt to find a form of the expression that may be computed in reusable pieces. We can consider

$$\mathfrak{A}_{\mathfrak{d}(su)_{01}w} = (-1)^{u_0+u_1+w} \Gamma \left[\frac{\mathfrak{h}_{\mathfrak{d}(s)_{01}}}{W} + w \right] \mathfrak{A}'_{(su)_{01}w}, \quad (\text{D.308})$$

$$\mathfrak{A}'_{(su)_{01}w} = \left[\sum_{v=u}^{\mathfrak{K}} \right]_{01} \delta_{w, v_0+v_1} [\mathfrak{A}''_{suv}]_{01}. \quad (\text{D.309})$$

The product of coefficients above has factors that depend only on information from a single state. We can write this single-state coefficient as so,

$$\mathfrak{A}_{suv}'' = \frac{1}{\sqrt{k!(k+l)!}} \frac{\mathfrak{K}!}{u!(v-u)!(\mathfrak{K}-v)!} \frac{(k+l)!}{(\mathfrak{a}-2u+v)!} \mathfrak{G}_{sv}, \quad (\text{D.310})$$

$$= \frac{1}{\sqrt{k!(k+l)!}} \binom{\mathfrak{K}}{u} \binom{\mathfrak{K}-u}{\mathfrak{K}-v} (k+l)^{\mathfrak{K}-v} \mathfrak{G}_{sv}, \quad (\text{D.311})$$

$$\mathfrak{G}_{sv} = \frac{(\mathfrak{a}-2u+v)!}{(\mathfrak{a}+v)!} \mathfrak{H}_{sv}^{(u)}, \quad (\text{D.312})$$

$$= (1-u) \mathfrak{H}_{sv}^{(0)} + u \frac{\mathfrak{H}_{sv}^{(1)}}{(\mathfrak{l}-\delta_{s\bar{1}}+v+1)(\mathfrak{l}-\delta_{s\bar{1}}+v)}. \quad (\text{D.313})$$

We have

$$\mathfrak{H}_{sv}^{(u)} = \sum_{t=\text{Max}[v-k, -2u]}^{u+\delta_{s\bar{1}}} h_{kl,st}^{(u)} \frac{(\mathfrak{K}-v)! \sqrt{(k+t)!(k+t+\mathfrak{a})!}}{(k+t-v)! \sqrt{k!(k+l)!}}. \quad (\text{D.314})$$

We also find it convenient to explicitly the separate the $u = 0$ (kinetic-off) and the $u = 1$ (kinetic-on) forms somewhat. I note that when the kinetic operator is "off" that $s = 0$ and that $v - k \leq 0$. This means

$$\mathfrak{H}_{sv}^{(0)} = 1. \quad (\text{D.315})$$

We also have

$$\mathfrak{H}_{sv}^{(1)} = \sum_{t=\text{Max}[v-k, -2]}^{1+\delta_{s\bar{1}}} g_{kl,st} \frac{(\mathfrak{K}-v)! \sqrt{(k+t)!(k+t+\mathfrak{l}+1-\delta_{s\bar{1}})!}}{(k+t-v)! \sqrt{k!(k+l)!}}. \quad (\text{D.316})$$

We note that we can treat this summation as a polynomial in $\mathfrak{K} - v$. Let us define

$$\mathfrak{H}_{sv}' = \sum_{t=0}^{\text{Min}[\mathfrak{N}, 3+\delta_{s\bar{1}}]} v^t \mathfrak{G}_{st}, \quad (\text{D.317})$$

$$\mathfrak{G}_{st} := g_{kl,s(1+\delta_{s\bar{1}}-t)} \frac{\sqrt{(k+1+\delta_{s\bar{1}}-t)!(k+\mathfrak{l}+2-t)!}}{\sqrt{k!(k+l)!}}. \quad (\text{D.318})$$

We may pull together these realizations and write

$$\mathfrak{A}''_{su(\mathfrak{K}-v)} = \frac{1}{\sqrt{k!(k+l)!}} \binom{\mathfrak{K}}{u} \binom{\mathfrak{K}-u}{v} (k+l)^v (1+u\mathfrak{J}_{sv}), \quad (\text{D.319})$$

$$\mathfrak{J}_{sv} = \frac{\mathfrak{H}'_{sv}}{(k+l+2-v)(k+l+1-v)} - 1, \quad (\text{D.320})$$

$$\mathfrak{H}'_{sv} = \sum_{t=0}^{\text{Min}[v, 3+\delta_{s\bar{1}}]} v^t \mathfrak{G}_{st}, \quad (\text{D.321})$$

$$\mathfrak{G}_{st} = g_{kl, s(1+\delta_{s\bar{1}}-t)} \frac{\sqrt{(k+1+\delta_{s\bar{1}}-t)!(k+l+2-t)!}}{\sqrt{k!(k+l)!}}. \quad (\text{D.322})$$

Please note that, in the interests of space and clarity, we did not make explicit the fact that $\mathfrak{A}''$ and the various basis indices and parameters within the expression should be indexed by the state identifiers 0 or 1.

It is generally less computationally intense to use the Kronecker delta to reduce the above double summation to a single summation to implement, rather than implementing a double summation with many zero terms. To accomplish this we let $v_0 \rightarrow \mathfrak{K}_0 - v$ and $v_1 \rightarrow w - \mathfrak{K}_0 + v$. Our expression becomes

$$\mathfrak{A}'_{(su)_{01}w} = \sum_{v=\text{Max}[0, \mathfrak{K}_0+u_1-w]}^{\text{Min}[\mathfrak{K}_0-u_0, \mathfrak{K}_0+\mathfrak{K}_1-w]} \mathfrak{A}''_{(su)_0(\mathfrak{K}_0-v)} \mathfrak{A}''_{(su)_1(w-\mathfrak{K}_0+v)}. \quad (\text{D.323})$$

Earlier we found some sequences of values we needed to compute to generate our matrix elements.

$$\tilde{\mathcal{G}}_{i(su)_{01}} := \mathcal{G}_{\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}; (u)_{01} \{ \Delta \tilde{\mathcal{Z}}_{i\mathcal{G}} \}}^{(W)}, \quad (\text{D.300})$$

$$\tilde{\mathcal{H}}_{i(s)_{01}w} := \mathcal{H}_{(\mathfrak{h}_{\mathfrak{d}_i(s)_{01}}+W)w \{ \tilde{\mathcal{Z}}_{i\mathcal{H}} \}}^{(W)}. \quad (\text{D.301})$$

We note that $\mathcal{G}$ and $\mathcal{H}$ above will simplify to less complicated forms depending on the values of $\tilde{\mathcal{Z}}$ which themselves depend on the functions $\tilde{\alpha}$, β and constant W . If both $\delta_{w1}\tilde{\alpha} \equiv 0$ and $\delta_{w2}\beta \equiv 0$ identically, then $\mathcal{Z}_{\mathcal{G}} \equiv 1$. We also see another simplification

when $\Delta \equiv 0$ identically as well. There is a third form when both conditions apply. To wit, we have

$$\mathcal{G}_{L;(u)_{01}\{\Delta(1)\}}^{(W)} = (-1)^{u_0} \Delta^{u_0+u_1}, \quad (\text{D.324})$$

$$\mathcal{G}_{L;(u)_{01}\{(0)\tilde{\mathcal{Z}}_{\mathcal{G}}\}}^{(W)} = (1 - \tilde{\mathcal{Z}}_{\mathcal{G}})^{u_0+u_1} \tilde{\mathcal{Z}}_{\mathcal{G}}^{\frac{1}{W}}, \quad (\text{D.325})$$

$$\mathcal{G}_{L;(u)_{01}\{(0)(1)\}}^{(W)} = \delta_{0,u_0+u_1}. \quad (\text{D.326})$$

We can also simplify $\tilde{\mathcal{H}}$. In the case when $W = 1$ and $\beta = 0$ we see that $\tilde{\mathcal{Z}}_{\mathcal{H}}$ becomes ill-defined. Returning to the original analytical integration we can see that in such a case, $\tilde{\mathcal{H}} \equiv 1$. This can also be shown as a limit to complex infinity, which we will denote as $\infty_{\mathbb{C}}$. Namely, we see $\lim_{\tilde{\mathcal{Z}}_{\mathcal{H}} \rightarrow \infty_{\mathbb{C}}} \tilde{\mathcal{H}} = 1$. In essence, as β tends to become identically zero, it is the case that $\tilde{\mathcal{H}}$ tends to become identically unity. A similar effect emerges when $W = 2$ and $\tilde{\alpha} \equiv 0$, but in that case $\tilde{\mathcal{Z}}_{\mathcal{H}} \equiv 0$. However, $\tilde{\mathcal{H}}$ becomes undefined for that value of $\tilde{\mathcal{Z}}_{\mathcal{G}}$. As before, examining the analytic continuation versus direct recalculation with $\tilde{\alpha} = 0$ yields the same result – as before, under these conditions $\tilde{\mathcal{H}} \equiv 1$.

These changes to the terms of the summation over u_0 , u_1 , and w significantly alters the structure of the summation. We present the following contractions of the main precalculated coefficients, namely

$$\mathfrak{A}_{\mathfrak{d}(s)_{01}uw} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \delta_{u,u_0+u_1} (-1)^{u_0} \mathfrak{A}_{\mathfrak{d}(s)_{01}w}, \quad (\text{D.327})$$

$$\mathfrak{A}'_{\mathfrak{d}(s)_{01}uw} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \delta_{u,u_0+u_1} \mathfrak{A}_{\mathfrak{d}(s)_{01}w}, \quad (\text{D.328})$$

$$\mathfrak{A}_{\mathfrak{d}(s)_{01}w} = \mathfrak{A}_{\mathfrak{d}(s0)_{01}w}. \quad (\text{D.329})$$

In instances when $\tilde{\mathcal{H}} \equiv 1$ we find an additional contraction along the w index, which

provides the following

$$\alpha_{\mathfrak{d}(s u)_{01}} = \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \mathfrak{A}_{\mathfrak{d}(s u)_{01} w}, \quad (\text{D.330})$$

$$\alpha_{\mathfrak{d}(s)_{01} u} = \left[\sum_{u=0}^{\mathfrak{R}} \right]_{01} \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \delta_{u, u_0+u_1} (-1)^{u_0} \mathfrak{A}_{\mathfrak{d}(s u)_{01} w}, \quad (\text{D.331})$$

$$\alpha'_{(s)_{01} u} = \left[\sum_{u=0}^{\mathfrak{R}} \right]_{01} \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \delta_{u, u_0+u_1} \mathfrak{A}_{\mathfrak{d}(s u)_{01} w}, \quad (\text{D.332})$$

$$\alpha_{(s)_{01}} = \sum_{w=u}^{\mathfrak{R}_0+\mathfrak{R}_1} \mathfrak{A}_{\mathfrak{d}(s 0)_{01} w}. \quad (\text{D.333})$$

As before, we note that the summations to generate these coefficients feature alternating signs, making this computations ill-posed to be done using standard floating-point computations. Additionally, the coefficients themselves can become quite large. These summations have problems with both overflow and underflow error when floating point numbers are used.

To work with this form within our numerical integrator, which will use standard floating-point numerical computation, we opted to compute the coefficients analytically and store the results as a floating point number. The analytic computation can be achieved and automated with a computer algebra system, and for this work we used the SymPy Python package. We can then store the results as floating-point numbers in a database that can be recalled as needed. More information is available in [3.2](#).

Such a database can be constructed at compile time, or may be referenced from previous runs of the software. As such, we call this set of coefficients the *precalculated coefficients*. Unfortunately, the database does prove to be quite large, making portability between systems an issue. Additionally, we find that pulling the needed coefficients from the database into runtime does add a non-negligible amount of time to add computation. However, this is anticipated to be a one-time expenditure of memory access time during a computation, one which is anticipated to scale linearly with

problem size.

In light of these simplifications and considerations we can attempt to rewrite our integral by assessing and partitioning certain expressions. Our goal with this formulation of the matrix elements is based on minimizing repeated computations as well as exploiting certain features of our subbases/superbasis structure. In particular, within a subbasis, we will have $\eta_0 = \eta_1 = \hat{\eta}$ and so we will also have $\mathfrak{f} = 0$, $\Xi_{\{\Omega\}} \equiv 1$, and quantities such as $\hat{\eta}/\eta_n = 1$. Additionally, when considering matrix elements with $q_0 = q_1 = z$ within the same subbasis, we find that $\vartheta_0 \equiv \vartheta_1$ which also permits further simplifications in the expressions we formulate below.

$$(\mathcal{M})_{\mathfrak{K},01}^{(\kappa',j)} := \mathfrak{E}^{u_0+u_1+s} \mathfrak{C} \mathcal{I}, \quad (\text{D.334})$$

$$\mathfrak{C} := \frac{\xi_0}{\left[\left(\frac{\eta}{\hat{\eta}} \right)^{\frac{s}{2}+2u} \right]_{01} \left(\frac{\hat{\eta}}{\mathfrak{r}} \right)^{2u_0+2u_1}}, \quad (\text{D.335})$$

$$\mathcal{I} := \int_{\mathcal{S}_0} d\Omega \left[\left(\frac{\vartheta_{\{\Omega\}}}{\hat{\vartheta}_{\{\Omega\}}} \right)^{\frac{s}{2}} Q_{m\zeta\{\phi_q\}} \right]_{01} \mathcal{S}_{\{\Omega\}}, \quad (\text{D.336})$$

$$\mathcal{S}_{\{\Omega\}} := \left[\sum_{s=-u}^{s=+u} \right]_{01} \sum_i \mathcal{P}_{(s)_{01}\{\Omega\}} \mathcal{X}_{\mathfrak{d}_i\{\Omega\}} \mathcal{V}_{i\{\Omega\}} \mathcal{R}_{i(s)_{01}\{\Omega\}}, \quad (\text{D.337})$$

$$\mathcal{P}_{(s)_{01}\{\Omega\}} := \left[\frac{k! \mathfrak{H}_{ml;\lambda,s}^{(\kappa)}}{\mathfrak{K}! \left(\frac{\vartheta_{\{\Omega\}} \hat{\mathfrak{r}}_{\{\Omega\}}}{\eta} \right)^{W \delta_{s1}}} P_{(l+2s)m;\lambda\{\chi_q\}} \right]_{01}, \quad (\text{D.338})$$

$$\mathcal{X}_{\mathfrak{d}\{\Omega\}} := \left(\frac{\hat{\eta}}{\mathfrak{r} \hat{\vartheta}} \right)^{\mathfrak{d}} (\Xi_{\{\Omega\}})^{-\hat{\mathcal{L}}_{\mathfrak{d}}}. \quad (\text{D.339})$$

Above we have defined the following constants,

$$\mathfrak{K} := k + u + \delta_{s1}, \quad (\text{D.340})$$

$$\hat{\mathcal{L}}_{\mathfrak{d}} := l_0 + l_1 - \delta_{W1}(u_0 + u_1) + \mathfrak{d} + 3. \quad (\text{D.341})$$

We can articulate the various conditions that affect the summation structure of $\mathcal{R}$.

There are three conditions that determine which form should be used. They are

$$\delta_{W1}\tilde{\alpha}_i + \delta_{W2}\beta_i = 0 \quad \Rightarrow \quad \tilde{\mathcal{E}}_i \equiv 1, \quad (\text{C1a})$$

$$\eta_0 = \eta_1 \text{ and } \lambda_0 = \lambda_1 \text{ and } q_0 = q_1 \quad \Rightarrow \quad \Delta \equiv 0, \quad (\text{C1b})$$

$$\delta_{W1}\beta_i + \delta_{W2}\tilde{\alpha}_i = 0 \quad \Rightarrow \quad \tilde{\mathcal{H}} \equiv 1. \quad (\text{C2})$$

If condition (C2) is not true, then the radially integrated part of the integrand will go as

$$\mathcal{R}_{i(s)_{01}\{\Omega\}} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(su)_{01}w} \mathcal{G}_{i(u)_{01}\{\Omega\}} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}, \quad (\text{D.342})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C1a})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(s)_{01}uw} \mathcal{G}_{u\{\Omega\}} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}, \quad (\text{D.343})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C1b})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \sum_{w=u}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}'_{\mathfrak{d}_i(s)_{01}uw} \tilde{\mathcal{G}}_{i(s)_{01}u\{\Omega\}} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}, \quad (\text{D.344})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C1a+C1b})} = \sum_{w=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{A}_{\mathfrak{d}_i(s)_{01}w} \tilde{\mathcal{H}}_{i(s)_{01}w\{\Omega\}}. \quad (\text{D.345})$$

However, if condition (C2) is true, then the radially integrated part of the integrand will go as

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C2})} = \left[\sum_{u=0}^{\mathfrak{K}} \right]_{01} \mathfrak{a}_{\mathfrak{d}_i(su)_{01}} \tilde{\mathcal{G}}_{(u)_{01}\{\Omega\}}, \quad (\text{D.346})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C2;C1a})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{a}_{\mathfrak{d}_i(s)_{01}u} \mathcal{G}_{u\{\Omega\}}, \quad (\text{D.347})$$

$$\mathcal{R}_{i(s)_{01}\{\Omega\}}^{(\text{C2;C1b})} = \sum_{u=0}^{\mathfrak{K}_0+\mathfrak{K}_1} \mathfrak{a}'_{\mathfrak{d}_i(s)_{01}u} \tilde{\mathcal{G}}_{i(s)_{01}u\{\Omega\}}, \quad (\text{D.348})$$

$$\mathcal{R}_{i(s)_{01}}^{(\text{C2;C1a+C1b})} = \mathfrak{a}_{\mathfrak{d}_i(s)_{01}}. \quad (\text{D.349})$$

where we employ sequences of the form

$$\tilde{\mathcal{G}}_{i(s)u\{01\}\{\Omega\}} := \tilde{\mathcal{G}}_{\mathfrak{h}_{\mathfrak{d}_i(s)01};(u)_{01}\{\Delta_{\{\Omega\}}\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}}\}}^{(W)}, \quad (\text{D.350})$$

$$\mathcal{G}_{u\{\Omega\}} := \Delta_{\{\Omega\}}^u, \quad (\text{D.351})$$

$$\tilde{\mathcal{G}}_{i(s)01u\{\Omega\}} := \tilde{\mathcal{G}}_{\mathfrak{h}_{\mathfrak{d}_i(s)01};u\{\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}}\}}^{(W)}, \quad (\text{D.352})$$

$$\tilde{\mathcal{H}}_{i(s)01w\{\Omega\}} := \tilde{\mathcal{H}}_{(\mathfrak{h}_{\mathfrak{d}_i(s)01}+Ww);\{\tilde{\mathcal{Z}}_{i\mathcal{H}\{\Omega\}}\}}^{(W)}. \quad (\text{D.353})$$

The sequence values may be calculated with the following constant

$$\mathfrak{h}_{\mathfrak{d}(s)01} := \hat{\mathcal{L}}_0 - W(\delta_{s_0\bar{1}} + \delta_{s_1\bar{1}}), \quad (\text{D.354})$$

and with the following accessory functions.

$$\tilde{\mathcal{G}}_{L;(u)_{01}\{\Delta\tilde{\mathcal{Z}}\}}^{(W)} := (1 - \tilde{\mathcal{Z}}(1 + \Delta))^{u_0} (1 - \tilde{\mathcal{Z}}(1 - \Delta))^{u_1} \tilde{\mathcal{Z}}^{\frac{L}{W}}, \quad (\text{D.355})$$

$$\tilde{\mathcal{G}}_{L;u\{\tilde{\mathcal{Z}}\}}^{(W)} := (1 - \tilde{\mathcal{Z}})^u \tilde{\mathcal{Z}}^{\frac{L}{W}}, \quad (\text{D.356})$$

$$\tilde{\mathcal{H}}_{M\{\tilde{\mathcal{Z}}\}}^{(W)} := \left(\delta_{W1} \frac{\sqrt{\pi}}{\Gamma\left[\frac{M+1}{2}\right]} \tilde{\mathcal{Z}}^M + \delta_{W2} \right) h_M\{\tilde{\mathcal{Z}}\}, \quad (\text{D.357})$$

$$\Delta_{\{\Omega\}} = \frac{\mathfrak{f} + \epsilon_{\{\Omega\}}}{1 + \mathfrak{f}\epsilon_{\{\Omega\}}}, \quad (\text{D.358})$$

$$\tilde{\mathcal{Z}}_{i\mathcal{G}\{\Omega\}} = \frac{1}{1 + \delta_{W1} \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right) + \delta_{W2} \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right)^2}, \quad (\text{D.359})$$

$$\tilde{\mathcal{Z}}_{i\mathcal{H}\{\Omega\}} = \frac{\delta_{W1} + \tilde{\alpha}_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right)}{\sqrt{\delta_{W2} + \beta_{i\{\Omega\}} \left(\frac{\check{r}_{\{\Omega\}}}{\mathfrak{r}} \right)^2}}. \quad (\text{D.360})$$

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